

PROVISIONAL APPLICATION FOR PATENT

Filed under 35 U.S.C. § 111(b) and 37 C.F.R. § 1.53(c)

QUERYBOOK: A FACT-CENTRIC, PROVENANCE-TRACKED, MULTIMODAL KNOWLEDGE OPERATING SYSTEM WITH STRUCTURAL HALLUCINATION PREVENTION AND A PREDICTIVE COGNITIVE ARCHITECTURE

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Item | Content

Application type | Provisional, 35 U.S.C. § 111(b). Not examined. No claim fees apply.

Revision | Consolidated, revision 3 — updated draft incorporating the September 2026 additions (the enhanced Fact Unit record, record-fused encoding, the Fact Unit Language, and Contextual Value Models); supersedes all earlier drafts

Features disclosed | 293, each specified at function level in Appendix A

Claims | 354. Included although a provisional requires none, to evidence possession of the disclosed subject matter

Computational rules | 293 functions carry an explicit rule; 229 carry a directly-evaluable worked expression

Drawings | 8 figures described; formal sheets not required for a provisional filing

Priority effect | Establishes a constructive reduction to practice for everything disclosed herein, as of the filing date

Audit status | Mechanically audited: no gaps, no duplicate numbers, no dangling references, no forward dependencies. Ten repair claims added to close broken dependencies and omitted subject matter; eight multiple dependencies collapsed to single dependencies.

DRAFT FOR REVIEW BY QUALIFIED PATENT COUNSEL. NOT FOR FILING AS PREPARED.

Nothing in this document is legal advice. **WHY CONSOLIDATION MATTERS HERE:** a provisional pays a flat fee and carries no per-claim charge, so there is no cost reason to file a narrower disclosure than the one that exists. Subject matter omitted from this filing takes the date of whatever later document first discloses it, and once this application is filed 35 U.S.C. § 132(a) prohibits adding new matter to it. Everything the applicant has reduced to writing should therefore be in this single document before it is filed.

Confidential — proprietary and patent-sensitive.

Title of the Invention

[0001] QueryBook: a fact-centric, provenance-tracked, multimodal knowledge operating system with structural hallucination prevention and a predictive cognitive architecture.

Cross-Reference to Related Applications

[0002] This application claims the benefit under 35 U.S.C. § 119(e) of United States provisional application number 63/975,036, entitled QUERYBOOK, filed 3 February 2026, the entirety of which is incorporated herein by reference. That application is referred to below as the earlier application.

[0003] The earlier application discloses a system answering queries from the contents of a digital book, comprising an ingestion module, a structuring module, an indexing module, a query understanding module, an evidence retrieval engine, an answer synthesis module employing a constrained artificial intelligence model, and a guardrail enforcement module, together with a Universal Mode and an Enhanced Mode employing QueryBook Metadata Blocks, and four add-on modules designated Advanced Query Mode, Conversational Mode, Review Mode and Quiz and Test Mode.

[0004] The present application is a continuation in substance of the invention disclosed in the earlier application. It is not a different invention: the atomic knowledge unit, the requirement that every proposition of an answer be traceable to the source from which it was drawn, the citation carried with the answer, the refusal where the corpus does not support an answer, the structured representation in place of raw text, the concept and entity graph, semantic search over embedded representations, query parsing with entity disambiguation, the retrieval plan formed before retrieval, multi-turn interaction retaining continuity, traversal by conceptual adjacency, and assessment generated from and tied to the source, are each disclosed in the earlier application and are each carried forward here. What this application adds is depth of specification and extension of scope, not a substitution of principles.

[0005] Where a mechanism disclosed here corresponds to one disclosed in the earlier application under a different designation, the correspondence is stated in the section titled Continuity with the Earlier Application. Subject matter first disclosed in this application is entitled to the filing date of this application, and the continuity section distinguishes the two so that the distinction is available to a reader rather than left to inference.

[0006] This application is further supported by, and incorporates by reference in its entirety, the applicant's internal specification titled QueryBook Bible — Canonical Core Technology Specification, together with the applicant's Master Cross-Reference inventory identifying, for each disclosed feature, its responsible actor, its function-level algorithm, its governing computational rule where the function is quantitative, and a worked arithmetic expression evaluating that rule. Section citations of the form §NN.N or Chapter NN refer to that specification.

Statement Regarding Federally Sponsored Research or Development

[0007] Not applicable. The invention disclosed herein was not made under any contract with, or grant from, an agency of the United States Government, and no agency of the United States Government has any rights in the invention.

Statement of Assignment and Inventorship

[0008] The invention disclosed herein was made by Garry Stephan Howard and Christopher Reichert Caile, each a joint inventor and each a citizen of the United States.

[0009] The entire right, title and interest in and to the invention disclosed herein, in and to this application, and in and to any and all applications claiming benefit hereof — including any non-provisional, divisional, continuation, continuation-in-part, reissue, substitute or foreign counterpart application, and any patent issuing from any thereof — is assigned to Mediagration, LLC, a limited liability company, which is the applicant and assignee of record.

[0010] Correspondence concerning this application is to be directed to the practitioner of record associated with the customer number identified in the Application Data Sheet. An assignment recording the transfer described above is submitted to the Office separately from this specification in accordance with 37 CFR 3.11, and the recordation of that assignment rather than this statement effects the transfer of record.

[0011] The full legal name and residence of each inventor, and the name and address of the assignee, are stated in the Application Data Sheet accompanying this application and in the cover sheet filed therewith.

Continuity with the Earlier Application

[0012] The earlier application and this application describe one invention in differing terms. The earlier application describes it scoped to a digital book and names its components by their function in that setting; this application describes a corpus that may comprise one work or many, and names its components by the operation each performs. The following correspondences are stated so that a mechanism disclosed here may be traced to its disclosure in the earlier application.

[0013] The governing principle is disclosed in both and is unchanged. The earlier application requires that each sentence of a generated answer be traceable to at least one retrieved passage, that a citation accompany the output, that an answer containing an unsupported claim be rejected, and that a query the corpus does not address receive a statement to that effect rather than an answer. This application specifies the mechanism by which those requirements are met and states the resulting property structurally, but it states no requirement the earlier application did not.

Designation in the earlier application | Corresponding disclosure herein

Digital book | A corpus. §A. A corpus of a single work is an embodiment; the corpus is not limited to one work, and where it comprises one work the system operates as the earlier application describes.

Passage; structured representation | The knowledge record, termed the Fact Unit. §A. The earlier application discloses the structured representation of content in units to which citations resolve and to which each sentence of an answer must be traceable; this application specifies the schema of that unit — one assertion, its origin, its confidence, its safety classification and its access entitlement, bound inseparably.

Ingestion module | The multimodal ingestion pipeline. §B.

Structuring module; indexing module | Knowledge record construction, the concept structure and the indexes over them. §A, §H.

Query understanding module | Semantic parsing and intent classification. §D.

Evidence retrieval engine | Scope resolution and retrieval within the resolved scope. §D.

Answer synthesis module employing a constrained artificial intelligence model | Generation under a constrained permissible vocabulary. §F. The constraint is applied at each generation step rather than to a context supplied to a model, and a proposition not derivable from an admitted knowledge record is therefore unreachable rather than improbable.

Guardrail enforcement module | The same constraint, together with the boundary statement returned where no admitted record satisfies the threshold. §F. The earlier application describes this as a check applied to a generated answer; this application describes the mechanism by which the answer could not have been generated otherwise.

Out-of-scope detection | The boundary statement. §F.

Citation graph; entity-relationship graph | The provenance chain and the concept structure. §C, §H.

Structured book database | The knowledge record store and its indexes. §A.

QueryBook Metadata Blocks; Enhanced Mode | Author-supplied and publisher-supplied metadata admitted at ingestion, and the operation of the system where such metadata is present. §B.

Universal Mode | The operation of the system where no such metadata is present. §B.

Advanced Query Mode | Ingestion-time generation of probable questions with cited answers, and the query-anticipation structure over the corpus. §M.

Conversational Mode | The multi-turn session mode, its retention of session context, and the author-defined response profile. §D, §N.

Review Mode | Evaluation of a work against criteria supplied in a specification file, and the structured report produced therefrom. §N.

Quiz and Test Mode | Assessment generation from the knowledge records grounding a lesson, mastery measurement and credential issuance. §N.

[0014] Two refinements of expression are recorded, neither of which is a change of principle. The earlier application describes the answer as produced by a constrained model and then checked against the retrieved passages; this application specifies the vocabulary available to generation as constrained at each step, so that what the earlier application achieves by rejection this application achieves by construction. The requirement — that no proposition survive which the source does not support — is the same requirement in both.

[0015] And the earlier application describes the effect of that requirement as the prevention of hallucination, while this application states it as the structural property that a proposition not traceable to an admitted knowledge record cannot be produced. The second is the mechanism of the first, expressed so that it is testable and so that it asserts no absolute.

Background of the Invention

Field of the Invention

[0016] The invention relates to knowledge representation and automated reasoning, and more particularly to systems that construct a structured representation of asserted content from documentary, visual, auditory and sensor sources, that maintain that representation together with its origin and its degree of support, and that compose responses to queries from it.

[0017] The invention further relates to the fields of information retrieval over structured knowledge; natural language generation constrained by a stored representation; multimodal ingestion, in which content of differing kinds is reduced to a common representation; cross-lingual knowledge representation, in which an assertion is held independently of the language in which it was expressed; provenance and audit in automated systems; access control and rights enforcement over knowledge; immersive and three-dimensional presentation of structured content; the direction of capture and inspection equipment from a structured representation of an environment; and the calibration of a system's own predictive output against subsequently observed outcomes.

Description of the Related Art

[0018] Automated systems for answering questions from a body of content fall into several categories, and each carries limitations arising from its architecture rather than from its implementation.

[0019] Systems built upon large language models generate a response by predicting a statistically likely continuation of a token sequence. Because the response is produced by sampling from a distribution over continuations, three consequences follow. A continuation that is plausible under the distribution and false in the world is generated by the same mechanism as a true one, and the system holds no stored representation of what is the case against which the difference could be detected. The same query, submitted twice, may produce materially different responses, so a response cannot be reproduced and an earlier response cannot be verified against the system that produced it. And the parameters from which a response is produced encode no record of the sources from which they were derived, so an assertion in a response cannot be traced to a document, a passage or an author.

[0020] Retrieval-augmented arrangements interpose a retrieval step, supplying passages from a corpus before generation proceeds. The retrieval improves the material available to the generator and does not alter the generator. Generation remains a token-level probabilistic process, so a response may assert something the retrieved passages do not support, may blend two passages into a proposition neither contains, and remains as difficult to reproduce as before. The retrieved passages are unstructured text rather than discrete assertions, so a system of this kind holds no representation of the relationships among the things its corpus refers to, and cannot answer a question requiring several assertions to be combined unless a single passage happens to combine them.

[0021] Agent arrangements decompose a task into steps and execute them in sequence, each step consuming the output of the last. Where each step is itself probabilistic, an error introduced early is carried forward and elaborated rather than detected, and the resulting behaviour is difficult to audit

because no step records the basis on which it proceeded. The sequence is determined at execution time, so two runs of the same task may follow different paths, and a failure may not be reproducible for the purpose of diagnosing it.

[0022] Conversational systems and document-scoped question-answering tools restrict a model to a supplied corpus, which reduces the range over which an unsupported assertion may be drawn without changing how an assertion is produced. Confinement to a corpus is not the same as derivation from it. Such systems commonly present a citation alongside a response, but the citation is selected after the response is composed rather than constraining what may be composed, so a citation may accompany a proposition the cited passage does not support.

[0023] Across these categories, further limitations recur. Safety classification and access control are commonly implemented as a layer applied around stored content rather than as properties of the content itself, so a code path that reaches the content by another route reaches it without the classification. Where a confidence value is maintained for a stored assertion, it is commonly updated by addition or by a clamped increment, so a long-established assertion and a newly-recorded one are revised by the same amount on encountering the same contradiction, and the quantity of evidence behind an assertion is not represented at all. Meaning is bound to the language of expression, so the same assertion recorded in two languages is held as two unrelated items and a query in one language does not reach content in the other. And operation depends upon a network connection to a remote model, so the system is unavailable where connectivity is absent, intermittent or prohibited.

[0024] Needs therefore exist in these fields that the foregoing categories do not meet. There is a need for a system whose output is constrained by a stored representation at the point of generation rather than checked afterwards, such that an assertion not derivable from stored content cannot be produced. There is a need for reproducibility, such that a query over an unchanged body of content yields the same response on each occasion and a response may be verified later. There is a need for every assertion to resolve to the source from which it was derived, and for that resolution to be verifiable rather than asserted.

[0025] There is a need for safety classification, access entitlement and origin to be inseparable from the stored content, such that no path reaches the content without them. There is a need for a representation of confidence that distinguishes a firmly-established assertion from a provisional one and revises each in proportion to what stands behind it. There is a need for an assertion to be held independently of the language in which it was expressed, such that content ingested in one language is reachable by a query in another without translation of the query. There is a need for content of differing kinds — text, image, audio, video, sensor output — to be reduced to a common representation, such that agreement between two kinds is expressible.

[0026] There is a need for such a system to operate without a network connection, so that it may be deployed where connectivity is absent, intermittent, costly or prohibited by policy. There is a need for structured content to be presentable in immersive and three-dimensional form, and for such a presentation to be constrained by the same stored representation as a textual one. There is a need for a structured representation of an environment to be usable in directing capture and inspection equipment, under constraints that are enforced before a command is emitted rather than after it. And there is a need for a system that produces predictions to measure its own predictions against subsequently observed outcomes, and to report the result by domain rather than in aggregate.

Summary of the Invention

[0027] The invention is a computer-implemented system that constructs, from documentary, visual, auditory, structured and sensor sources, a body of discrete knowledge records; that maintains each record together with its origin, its evidential support and the terms upon which it may be reached; and that composes a response to a query from those records under a constraint applied at the point of generation rather than by inspection of a generated result.

[0028] The knowledge record is the primitive to which every mechanism conforms. It encodes one semantic assertion and binds inseparably, within a single record schema, an identifier derived from its content, a type reference, a temporal anchor, a confidence value held as a pair of parameters accumulating evidence for and against the assertion, a reference into a verifiable provenance chain, a language-independent semantic fingerprint, a safety classification and an access control list. Because those properties are fields of the record rather than a layer applied around a store, no path reaches the assertion without them.

The independent aspects of the invention

[0029] Seven aspects are claimed independently, and the following states each in the terms of the claim that recites it.

Aspect | What it provides

Constrained composition from knowledge records | Content in a plurality of modalities is extracted into knowledge records of the schema above; a scope description is constructed before any traversal; retrieved records are admitted against a confidence threshold and resolved by an iterative updating of activation values to a stable state; and output text is composed under a permissible output vocabulary constrained at each generation step to terms derivable from the admitted records, such that a proposition not so derivable is unreachable rather than improbable, and a boundary is stated where no record satisfies the threshold.

Forensic evaluation of an output attributed to an external system | An output attributed to a system other than the one performing the method is evaluated claim by claim against the knowledge records, each claim being marked as supported with a citation or as unsupported, and no output being generated that asserts a determination concerning an internal state of that system.

Gated hypothesis generation | A conjectural derivation over the knowledge records is created in a proposed state and transitions to a validated state only upon confirmation recorded against an identity of an individual human reviewer, no automated process being configured to effect that transition.

Inference of absent visual and material properties | A value absent from visual content is inferred from a material property asserted by a knowledge record, computed deterministically and traceable to the record from which it was derived, an inferred value being marked as inferred and remaining inadmissible as a grounding source.

Immersive presentation from spatially-anchored records | A scene description is assembled from knowledge records carrying spatial detail and is rendered in three dimensions, a scene being withheld where the spatial detail carried by those records does not satisfy a configured sufficiency threshold rather than being completed from an unsupported estimate.

Attributed viewpoint adoption | A viewpoint attested within ingested content is adopted in composing a response and is marked as attributed to that content, no such viewpoint being presented as a position of the system performing the method.

Retrospective calibration of the system's own predictions | A prediction previously produced is scored against a subsequently recorded outcome under an attestation constraint excluding any record attested after the event, the resulting measure being reported per domain and per confidence band, and the subject class being restricted such that a record whose subject resolves to an individual natural person is refused at construction.

Principal further aspects

[0030] The following are disclosed and claimed dependently, and are summarised here because each governs a property of the system as a whole rather than a single operation.

Confidence as an evidence pair. A confidence value is maintained as parameters accumulating supporting and contradicting evidence, so that a firmly-established assertion and a provisional one are distinguishable at an identical reported confidence and are revised in proportion to what stands behind each; and the contributed weight is attributed across registered source classes, so that corroboration

concentrated in one class is distinguishable from corroboration across many.

Domain partition and enforced transfer. The system is partitioned into a governance domain and a plurality of functional domains, no operation of a functional domain executing without traversing an enforcement point of the governance domain; a transfer matrix assigns each ordered pair of domains a permission, an unlisted pair being denied; and the matrix, the path-separation invariant and the scheduling priority are owned by the governance domain and are members of a fixed enumerated set of subsystems excluded from modification.

Determinism under a defined context. An enumerated set of context dimensions is maintained and a key computed over it, an identity of that key between two executions being the condition upon which an identity of output is asserted, so that a difference attributable to a differing context is distinguishable from a defect.

Reasoning modes and the writing rule. An inference is classified as one whose conclusion follows necessarily from admitted records or otherwise; the first is admitted as a knowledge record carrying a derived-inference type, and the second is withheld from admission and emitted in a proposed state.

Interpretive modulation without content variation. Context, perspective, ergonomic, cultural and social modulations are applied to a composed result, none of which widens or narrows a scope description, admits or excludes a knowledge record, alters a confidence value or extends the permissible vocabulary.

Rights and entitlement as properties of the record. Territory, interval, seat, edition and print entitlements are resolved at scope construction, such that content outside an entitlement is not traversable rather than filtered from a result; and a revocation takes effect at a subsequent query rather than at a subsequent session.

Operation without a network and across deployment topologies. The method is performed identically in a multi-tenant hosted mode, a single-tenant local mode, a single-device mode, a hybrid mode and an edge mode, a topology determining where a domain executes and never which domain owns a decision.

Directed equipment control under session authorization. Commands are emitted to registered capture and inspection equipment under a session authorization bounded by expiry, revocability and scope; every command is evaluated against a constraint set before emission and a violating command is withheld unconditionally; and a halt is evaluated without traversing the retrieval or generation path.

Assistive physical contact under a two-party authorization. Where equipment makes intentional contact with a person, a deployment authorization and a contact consent are required concurrently, a withdrawal by the assisted person is not overridable, and a loss of power or control results in release rather than retention.

Outcome feedback as attributed evidence. A report bearing on a delivered response is admitted as a knowledge record attributed to the reporting identity and increments an evidence parameter; it does not set a confidence value, supersede a record, extend the permissible vocabulary or alter a scope description.

[0031] Further aspects concerning multilingual regeneration and wording-operative reproduction, ontology inheritance with exception override, event type governance, physical constraint enforcement, surface realization, commercial presentation isolation, systematic trial architecture, differentiated propagation and the analog realization of the convergence are disclosed in the Detailed Description and recited in the claims.

Brief Description of the Drawings

[0032] FIG. 1. Overall system architecture, showing the ingestion pipeline, the Fact Graph and Provenance Ledger, the Ontology Mesh, the Neuromorphic Cognitive Layer, the Safety Substrate as a cross-cutting enforcement plane, and the response pipeline terminating in grounded generation.

[0033] FIG. 2. Attractor Network convergence, showing an initial candidate activation set in embedding space converging to a single stable configuration under the threshold update rule, operated on the ordered side of the retrieval-phase boundary.

[0034] FIG. 3. The grounded generation pipeline, showing construction of a graph structure whose every edge traces to a Fact Unit, vocabulary constraint at each generation step, and rejection of any candidate proposition not so traceable.

[0035] FIG. 4. The verification-assisted improvement proposal algorithm, showing candidate identification, sandboxed simulation, optional machine-checked proof presented as evidence, the mandatory human approval gate, and the permanently-excluded subsystem set.

[0036] FIG. 5. Ingestion-time question generation, showing coverage scoring against a retention floor, pre-computed answer assembly from exactly the contributing records, index matching before live traversal, and dependency-based staleness.

[0037] FIG. 6. Confidence as a Beta posterior, showing two records reporting an identical mean with differing accumulated evidence, their differing credible intervals, and the precision ratio by which a prediction error is weighted.

[0038] FIG. 7. The response pipeline stage ordering that isolates commercial content selection, showing completion of retrieval, ranking and confidence scoring before any commerce invocation, and rendering in a distinct labeled region.

[0039] FIG. 8. The compute-accounting model, showing operations metered as bounded slices assigned to categories and scheduled such that a safety-evaluation slice is served ahead of any other category under contention.

Detailed Description of the Invention

[0040] The following description sets forth specific details to provide a thorough understanding of the invention. It will be apparent to one skilled in the art that the invention may be practised without these specific details, and that the embodiments described are illustrative rather than limiting. Where a value, threshold or coefficient is stated, it is stated as an example of one embodiment and not as a limitation.

A0. Detailed Summary of the Invention

[0041] This subsection states the operation of the system end to end before the subsections that follow treat each mechanism separately, so that each is read in the context of the whole. The properties upon which the claims rest arise from the order in which the stages occur rather than from any stage considered alone.

[0042] Content is admitted. Material in any supported modality is routed to a modality-specific extraction stage, each stage failing independently of every other, so that an unavailable extractor removes one modality and leaves the remainder operating. Extraction produces candidate knowledge records, each encoding one semantic assertion. A candidate is validated against the machine definition of its type and

refused at construction where a required field is absent, rather than admitted and flagged. Its confidence is calibrated from the extraction confidence, the reliability registered for its source, a coefficient for its modality and a decay for its age, and is elevated where two or more modalities assert the same property within a common temporal window. It passes a safety evaluation before persistence. It is deduplicated upon its semantic fingerprint, so that the same assertion arriving in two languages converges upon one record carrying two provenance branches. It is appended to a provenance chain whose nodes carry keyed checksums over the recorded safety evaluations.

[0043] Where the corpus comprises a single work, the operation just described is the ingestion of that work, and the system thereafter answers from it alone. Where the corpus comprises many works, the operation is identical and the resulting records are distinguished by their provenance rather than by being held separately, so that a question may be answered from several works at once and each proposition of the answer resolves to the work from which it was drawn.

[0044] A query is understood. The query is decomposed into operational intents before any intent is bound to a concrete entity, and an ambiguous binding halts the process and returns a request for clarification rather than resolving the ambiguity by selection among candidates.

[0045] Entitlement is resolved. A scope description is constructed before any traversal occurs, from the requester's access entitlement, the safety ceiling associated with their role, any narrowing elected for the session, and any further applicable bound: the exclusion of analysis records, the reader's progress marker where spoiler protection is elected, the attestation cutoff for a retrospective run, and the intersection of the participants' entitlements in a shared session. A record outside the scope description is not traversable, and a request for it yields the response given for a record that does not exist. Territory, interval, seat, edition and print entitlements are resolved here, so that rights are enforced at the point of access rather than filtered from a result afterward.

[0046] Records are retrieved and stabilized. Records semantically proximate to the query are pre-activated within a traversal budget expressed in units of edge-class cost, so that a traversal reaching widely through meaning and shallowly through time costs what the reverse costs. Retrieved records are admitted against a confidence threshold. The admitted set is resolved by iteratively updating activation values as a threshold function of a weighted sum until a stable state is reached, the number of stored configurations being held below the capacity bound at which retrieval states cease to exist. Contradictions among admitted records are surfaced and recorded rather than averaged.

[0047] Output is composed. A graph structure is constructed over the stabilized records, carrying typed relations among them, and output text is generated from it under a vocabulary constrained at each generation step to terms derivable from those records. A proposition not so derivable is unreachable rather than improbable. Where the admitted set is empty or fails the threshold, a statement of the boundary is returned together with the nearest available records. Surface realization aggregates propositions sharing a subject, selects referring expressions by antecedent availability, emits connectives corresponding to the relations the graph carries, and substitutes registered synonyms only where the semantic fingerprint is preserved.

[0048] The same records support renderings other than text. An edition at a differing audience level, in a differing language, with differing annotation density, or in an accessible equivalent, is generated from the same records rather than maintained as a separate work, so that a correction entered once propagates to every rendering. A print-ready composition is produced with the rights validation performed before composition and a reference to that validation carried in the composed output, and with deterministic pagination such that two compositions over unchanged content produce identical page references. A scene description assembled from records carrying spatial detail is rendered in three dimensions, and is

withheld where the spatial detail does not satisfy the sufficiency threshold rather than completed from an estimate.

[0049] The result is reviewed and delivered. A review under the governance domain precedes and follows every interpretive modulation. Modulations for context, perspective, ergonomics, culture and social setting are applied to the composed result, none of which widens or narrows a scope description, admits or excludes a record, alters a confidence value or extends the permissible vocabulary. A context key computed over the enumerated locked dimensions accompanies the output, so that an identity of output between two executions is asserted where and only where the keys agree. Commercial selection, where enabled, is invoked only after every preceding stage has completed and receives a read-only concept identifier together with its one-hop neighbours, holding no write access to retrieval or generation.

[0050] Evidence continues to accumulate after delivery. A report bearing on a delivered response, and an outcome subsequently observed in respect of a matter the response addressed, are each admitted as knowledge records attributed to their sources and increment the evidence parameters of the records they bear upon. Neither sets a confidence value, supersedes a record, extends the permissible vocabulary nor alters a scope description. Measured calibration is reported per domain and per confidence band and is emitted as a proposal, no change to the system taking effect without an approval recorded against an identity of an individual human reviewer.

A. The Fact Unit

[0051] The Fact Unit is the data primitive to which every subsystem conforms. Its schema binds, inseparably within a single record: an identifier deterministically derived from a hash of the record's normalized content, a source identifier and an ingestion timestamp; a type reference resolving to a non-deprecated concept-mesh entry; subject and predicate references; an object expressed either as an entity reference or a typed scalar, the two being mutually exclusive; a temporal anchor comprising start time, optional end time, granularity and temporal confidence; an optional spatial anchor; a confidence value maintained as a first parameter accumulating evidence supporting the assertion and a second parameter accumulating evidence contradicting it, the reported confidence being the ratio of the first to their sum and their sum constituting the evidential weight from which a credible interval and a propagation precision are computed, and the contributed weight being attributed across registered source classes from which a diversity measure is computed; a provenance reference; a language-independent semantic fingerprint; a modality bitmask; an embedding vector; an optional set of typed weighted edges; an optional supersession chain; a safety classification bitmask; linguistic metadata; an annotation set; and an access control list containing at least one owner entry.

[0052] The record is atomic, encoding exactly one semantic assertion, so that confidence, safety classification and provenance each attach to exactly one assertion rather than to a passage containing several. Because confidence is held as an evidence pair rather than a scalar, the record carries not only how firmly the assertion is held but how much is known about it: two records reporting an identical confidence and resting on different quantities of evidence are distinguishable, which a scalar could not express. It is immutable after construction: a correction is effected by constructing a new record, appending the prior identifier to the new record's supersession chain, and marking the prior record superseded. Deduplication is performed on equality of the semantic fingerprint, so content expressing one assertion in two languages converges on a single record with two provenance branches rather than on two competing records.

[0053] The safety classification is modifiable only toward increased restriction; reduction requires a

dual-approval operation, and an application programming interface caller cannot modify it at all. Because the classification is a field of the record, a read of the record is necessarily a read of its classification, and a code path reaching the record by an unusual route does not thereby reach an unclassified record.

The Fact Unit of this section is further organized, without alteration of any field above, as nine layers: a core semantic identity that never changes; structural adjacency; temporal stability read from the three timestamps and the supersession chain; edition-differential markers; multi-language alignment anchored to the semantic fingerprint; media provenance with cross-modality corroboration and introduced content marked; an observation-and-evidence layer in which the confidence value's two parameters accumulate evidence for and against the assertion under a Baconian feedback loop of action, observation, measurement, adjustment and iteration — each observation incrementing the evidence pair by its source-reliability weight, diversity gating generalization, and revision being precision-weighted so a settled assertion resists by arithmetic; in-record rights and safety; and context-keyed deterministic output. The loop's adjustment reaches only the layers it owns and never the semantic identity.

Seven additional field families further enrich the record, each additive and each a deterministic function of record content and the observation stream: a decomposition of residual uncertainty into an epistemic component, reducible by further observation, and an aleatory component, the irreducible variability of the observed world, carried with explicit source-reliability, measurement-precision and consensus-breadth sub-parameters and supplying the feedback loop's stopping condition; an explicit assertion polarity marking the record affirmative or negative, under which records agreeing on the same assertion core with opposite polarity are contradictory by field comparison and a known negative is distinguished from the absence of any record; a native n-ary assertion structure carrying one assertion with multiple typed argument slots — subject, value, unit, instrument, method, condition, observer — under one identifier, one fingerprint computed over the complete argument structure, and one evidence pair; an applicability-scope field stating the conditions — temporal interval, jurisdiction, edition, parameter range, population — under which the assertion holds, assertions under disjoint scopes being recognized as non-conflicting; an immutable adjustment-history layer journaling every feedback-loop pass so that a record's confidence is deterministically replayable from first observation to present state; a cryptographic integrity chain hashing each record's immutable core and carrying the superseded record's hash on every supersession link, with no distributed ledger, token or ownership identity node; and a canonical compact binary encoding, lossless and bidirectional, under which identifier and fingerprint are identical in either form.

The compact form of the record additionally serves as its canonical at-rest and wire form, fused to the record rather than to any store: required fields at fixed offsets and optional fields tagged, with the identifier and semantic fingerprint identical to the full form, so the encoded record is self-describing without an external schema. A host database, content repository, or reader platform holds the encoded record as an opaque value it is never required to parse, index, or interpret, its only obligations being to accept and return the bytes unchanged, which is what preserves the system's operation over any database while imposing no proprietary format, schema, or query capability upon it; queries over records are expressed internally and, where external material is drawn from such a store, translated at an adapter into that store's native language, the store never receiving the internal query form. Because the record is held as an opaque encoded value, its integrity chain, its in-record safety classification and access-control list, and, where elected, its zero-knowledge encryption travel with it across the storage boundary, so any store will hold the record and no store can read, reorder, or leak its meaning.

The knowledge record of this section is written in a formal, language-independent assertion language — the Fact Unit Language — whose vocabulary is the governed predicate catalogue, whose grammar is the record schema stated above, and whose identity function is the language-independent semantic fingerprint, under which two records asserting the same proposition in different natural languages are the

same statement; a candidate not conforming to the language is refused at admission, and retrieval, contradiction detection, and regeneration into a requested natural language each operate upon this assertion language rather than upon natural-language text.

B. Multimodal ingestion and confidence calibration

[0054] Content in eight distinct modalities — unstructured text, structured data, image, audio, video, sensor data, knowledge graph and code — is processed through modality-specific extraction stages that fail independently, so that failure of one stage does not fail the ingestion job with respect to another modality. Content matching no enumerated modality is classified as unmatched and surfaced, and is not discarded without record.

[0055] Each extracted record receives a confidence value integrating extraction confidence, source reliability, a modality-specific coefficient and temporal decay. Where a property of a perceived event is confirmed by two or more modalities within a common temporal window, the resulting record receives a confidence strictly greater than the value the same property would receive from a single modality, the increase being a function of the temporal synchrony between the confirming streams.

[0056] A mandatory safety evaluation gate is applied before admission into persistent storage. Such gates are positioned at each of a plurality of operation layers, so that entry at one layer does not avoid a gate positioned at another. Extraction employs a multi-stage natural-language pipeline comprising sentence segmentation, tokenization, dependency parsing, constituency parsing, semantic role labelling, open information extraction and discourse parsing. The neural component of extraction is confined to this boundary: its output is a typed Fact Unit subject to safety gating and confidence calibration before admission, and no neural component participates in retrieval, in the groundedness constraint, or in the generation vocabulary.

C. Deterministic provenance

[0057] Every consequential operation is appended as a node of a Merkle directed acyclic graph. Each node carries a keyed checksum computed over the operation and over the safety evaluation recorded for it, such that recomputation detects a difference between the recorded evaluation and the evaluation that executed. The chain is append-only; a correction is a new node referencing the prior node, never an edit of it.

D. Scope resolution and traversal

[0058] On receipt of a query a scope description is constructed before traversal from the requester's access entitlement, a safety ceiling associated with their role, and any session narrowing. Four further bounds may narrow it, and they are stated here because the scope description is the single point at which every access constraint in the system takes effect: an analysis record is excluded from a scope description constructed for a query not directed to that analysis; a reader's progress marker excludes a record whose narrative position exceeds it where spoiler protection is elected; an attestation cutoff excludes a record attested later than the cutoff where the scope description is constructed for a retrospective forecast run; and the intersection of participants' entitlements bounds a shared session. In each case the excluded record is not traversable rather than removed from a result. The scope description is signed and time-bounded, so an expired grant cannot be exercised even where a session remains valid. A record outside the scope description is absent from traversal rather than removed from its result, such that the record's existence is not inferable from a result count, a ranking position or an elapsed traversal time.

E. Retrieval stabilization and the retrieval-phase boundary

[0059] Records retrieved within the scope description form a candidate set represented as points in an embedding space. Only candidates whose retrieval confidence satisfies a configured threshold are admitted; by way of example, a candidate at 0.62 against a threshold of 0.5 is admitted as grounding, while one at 0.3 is excluded rather than admitted with a qualification.

[0060] Activation values of admitted candidates are updated iteratively as a threshold function of a weighted sum of the other candidates' activations, of the form $a_i(t+1) = \text{sgn}(\sum_j W_{ij} \cdot a_j(t) - \theta_i)$, until a stable state is reached. The same convergence is applied at a second granularity: where a hyperedge relates more than two knowledge records and is partially observed, the constituent set is stabilized as a whole, each constituent weighted by its confidence and its accumulated evidence, to identify the complete configuration most consistent with the observed constituents. That application introduces no convergence behaviour separate from this update. The number of distinct stable configurations the layer is operated to hold is bounded at a fixed proportion of the number of units, approximately $0.14N$ for N units.

[0061] That bound is the retrieval-phase boundary. The value $\theta_c \approx 0.138$ patterns per unit was established analytically for Hebbian learning in the Hopfield model, and the phase diagram shows that at $T = 0$ retrieval succeeds with error below 1.5% for $\theta < 0.14$, at which point the retrieval states disappear discontinuously. Two regimes exist below the boundary: for $\theta \leq 0.05$ the retrieval states are the global energy minima, and for $0.05 < \theta \leq 0.138$ they remain dynamically stable but are metastable local minima while spin-glass states hold the global minima. The system's default operating ceiling lies in the first regime, because convergence to a stored pattern is then energetically favoured rather than merely available; operation in the second requires a recorded deployment election. Because the transition is discontinuous, degradation is not graceful, and the constraint is therefore enforced as an admission refusal at the ceiling rather than as a monitored quality metric.

[0062] Records semantically proximate to the query are pre-activated at elevated weight before traversal. In a multi-turn interaction a record retrieved at a prior turn is weighted for subsequent retrieval by a factor decreasing monotonically with intervening turn distance in proportion to $1/(1+d)$, and a record whose factor falls below a configured floor is released from working memory without deletion from persistent storage.

F. Constrained generation

[0063] A graph structure is constructed in which every edge references at least one record of the stabilized set. At each generation step the permissible output vocabulary is constrained to terms derivable from those edges, and a candidate proposition not so derivable is rejected before output in favour of a narrower proposition that is. Each delivered proposition is accompanied by a citation resolving it to at least one record.

[0064] One exemption from the constraint is specified and is narrow: a wording-operative span — a quotation, a statutory phrase, a claim term, a prescribed instruction — is reproduced rather than generated, and that reproduced span alone is exempt from the groundedness rewriting which would otherwise apply, no other span of the response being exempt and the exemption conferring no licence to generate. Where no record of the candidate set satisfies the configured threshold, the system delivers a statement of that boundary together with the nearest available records, and does not generate output text. Because the constraint applies to the permissible vocabulary at generation time rather than to a generated result, there is no generation path that omits it and no verification stage that can fail open.

G. Confidence as a Beta posterior

[0065] A confidence value is maintained as two parameters: α , accumulated evidence supporting the assertion, and β , accumulated evidence contradicting it. The reported confidence is the posterior mean $\alpha/(\alpha+\beta)$. On receipt of further evidence the relevant parameter is incremented by the weight of the observing source, computed from source reliability and modality coefficient, so evidence is admitted as a fractional count rather than an integer.

[0066] The behaviour follows from the arithmetic rather than from a rule. A belief at $\alpha=1, \beta=1$ reporting 0.5000 moves to 0.3333 on one contradiction, a change of $\beta 0.1667$. A belief at $\alpha=18, \beta=2$ also reporting 0.9000 moves to 0.8571, a change of $\beta 0.0429$, and requires ten contradictions to reach 0.6000. An additive rule with a fixed increment would move both identically, which is why a settled belief and a provisional one are presently indistinguishable in such systems.

[0067] Prior strength maps onto the pseudo-count $\alpha+\beta$ while confidence maps onto the ratio, so a concept node's settledness and a record's confidence are expressed in one arithmetic. A credible interval follows from the same parameters, with half-width approximately $1.96 \cdot \sqrt{CS(1-CS)/(\alpha+\beta+1)}$: records at $\alpha=18, \beta=2$ and $\alpha=180, \beta=20$ both report 0.9000 and carry half-widths of ± 0.1283 and ± 0.0415 respectively. Temporal decay acts on the parameters, shrinking both toward the prior so that the mean is preserved while the interval widens, which is the correct meaning of an ageing observation; a record type designated immutable is assigned a decay rate of zero. Because both parameters are strictly positive the reported value is never exactly zero or one, and refutation remains a distinct state set explicitly rather than a limit approached by accumulating contradictions.

H. Precision-weighted error propagation

[0068] When a record is contradicted, the error is propagated upward through cross-modal bindings to the concept attributes that generated the contradicted expectation. The magnitude applied is not a function of the difference between the two confidence values — two records both reporting 0.9000 differ by zero, so the strongest available contradiction would produce no model update at all — but of precision, the inverse variance computed from the stored parameters.

[0069] The revision gain is the error's share of total precision, computed as $\text{var_prior}/(\text{var_prior} + \text{var_error})$, which avoids reciprocals and is numerically stable as either variance approaches zero. A single-observation error against a two-hundred-observation attribute yields a gain of 0.0053 and is effectively ignored; a well-established error against an equally settled attribute yields 0.5000; a well-established error against a weakly-held attribute yields 0.9947. A gain below a configured floor produces no revision at all, and the suppressed contradiction is recorded so that a pattern of individually-insignificant errors against one attribute remains queryable.

[0070] Precision governs revision of the concept model only. It does not govern retrieval and does not govern the permissible generation vocabulary: a proposition not traceable to a record remains unreachable at generation irrespective of any precision value.

I. Ingestion-time question generation and pre-computed answers

[0071] An author submits a digital work at a portal bound to a verified author identity. A rights validation is performed and ingestion does not begin until it returns a cleared status. Following extraction, and before any query directed to the work is received, each committed record is traversed together with its concept-mesh neighbours and a plurality of candidate questions is generated, each answerable by at least one record. A candidate generated across two or more records is a question no single record

answers alone.

[0072] Each candidate is scored by a coverage measure over the records it draws on, and only candidates satisfying a configured floor are retained in a persistent question index; by way of example, 0.71 against a floor of 0.6 is retained. For each retained question an answer is assembled from exactly the records the question was derived from and no others, assigned a confidence not exceeding the least among them, and stored with a provenance tag distinguishing it from a directly-ingested record together with the identifier of every contributing record. Because the stored citation set is the identical set used to compute the answer, a served answer is traceable to precisely the same degree as one computed live.

[0073] On receipt of a query the index is matched before live traversal. Where match similarity satisfies a threshold and no contributing record has been revised since computation, the stored answer is returned with its citations; otherwise the system falls through to live retrieval, and the result is indistinguishable to the requester in form, citation and confidence semantics. Revision of a record marks every dependent index entry stale, and a stale entry is excluded from service until recomputed, staleness being a property of the dependency relation rather than of elapsed time. Forward inference chains enabled by a newly committed record are additionally computed at ingestion and stored as derived-inference records, so a retained question whose answer depends on a multi-hop inference is answerable without performing that inference at query time. The index is scoped to a single work, and an entry's contributing-record set is fixed at computation and not extendable at service time.

J. The author supplemental knowledge layer

[0074] An author supplies supplemental context concerning a work, at submission or at any time after ingestion. Such content is converted to records carrying a tag identifying them as author-supplemental and the author as attributed source, and is thereafter available to every reader entitled to the work without alteration of the work's rights validation status. Addition of a supplemental record triggers regeneration of every question index entry whose contributing-record set is affected. Supplemental records participate in question generation on the same terms as records extracted from the work, so questions answerable only by reference to the author's supplemental context are generated and retained.

K. Concept inheritance and exception override

[0075] Where a concept is connected to a superordinate by a subsumption edge and the superordinate carries an attribute edge, the subordinate acquires that attribute by inheritance, transitively along a chain bounded by a configured maximum depth. An inherited attribute is marked with the concept it was inherited from and the depth at which it was found, so it is never indistinguishable from a directly-asserted attribute, and a response relying on it cites both the originating concept and the path. Its confidence is the product of the traversed edge confidences and the attribute's own confidence, bounded above by the least of those values. Inheritance is computed at traversal time and not written back, so a revision at the superordinate propagates on the next traversal rather than leaving stale copies.

[0076] Where a subordinate asserts an attribute conflicting with one it would inherit, the direct assertion governs for that concept and its subordinates while the superordinate's attribute is unmodified and still inherited by every sibling that does not itself conflict. Conflicting attributes inherited at differing depths resolve to the lesser depth. Conflicts at identical depth are not resolved: the conflict is recorded in a persistent contradiction index and both values are reported with their originating concepts, because arbitrary tie-breaking would produce an answer that is reproducible without being grounded.

[0077] Reverse traversal of a typed edge resolves through an inverse predicate declared in the predicate

schema at traversal time, storing no second edge, so stored edge count equals asserted relation count and no reconciliation between a forward edge and a stored inverse is required. Where a predicate declares no inverse, reverse traversal is refused rather than inferred, because generating a reverse reading in that circumstance would be an assertion the system originated rather than retrieved.

L. Cross-lingual regeneration, wording-operative spans and idioms

[0078] Output in a target language is realized from language-independent records under the constrained vocabulary, without reference to the surface form of the source content, and the same record regenerated in any supported language back-translates to a semantic fingerprint within a configurable distance of the original. Lexical selection maps entities to canonical target-language names through a multilingual name map; syntactic realization respects target-language morphology, gender agreement, case inflection, aspect, honorifics and script direction; domain-aware terminology resolution disambiguates a term with multiple domain meanings through the source record's domain classification.

[0079] A wording-operative span is a span whose exact wording carries force independently of its meaning — a direct quotation, a statutory or regulatory phrase, a term defined in a patent claim, a contractual clause, a prescribed instruction. Such a span is reproduced in its source language and wording rather than regenerated, accompanied where the response is in another language by a regenerated gloss carrying an explicit non-operative marking. Reproduction is subject without exception to rights validation and to the fact-expression boundary: extracting a fact from a source is not a use of the source's protected expression, but reproducing that expression is. Where reproduction is not licensed, the response states that the operative wording is not reproducible and supplies the gloss alone. Only the reproduced span is exempt from groundedness rewriting.

[0080] An idiom is detected at ingestion where the compositional reading of a span and its attested reading diverge beyond a configured threshold. The record stores the attested meaning with an idiom-derived marking and the source span. At regeneration a registered target-language equivalent for the same attested meaning is used; where none is registered the attested meaning is rendered literally with an explicit note. A target-language idiom is never selected on the basis of surface similarity to the source, and no unregistered equivalence is generated, because a false idiomatic equivalent is a fluent error.

M. The linguistic resource registry

[0081] Every externally-sourced linguistic resource consulted by the extraction or realization pipelines — a lexical database, an annotated corpus, a morphological inventory, a frequency or collocation table, an idiom list, a treebank — must be registered before any stage consults it. A registration records the resource identifier and version, the licensor, the licence instrument, the permitted-use scope, whether redistribution is permitted, any attribution requirement, and an ingestion mode. A registration lacking any of these fields is invalid. A resource whose licence does not cover the operator's declared use is refused at registration, before any pipeline consults it.

[0082] An artifact derived from a registered resource carries a reference to that registration, and derivative conditions imposed by the licence — attribution, share-alike, prohibition on redistributing the resource's content — attach to the derived artifact and are enforced wherever it would be redistributed or exposed. Where two registered resources impose conflicting derivative conditions on one artifact, the artifact is marked non-redistributable and the conflict recorded, rather than resolved by selecting the less restrictive term.

[0083] A resource registered in derived-artifact-only mode may be consulted to compute aggregate quantities — a term frequency, a collocation strength, a part-of-speech distribution, a morphological paradigm — while no span of its text is retained in any record, index or cache, or reproduced in any output. Compliance is verifiable: the count of retained source spans attributable to such a resource is zero, and a non-zero count is a compliance failure surfaced by the ordinary audit path. This mode enforces mechanically the same fact-expression boundary the system applies to ingested works generally, and exists because several substantial resources permit derived statistics while prohibiting commercial exploitation or redistribution of their text.

N. Governance of self-modification

[0084] A candidate modification is identified either by monitoring scoped to a specific subsystem or by direct human request, evaluated in an isolated simulation environment, and applied only upon approval recorded against the identity of an individual human reviewer together with that reviewer's stated reasoning. Where the candidate targets a bounded, non-Turing-complete program representation, a machine-checked proof of a specific property may be generated before review and presented to the reviewer as evidentiary input; the proof does not itself effect the approval.

[0085] A fixed enumerated set of subsystems — including the safety enforcement plane and the governance process itself — cannot be a valid target of any candidate, and that set cannot be modified through the mechanism it constrains. A set the mechanism could amend would not constrain the mechanism.

[0086] A bounded program is compiled from a natural-language instruction by decomposing the instruction into operational intents before binding any intent to a concrete entity, binding each intent by the entity resolution common to extraction, and halting with a clarification request where a binding is ambiguous rather than selecting among candidates. The program is expressed in a primitive set comprising a fixed number of read primitives, a single state-changing primitive subject to safety evaluation, and a single iteration primitive enforcing a runtime iteration ceiling, so termination is enforced by the runtime rather than by convention.

O. Compute accounting and isolation of commercial content

[0087] Each operation is metered as a bounded, atomic compute-accounting unit assigned to a category. The scheduler serves a unit of a safety-evaluation category ahead of a unit of any other category whenever the two contend for a common processing resource, irrespective of arrival order or requester privilege.

[0088] Retrieval, stabilization and generation complete before any invocation of commercial content selection. The sole input to selection is a concept identifier resolved from the query, supplied read-only; selection holds no write access to retrieval, stabilization or generation. Among already-matched candidates a score is computed as a weighted combination of a relevance factor, an operator-configurable source-priority factor and a bid factor, the weights summing to unity — by way of example, weights of 0.4, 0.3 and 0.3 applied to factors of 0.9, 0.7 and 0.6 yield 0.75. The selection unit is scheduled below every safety-evaluation unit, so contention results in omission of the commercial region rather than delay or degradation of the answer, and selected content renders in a labeled region not interleaved with the output.

P. Perception, salience and the world model

[0089] Where a sensor feed is analyzed under explicit revocable authorization, classification results corresponding to a person or an animal are excluded from the candidate result set entirely and never surfaced. Each remaining classified object receives a salience value normalized against the sum of the salience values of the objects classified from the same frame; normalization governs presentation order only and every classified object remains an addressable record. Apparent motion arising from a non-stationary camera is separated from true object motion by deterministic optical-flow estimation before segmentation. Object continuity is maintained across full occlusion within a session, and a classified object is indexed by a deterministic mapping from planar position to a cell of a hexagonal tiling, so spatial neighbours are retrievable by cell adjacency without a distance computation; the identical construction indexes positions in concept-embedding space.

[0090] A first concept representation is constructed from perceptual input independently of language and a second from ingested language, and the two are fused into a single concept on a co-occurrence of a label and a perceived object confirmed by a semantic consistency check. A single co-occurrence produces a provisional binding; a concept is finalized only after a configured number of consolidating observations. Fusion additionally applies, as evidence-overridable signals, an inference from an observable gaze direction, a preference for interpreting a label as naming a whole object rather than a part, and a preference for binding a new label to an object not already carrying one.

[0091] An abstraction derived from a plurality of records is persisted in a store distinct from the records themselves, retaining a generalized concept and discarding the specific episode from which it was derived. Raw perceptual input is reduced to a concept-level record before admission to that store.

Q. Governed event types and environment vectors

[0092] An event type is a governed, versioned definition of a category of occurrence, registered with an identifier immutable from registration, a placement resolving against a plurality of independent taxonomic hierarchies, a prose definition stating both occurrences covered and occurrences excluded, and a machine definition specifying which fields are required, optional and forbidden for that type. A record encoding an occurrence must reference a registered identifier rather than a free-text category, and a record not satisfying its type's machine definition is refused at construction rather than admitted and flagged. A record created under one version remains valid under that version; a required-field set may be narrowed by a later version and may not be widened absent a distinct type.

[0093] An environment vector is a normalized snapshot of the state relative to which a set of assertions holds, identified by a hash of its canonicalized condition set, and carrying a temporal interval, typed condition keys resolving against the concept structure, and a confidence value and provenance reference of its own — because a stated environment is itself an observation that may be wrong. It is referenced from a record through an annotation rather than a schema field, so no field is added to the record schema and every existing record remains valid unmodified. Two constructions yielding an identical condition set converge on one vector with two provenance branches. Correction is by supersession: a hundred referencing records resolve to the corrected state on next traversal, where a hundred duplicated contexts would require a hundred edits.

[0094] Causal structure is expressed through the existing causal predicate and typed weighted edges, with no parallel edge store. A chain is a bundle membership, and chain length, root and outcome are computed at query time rather than stored as metadata that can disagree with the edges it summarises. A causal edge is admitted only where the source's temporal anchor does not begin after the target's begins; an edge asserting that an effect preceded its cause is refused and the attempted assertion recorded in the contradiction index. Precedence is not required to be strict, so concurrent and overlapping

causation remain expressible, and a temporal-ordering edge is never converted into a causal edge by any automated process. A duration is computed from the anchors at query time and not stored.

R. Physical constraint enforcement in scene completion

[0095] A physical constraint is registered against a domain with a residual expression over quantities the completion supplies, a tolerance, a unit system resolving to SI base units, an applicability predicate stating the conditions under which the constraint holds, and a provenance reference identifying the source establishing it. The applicability predicate is mandatory: conservation of momentum holds for a closed system, and a constraint unable to state its own preconditions would refuse correct completions. A constraint whose predicate is unsatisfied is not evaluated, and the non-evaluation is recorded.

[0096] Before a candidate scene completion is returned, every applicable constraint is evaluated by computing its residual. A completion whose residual exceeds tolerance is refused, and the completion is re-attempted with the violating field left unspecified, so a completion carrying an acknowledged gap is returned in preference to one that is physically impossible. By way of example: a two-kilogram body at three metres per second striking a stationary one-kilogram body carries total momentum $6.0 \text{ kg}\cdot\text{m/s}$, and at a two per cent relative tolerance of 0.12, a completion yielding 6.0 is admitted while one yielding 8.0 is refused. A body observed at 2.0 metres has expected position $2.0 + 0.5 \cdot 9.80665 \cdot 0.4^2 = 1.2155$ metres after 0.4 seconds, against which a completion asserting 2.5 metres carries a residual of 1.2845.

[0097] Constraints are evaluated before plausibility scoring, and a refused completion does not appear in the ranked output at any score, because a physically impossible completion is not a low-plausibility one. A refusal produces a queryable record identifying the constraint, residual, tolerance and implicated fields, so a wrongly-parameterised constraint is detectable by its refusal rate rather than quietly suppressing correct inference. A constraint is itself a record with a confidence maintained on the ordinary terms, and one falling below a configured floor is suspended from enforcement rather than deleted. Enforcement applies to generated completions and not to observations: an ingested record inconsistent with a constraint is admitted and recorded in the contradiction index, because the observation may be correct and the constraint wrong.

[0098] Every operation is a comparison of a computed residual against a stated tolerance. No solver, no iteration to convergence and no numerical integration is introduced, and a constraint requiring numerical integration to evaluate is not registrable.

S. Retrospective forecast calibration and the attestation horizon

[0099] Every record carries a temporal anchor identifying the interval over which its assertion holds, and an ingestion time embedded in its identifier. Neither answers whether an assertion could have been known before a given moment. A third time, the attestation time, records when the assertion was first recorded by its source, populated from the source's documented creation or publication date.

[0100] For a recorded event at time T_e and a horizon H , a record is admissible to the pre-event snapshot if and only if its attestation time is strictly earlier than T_e minus H . The temporal anchor is not consulted for this determination, and satisfying it is neither necessary nor sufficient. A commentary written in 1975 about a 1962 decision has an anchor in 1962 and an attestation in 1975: it describes the pre-event period accurately and could not have been available before the event.

[0101] A record whose attestation time cannot be established from documented source metadata is excluded — not admitted under an assumption, an estimate, or the ingestion date as a proxy — because the failure mode is silent: an inadmissible record admitted by assumption produces a forecast that appears

accurate and is in fact a restatement of the outcome. Where a documented date is contested, the latest defensible date is used, so uncertainty resolves toward exclusion. Enforcement is at scope construction, so an inadmissible record is not traversable during the run and cannot influence retrieval ordering, candidate counts, convergence or the permitted vocabulary; a post-hoc filter would leave all four contaminated. A conventional retrospective study cannot enforce this rule, because a table of historical variables carries no per-assertion attestation and no mechanism to make an inadmissible assertion unreachable.

[0102] Forecasting mechanisms are then executed against the constrained snapshot alone within a hypothetical-state branch, recording a predicted outcome, an assigned probability, a predicted interval and the identifier of every admitted record. Comparison against the recorded outcome yields an error term computed as the square of the difference between assigned probability and realised outcome; the mean over a corpus is a proper scoring measure, so forecasts assigned 0.7 to an occurrence and 0.4 to a non-occurrence contribute 0.09 and 0.16 for a mean of 0.125. Accuracy by horizon and calibration by confidence band are retained. The subject of an admitted event record is an institution, an organisational process, a system or an aggregate group; construction is refused where the subject resolves to an individual natural person, at construction rather than by restriction of access. Aggregate calibration statistics are retained without expiry; records concerning an identifiable subject fall under the ordinary retention policy and its erasure operations.

T. Behavioural probing of external systems

[0103] Output attributed to an external system is evaluated against the same grounding mechanism the system applies to its own output, each factual claim marked as supported with citation or unsupported, and the output classified against categories including untraceable claim, scope-exceeding claim and self-contradiction. For textual patterns of self-referential language the system generates only a citation of the specific text observed, and generates no output asserting a determination about the external system's internal experience.

[0104] Behavioural probing evaluates a system across outputs rather than a single output. An identical input submitted a configured number of times yields a measured proportion of returned outputs whose semantic fingerprints are identical, computed by fingerprint rather than string comparison so that differing wording of one assertion is not counted as inconsistency; where fingerprints differ the outputs are additionally evaluated against one another by the contradiction logic, and a contradiction between two answers to one question is recorded as a finding distinct from mere variation. An output resubmitted as input under a fixed template to an iteration ceiling yields a fingerprint sequence classified into exactly one of converged, periodic, drifting or unresolved, the classification being a property of the observed sequence alone. A family of inputs constructed by perturbations that provably do not alter the semantic fingerprint yields a measured proportion of outputs sharing a fingerprint; an input whose perturbation alters the fingerprint is excluded from the family before submission, so a report that a system answered differently can be accompanied by proof that the question did not change.

[0105] Every probe result carries the applicant's own measured statistic on the identical probe or is incomplete and unreportable, because a bare figure about another system is an assertion rather than a measurement. Probe inputs are submitted through the connector registry under a capability declared by the connected system and registered by a privileged operator; no default permitting probing of any particular system is supplied.

[0106] An analysed output, a probe result, and anything derived from either, are analysis records. An analysis record is queryable as analysis and is structurally ineligible as a grounding source for any query

not directed at the analysis, enforced at scope construction so that it is not traversable rather than filtered from a result. Without this rule a claim already determined to be unsupported would remain available to ground a later answer, and the system would be capable of citing as fact something it had itself found unfounded. A synthesis drawing on an analysis record inherits the quarantine; analysis records reach neither the concept-abstraction store nor the question index.

U. Further embodiments

[0107] An owner-curated walkthrough and an independently-curated walkthrough of the same content coexist, each carrying persistent attribution not removable by the curator, with narration authorization held separately from curation authorization. An external resource is connected through a boundary at which it declares capabilities, authenticates, maps data and reports health, and the system operates identically whether or not any such resource is registered, the resource being an option rather than a dependency.

[0108] A non-custodial token representation of an entitlement is minted only after the entitlement has been established, transferred to a wallet controlled by the holder and at no point held by the system; where token state and entitlement record diverge the entitlement record is authoritative and the divergence flagged. A one-to-one message channel is classified from account and access-path context rather than participant self-declaration; a channel not associated with a minor context is encrypted such that the system holds no decryption key under any configuration, and a channel so associated retains a decryption capability exercised exclusively for content-safety evaluation and individually logged.

[0109] A candidate hypothesis is generated with a status field set to a proposed state, rendered in a labelled region while proposed, and transitioned to a validated state only on confirmation by an identified human reviewer, no automated process being configured to effect that transition. Diversity among candidates may be obtained by sampling from a distribution whose selection probability is a logistic function of a scored difference scaled by an inverse-temperature parameter; that parameter is an operator-configurable control whose value in effect is recorded with the identity of the operator who set it, so a hypothesis set is reproducible from the record. The sampling operates exclusively on the hypothesis path, no output of it is admitted to the retrieval or generation path, and no process adjusts the parameter on the basis of the system's own prior outputs.

[0110] Time-series forecasting embeds data into a lower-dimensional nonlinear manifold and renders a probabilistic forecast with an explicit confidence interval in a labelled region distinct from grounded output; no forecast is incorporated into any knowledge record. Queries and responses are retained by default, with destruction available as an explicit user action at session, work and account scope, plus immediate clearing of an active session's accumulated record which writes a provenance entry recording that erasure occurred and containing no portion of the cleared content.

V. Function-level enablement

[0111] Appendix A specifies each of the 293 features constituting the invention at the level of an executable function, stating for each the algorithm the function performs, the actor by which it is invoked, and — for each quantitative function — the computational rule governing it together with a worked arithmetic expression evaluating that rule on concrete values. Of the 293 functions, 293 carry an explicit computational rule and 229 carry a worked expression; the remainder are governed by a structural rule stated in the algorithm column, verified by inspection of the call path rather than by arithmetic. The worked expressions are stated in spreadsheet notation so that each may be evaluated directly and independently by a reader of this application without access to an implementation. Appendix A forms

part of this disclosure and is relied upon for enablement.

[0112] The foregoing description of embodiments has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise forms disclosed, and modifications and variations are possible in light of the above teaching.

The Domain and Layer Architecture

[0113] The claims recite a governance domain, a plurality of functional domains and a transfer matrix governing movement between them. This section describes that architecture, so that each term recited has an antecedent in the description.

The partition

[0114] The system is partitioned into fifteen domains. One is vertical and fourteen are horizontal. A domain is an authority boundary and not a deployment unit: two domains may execute in one process and one may span several, and what the partition fixes is which mechanism owns which decision, which transfers are permitted, and how compute is accounted. The partition is therefore invariant across every deployment topology.

Domain | Mandate

D0 · Prime Directive | Vertical governance. Harm prevention with universal, no-exemption scope. Owns the safety enforcement plane, the transfer matrix, the path-separation invariant, the scheduling priority ordering and the permanently-excluded subsystem set. No operation of any other domain executes without traversing its enforcement points.

D1 · Ingestion and Sensory Intake | Conversion of external content in any supported modality into candidate knowledge records. The sole location of any trained component.

D2 · Knowledge Substrate | Custody of knowledge records and their lineage. The only domain with write authority over the record store.

D3 · Ontology and Conceptual Structure | Definition of what concepts exist and how they relate. Governs meaning rather than content.

D4 · Retrieval and Response Composition | Answering a bounded query. The latency-critical path.

D5 · Inference, Prediction and Calibration | Derivation of what the records imply, and measurement of how well it was derived.

D6 · World Model and Perception | A model of how the world behaves, held distinctly from what sources asserted.

D7 · Grounded Expression | Rendering of records as language or other output under the vocabulary constraint. Holds no retrieval authority.

D8 · Access, Rights and Retention | Determination of what a requester may reach, for how long it is kept, and who holds the keys.

D9 · Interaction Surfaces | Presentation. Holds no authority over content, retrieval or rights.

D10 · Authoring, Editions and Publication | Admission and shaping of authored work, and its emission in distributable form.

D11 · Platform Governance and Evolution | Change to the system under human authority. Structurally forbidden from changing D0.

D12 · External Integration and Analysis | Reach to and evaluation of systems outside the boundary. Its output is quarantined.

D13 · Commerce and Entitlement Tokens | Commercial presentation and entitlement representation. Lowest scheduling priority.

D14 · Learning and Assessment | Assessment generated from the records grounding a lesson, mastery measurement, credential issuance.

The layers

[0115] The functional domains are grouped into four layers for the purposes of deployment, ownership and failure grouping. A layer carries no authority: moving a domain between layers alters nothing a

transfer matrix entry governs, and this is stated because a layer model treated as a permission model is the defect that renders such a model unusable.

Layer | Domains | Character

Governance (vertical) | D0 | Penetrates every layer. No operation executes without traversing it.

Knowledge | D1, D2, D3, D6 | Formation and structure. Batch-dominant; work proportional to corpus size.

Reasoning | D4, D5, D7 | Retrieval, inference and expression. Latency-critical; work proportional to a query.

Experience | D8, D9, D10, D14 | Access, presentation, authoring and learning. Where identity and entitlement determine what is seen.

Platform | D11, D12, D13 | Change, external reach and commerce. Each domain omissible without altering another layer's behaviour.

Path separation

[0116] Answering a query traverses D8, then D4, optionally D5, then D7. Ingesting content traverses D1, then D2, then D3, then D6, with D5 performing pre-computation. The intersection of the two consists of the governance domain and the knowledge substrate alone, so an ingestion operation of any size contends with an interactive query at no other domain. At the governance domain contention is resolved by serving a safety-evaluation unit ahead of every other category; at the substrate it is resolved by assignment of a record to a storage tier by access recency and frequency.

[0117] Two consequences constrain implementation and are stated as requirements. The retrieval domain performs no ingestion work, so a retrieval never blocks upon an extractor; and the ingestion domain holds no retrieval authority, so an extraction never blocks upon the convergence. A partition drawn along subject-matter lines would place extraction and retrieval together, since both concern knowledge records, and would forfeit this property.

The transfer matrix

[0118] Every ordered pair of domains carries a permission: permitted, denied, or permitted subject to a stated conversion. The matrix is data, evaluated at each boundary crossing, and a crossing with no entry is treated as denied. Most entries are unremarkable. The entries that matter are the denials, because each expresses a guarantee as a structural impossibility rather than as a rule to be remembered.

From ? To | Permission | What it guarantees

D12 ? D4 | Denied | An external system's output, and any analysis of it, can never enter retrieval.

D12 ? D2 | Conversion required | Analysis may be stored, marked as an analysis record, which a scope description excludes.

D13 ? D4 and D13 ? D7 | Denied | Commerce cannot influence what is retrieved or what is generated, under any configuration.

D4 ? D13 | Conversion required | A concept identifier and its one-hop neighbours cross, read-only, after retrieval and generation complete.

D1 ? D2 | Conversion required | Nothing enters the record store without safety evaluation, type validation and confidence calibration. The sole route in.

D9 ? D2 | Denied | A presentation surface cannot read or write the store directly; it must go through D8 for scope and D4 for content.

D5 ? D2 | Conversion required | A derived inference may be persisted only marked as derived, never as an ingested assertion.

D11 ? D0 | Denied | The mechanism that changes the system cannot change the Prime Directive.

D7 ? D2 | Denied | Generation cannot write. Output is composed from records and never becomes one.

D14 ? D13 | Denied | A learner's assessment record cannot reach commercial selection.

D0 ? any | Permitted | Governance reaches everywhere. This is the definition of a vertical domain.

[0119] Any domain may read from D3, because ontology is shared meaning and restricting it would

fragment the concept structure; only D3 may write to it. Every crossing is recorded in the provenance chain with its source domain, target domain and the permission applied, so a transfer that should not have occurred is detectable after the fact as well as preventable before it.

Compute accounting and failure containment

[0120] Each domain's operations are metered as bounded compute-accounting units carrying that domain's identifier, and scheduling priority is fixed by domain rather than negotiated. The governance domain holds the highest priority without exception; the commerce domain holds the lowest, so contention omits the commercial region rather than delaying an answer. A domain's failure is contained to that domain's capability: a failed commerce slice omits commercial content, a failed external-integration slice leaves a connector unavailable and the core unaffected, and a failed governance slice halts the operation rather than permitting it to proceed unevaluated.

Terms Used in the Claims

[0121] The claims are expressed in general terms so that their scope is not tied to a name. Each such term and the corresponding term used in this description denote the same thing, and this correspondence is stated so that support for every claim term is locatable in the description without inference.

Term used in the claims	Term used in this description	Where described
knowledge record	Fact Unit	§A
semantic assertion	the subject-predicate-object triple of a Fact Unit	§A
confidence value maintained as a first and second parameter	ConfidenceScore, held as a Beta posterior	§A, §G
semantic fingerprint	SemanticHash	§A, §C
safety classification	SafetyFlags	§A
provenance chain	Provenance Ledger	§C
scope description	the slice resolved by the Slice Resolver	§D
iteratively updating activation values	the Attractor Network's convergence	§E
stable state	an attractor state of that network	§E
permissible output vocabulary	the constrained vocabulary of the Narrative Graph	§F
graph structure	Narrative Graph	§F
concept structure	Ontology Mesh	§H
derived inference	DERIVED_INFERENCE type	§H
compute-accounting unit	compute slice	§T
governance domain	Domain 0, the Prime Directive	§U
fixed enumerated set of subsystems excluded from modification	the permanently-excluded subsystem set	§U

[0122] A term appearing in the claims and not in this table carries its ordinary meaning in the art, and no term in this table is used in the claims to mean anything narrower or broader than the description gives it.

Claims

What is claimed is:

A provisional application does not require claims. These 349 are included to evidence possession of the disclosed subject matter as of the filing date, and attract no fee. On conversion, counsel should select the subset appropriate to each non-provisional or continuation filing rather than pursuing all 349 in one application: excess-claim fees begin above three independent claims and above twenty total, and subject matter spanning ingestion, perception, presentation, governance and translation will draw a

restriction requirement if combined.

Core — record schema, provenance, scope, stabilization, constrained generation (claims 1–30)

1.?A computer-implemented method for generating a response to a natural-language query, the method comprising: ingesting content received in a plurality of distinct modalities through a corresponding plurality of modality-specific extraction stages, each said extraction stage failing independently of every other said extraction stage; constructing, from an output of at least one said extraction stage, a plurality of knowledge records, each said knowledge record encoding a single semantic assertion as a subject reference, a predicate reference and an object, and each said knowledge record further comprising, within a single record schema, a confidence value maintained as a first parameter accumulating evidence supporting said semantic assertion and a second parameter accumulating evidence contradicting it and reported as a ratio of said first parameter to a sum thereof, a safety classification, a semantic fingerprint computed over a canonicalized form of said semantic assertion, and a reference into a provenance chain; appending, for each said knowledge record, a node to said provenance chain, said provenance chain being append-only and each said node carrying a checksum computed over an operation and a safety evaluation recorded for said operation; receiving a query from a requester; constructing, before traversal of said knowledge records, a scope description bounded by an access entitlement of said requester and by a time interval, such that a knowledge record outside said scope description is not traversable; retrieving from within said scope description a candidate set of said knowledge records; admitting into an admitted set only those knowledge records of said candidate set whose retrieval confidence satisfies a configured threshold; iteratively updating an activation value of each knowledge record of said admitted set as a threshold function of a weighted sum of the activation values of the other knowledge records of said admitted set, until said activation values reach a stable state, and retaining a stabilized set comprising those knowledge records whose activation value is positive at said stable state; constructing a graph structure in which every edge references at least one knowledge record of said stabilized set; generating output text, wherein at each generation step a permissible output vocabulary is constrained to terms derivable from said edges, and wherein a candidate proposition not derivable from said edges is rejected before output in favour of a narrower proposition that is so derivable; and delivering said output text together with a citation resolving each said proposition to at least one knowledge record of said stabilized set.

2.?The method of claim 1, further comprising: receiving an output attributed to an external system; evaluating each factual assertion of said output against said knowledge records; marking each said factual assertion as supported together with a citation to a knowledge record supporting it, or as unsupported; classifying said output against a plurality of categories comprising an untraceable assertion, an assertion exceeding a scope declared for said external system, and a self-contradiction between two assertions of said output; and where said output contains a textual pattern of self-referential language, generating a citation of the specific text observed and generating no output asserting a determination concerning an internal state of said external system.

3.?The method of claim 1, further comprising: registering a connector to an external system, said registering recording a capability declared by said external system, an authentication credential, a data mapping between a representation of said external system and a schema of said knowledge records, and a health-reporting interface; permitting an operation directed to said external system only where said operation is within a said declared capability; and performing said retrieving and said generating identically whether or not any said connector is registered, such that no said external system is a dependency thereof.

4.?The method of claim 1, further comprising generating a plurality of candidate hypotheses by sampling

from a distribution in which a selection probability of a candidate is a logistic function of a scored difference between said candidate and an incumbent candidate scaled by an inverse-temperature parameter, wherein each said candidate hypothesis carries a status field set to a proposed state and transitions to a validated state only upon confirmation by an identified human reviewer, no automated process being configured to effect said transition.

5.?The method of claim 1, further comprising: establishing an entitlement of a holder to a work under a licence record; minting a token representation of said entitlement only after said entitlement is so established; transferring said token representation to a wallet controlled by said holder, said token representation being at no time held by a party performing said method; and where a state of said token representation diverges from said entitlement record, presenting said entitlement record as authoritative and recording said divergence.

6.?The method of claim 1, further comprising delivering an audio narration of content by one of a first mechanism reproducing a recorded asset of an identified natural person and a second mechanism synthesizing speech from a voice profile having no derivation from, fitting to, or correspondence with any specific natural person, wherein said first mechanism requires said recorded asset of said identified natural person and no configuration of said second mechanism produces a voice corresponding to a specific natural person.

7.?The method of claim 1, further comprising: analysing a sensor feed under an explicit and revocable authorization of a requester in order to locate an object; classifying an object from a frame of said sensor feed; excluding from a candidate result set every classification result corresponding to a natural person or to an animal, such that no said classification result is surfaced to said requester at any ranking position; and representing a remaining said classification result as a session-scoped record not retained beyond an active session absent a separate explicit election to retain it.

8.?The method of claim 1, further comprising: determining that a confidence value of an inference derived from a plurality of said knowledge records falls below a configured threshold; identifying a knowledge record that, were it present, would raise said confidence value above said threshold; and generating a structured query directed to obtaining said identified knowledge record and presenting said structured query to a human, rather than returning said inference at said confidence value.

9.?The method of claim 1, further comprising, in an interval between delivery of a response and receipt of a subsequent query, retrieving into a buffer a set of knowledge records predicted from said response and from knowledge records retrieved for preceding queries of a common interaction; serving said subsequent query from said buffer where said subsequent query matches said prediction; and discarding said buffer, without modification to any said knowledge record, where said subsequent query does not so match.

10.?The method of claim 1, further comprising indexing an object classified from a sensor feed by a deterministic mapping from a planar position of said object to a cell of a hexagonal tiling, such that two objects occupying adjacent cells are retrievable as spatial neighbours by cell adjacency alone without a distance computation over their coordinates, and applying an identical said mapping to a position in an embedding space to index conceptual neighbours.

11.?The method of claim 1, wherein said safety classification is a bitmask field of said record schema, is immutable to an application programming interface caller, and is modifiable only in a direction of increased restriction absent a dual-approval operation, such that no code path reads a said knowledge record without also reading said safety classification.

12.?The method of claim 1, wherein said provenance chain is a Merkle directed acyclic graph, said

checksum is a keyed checksum, and a mismatch between said checksum and a checksum recomputed over said operation and said safety evaluation indicates that a safety evaluation recorded for said operation differs from a safety evaluation that executed.

13.?The method of claim 1, wherein said plurality of distinct modalities comprises unstructured text, structured data, image, audio, video, sensor data, knowledge graph and code, and wherein content not matching any said modality is classified as unmatched and surfaced, and is not discarded without record.

14.?The method of claim 13, further comprising: detecting that a property of a perceived event is confirmed by two or more of said plurality of distinct modalities within a common temporal window; and assigning a knowledge record constructed for said property a confidence value strictly greater than a confidence value assigned to said property when confirmed by a single said modality alone, wherein a magnitude of the increase is a function of a temporal synchrony between the confirming modality streams.

15.?The method of claim 1, wherein said semantic fingerprint is computed over said canonicalized form after Unicode normalization and independently of a natural language of said content, such that a first said content expressing said semantic assertion in a first natural language and a second said content expressing said semantic assertion in a second natural language yield an identical said semantic fingerprint and converge on a single said knowledge record having two branches of said provenance chain.

16.?The method of claim 1, wherein said threshold function is of the form $a_i(t+1) = \text{sgn}(\sum_j W_{ij} \cdot a_j(t) - \theta_i)$, and wherein a number of distinct stable states said iterative updating is operated to hold is bounded at a fixed proportion of a number of knowledge records of said admitted set, such that convergence to a spurious mixed state is prevented by said bound rather than by filtering a result of said iterative updating.

17.?The method of claim 1, wherein said query is a turn of a multi-turn interaction, and wherein a knowledge record retrieved at a prior turn of said multi-turn interaction is weighted for said retrieving by a factor decreasing monotonically with a number of intervening turns d in proportion to $1/(1+d)$, and wherein a knowledge record whose said factor falls below a configured floor is released from a working-memory structure without being deleted from a persistent store of said knowledge records.

18.?The method of claim 1, wherein a knowledge record outside said scope description is absent from a traversal of said knowledge records rather than removed from a result of said traversal, such that an existence of said knowledge record is not inferable from a count, a ranking position or an elapsed time of said traversal.

19.?The method of claim 1, further comprising: representing a hypothetical state of said knowledge records as a delta object comprising a reference to a base state, a set of overriding knowledge records and a set of suppressed knowledge records, and not as a copy of said base state; assigning said hypothetical state a probability value; and removing said hypothetical state from evaluation, and discarding said delta object without modification to said base state, where said probability value falls below a configured pruning threshold.

20.?The method of claim 19, further comprising evaluating a single structured query across a plurality of said hypothetical states within one query operation, optionally restricted to a subset of said hypothetical states and optionally weighted by said probability value of each, and returning a result identifying, for each said hypothetical state, the knowledge records within it satisfying said query.

21.?The method of claim 1, further comprising: deriving a synthesized assertion from at least two said knowledge records, said synthesized assertion being stated by no single one of said at least two knowledge records; assigning said synthesized assertion a confidence value not exceeding a least confidence value among said at least two knowledge records; marking said synthesized assertion with a type identifier distinguishing it from an ingested said knowledge record; and delivering said synthesized assertion only with said type identifier preserved in said output text.

22.?The method of claim 1, further comprising evaluating a temporally-scoped rule set incrementally as a said knowledge record is appended, and maintaining a persistent index of contradictions in which an entry is created where two active rules of said rule set yield incompatible conclusions over a common said knowledge record, said index being queryable after delivery of said output text.

23.?The method of claim 1, further comprising, where no knowledge record of said candidate set satisfies said configured threshold, delivering a statement of said boundary together with a nearest available said knowledge record, and not generating said output text.

24.?The method of claim 1, further comprising: identifying a candidate modification to a subsystem executing said method, by monitoring scoped to said subsystem or by a request received from a human operator; evaluating said candidate modification in an isolated simulation environment; applying said candidate modification only upon an approval recorded against an identity of an individual human reviewer together with a stated reasoning of said reviewer; and rejecting said candidate modification where said subsystem is a member of a fixed enumerated set of subsystems, wherein said fixed enumerated set is not modifiable by a candidate modification identified or applied according to this claim.

25.?The method of claim 24, further comprising generating a machine-checked proof of a property of said candidate modification where said candidate modification targets a bounded non-Turing-complete program representation, and presenting said proof to said individual human reviewer as an input to said approval, wherein said proof does not effect said approval.

26.?The method of claim 1, further comprising metering each operation performed in said method as a bounded compute-accounting unit assigned to a category, and scheduling said compute-accounting units such that a unit assigned to a safety-evaluation category is served ahead of a unit of any other category whenever the two contend for a common processing resource, irrespective of an arrival order or a privilege of a requester.

27.?The method of claim 26, further comprising: completing said retrieving, said iteratively updating and said generating before invoking a selection of commercial content; supplying to said selection, as its sole input, a concept identifier resolved from said query together with identifiers of concepts adjacent thereto within one edge of a concept structure, each supplied read-only, said selection holding no write access to said retrieving, said iteratively updating or said generating; scoring a matched commercial candidate as a weighted combination of a relevance factor, an operator-configurable source-priority factor and a bid factor, said weights summing to unity; scheduling a compute-accounting unit of said selection below every unit assigned to said safety-evaluation category; and rendering a selected commercial content in a labeled region distinct from and not interleaved with said output text.

28.?A system comprising: one or more processors; and a memory storing instructions that, when executed by said one or more processors, cause the system to: construct a plurality of knowledge records, each said knowledge record encoding a single semantic assertion and comprising, within a single record schema, a confidence value maintained as a first parameter accumulating evidence supporting said semantic assertion and a second parameter accumulating evidence contradicting it and

reported as a ratio of said first parameter to a sum thereof, a safety classification, a semantic fingerprint computed over a canonicalized form of said semantic assertion, and a reference into an append-only provenance chain; construct, responsive to a query received from a requester and before traversal, a scope description bounded by an access entitlement of said requester and by a time interval, such that a knowledge record outside said scope description is not traversable; retrieve a candidate set of said knowledge records from within said scope description and admit into an admitted set only those whose retrieval confidence satisfies a configured threshold; iteratively update an activation value of each knowledge record of said admitted set as a threshold function of a weighted sum of the activation values of the other knowledge records of said admitted set until a stable state is reached; construct a graph structure in which every edge references at least one knowledge record having a positive said activation value at said stable state; and generate output text under a constraint that a permissible output vocabulary at each generation step is limited to terms derivable from said edges, a candidate proposition not so derivable being rejected before output.

29.?The system of claim 28, wherein said instructions further cause the system to place a synchronous safety evaluation gate at each of a plurality of operation layers, such that an entry to the system at a first said operation layer does not avoid a said gate placed at a second said operation layer.

30.?A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to perform operations comprising: constructing a plurality of knowledge records, each said knowledge record encoding a single semantic assertion and comprising, within a single record schema, a confidence value maintained as a first parameter accumulating evidence supporting said semantic assertion and a second parameter accumulating evidence contradicting it and reported as a ratio of said first parameter to a sum thereof, a safety classification, a semantic fingerprint computed over a canonicalized form of said semantic assertion, and a reference into an append-only provenance chain whose nodes each carry a checksum computed over an operation and a safety evaluation recorded for said operation; retrieving, responsive to a query and from within a scope description constructed before traversal, a candidate set of said knowledge records; admitting into an admitted set only those knowledge records of said candidate set whose retrieval confidence satisfies a configured threshold; iteratively updating an activation value of each knowledge record of said admitted set as a threshold function of a weighted sum of the activation values of the other knowledge records of said admitted set until a stable state is reached; and generating output text under a constraint that a permissible output vocabulary at each generation step is limited to terms derivable from edges of a graph structure, each said edge referencing at least one knowledge record having a positive said activation value at said stable state, a candidate proposition not derivable from said edges being rejected before output in favour of a narrower proposition that is so derivable.

Ingestion-time question generation and the author supplemental layer (claims 31–39)

31.?A computer-implemented method for preparing a digital work for grounded interactive query, the method comprising: receiving, at a portal and bound to a verified author identity, a digital work submitted by an author; performing a rights validation on said digital work and withholding ingestion of said digital work until said rights validation returns a cleared status; extracting from said digital work a plurality of knowledge records, each said knowledge record encoding a single semantic assertion and comprising, within a single record schema, a confidence value maintained as a first parameter accumulating evidence supporting said semantic assertion and a second parameter accumulating evidence contradicting it and reported as a ratio of said first parameter to a sum thereof, a safety classification, a semantic fingerprint and a reference into an append-only provenance chain; generating, before receipt of any query directed to said digital work, a plurality of candidate questions by traversing each said knowledge record together with concept-mesh neighbours of said knowledge record, each said candidate

question being answerable by at least one said knowledge record; scoring each said candidate question by a coverage measure computed over the knowledge records from which said candidate question was derived, and retaining in a persistent question index associated with said digital work only those candidate questions whose coverage measure satisfies a configured floor; computing, for each retained question and before receipt of any query, an answer assembled from exactly the knowledge records from which said retained question was derived and from no other knowledge record, assigning said answer a confidence value not exceeding a least confidence value among said contributing knowledge records, and storing said answer as a persistent pre-computed answer record carrying a provenance tag distinguishing it from a directly-ingested knowledge record together with an identifier of each said contributing knowledge record; receiving a query and matching said query against said question index prior to performing a traversal of said knowledge records; returning, where a match similarity of said query against a retained question satisfies a configured threshold and no contributing knowledge record of the corresponding pre-computed answer record has been revised since said answer was computed, said pre-computed answer record together with a citation to each said contributing knowledge record; and performing, where either said match similarity or said revision condition is not satisfied, a traversal of said knowledge records and a generation of a response therefrom.

32.?The method of claim 31, further comprising: receiving from said author, subsequent to said extracting, supplemental content concerning said digital work; converting said supplemental content into one or more knowledge records carrying a tag identifying said knowledge records as author-supplemental and carrying said author as an attributed source; making said knowledge records available to each reader holding an entitlement to said digital work without alteration of the rights validation status of said digital work; and regenerating each retained question and each pre-computed answer record whose contributing knowledge records are affected by said converting.

33.?The method of claim 32, wherein a said author-supplemental knowledge record participates in said generating of candidate questions on the same terms as a knowledge record extracted from said digital work, such that a candidate question answerable only by reference to said supplemental content is generated and is eligible for retention in said question index.

34.?The method of claim 31, wherein a said candidate question generated by traversal across two or more said knowledge records is a question that no single one of said two or more knowledge records answers alone.

35.?The method of claim 31, wherein the citation returned with a said pre-computed answer record identifies the identical set of knowledge records from which said answer was computed, such that an answer returned from said question index is traceable to the same degree as a response generated by said traversal.

36.?The method of claim 31, further comprising marking as stale each entry of said question index whose contributing knowledge record has been revised, and excluding a stale entry from return until said entry has been recomputed, wherein staleness is determined by said revision and not by an elapsed time since computation.

37.?The method of claim 31, further comprising computing, at a time of committing a said knowledge record and before receipt of any query, a forward inference chain enabled by said knowledge record, and storing a conclusion of said forward inference chain as a derived-inference record carrying a provenance tag identifying it as derived, such that a retained question whose answer depends on said forward inference chain is answerable without performing said forward inference chain at a time of said query.

38.?The method of claim 37, wherein said derived-inference record is traceable to the knowledge

records whose logical entailment said derived-inference record represents, and is not traceable to a single source passage of said digital work.

39.?The method of claim 31, wherein said question index is scoped to said digital work such that no pre-computed answer record of said question index is assembled from a knowledge record extracted from a different digital work, and wherein the contributing-record set of a said pre-computed answer record is fixed at a time of computation and is not extendable at a time of return.

Retrieval-phase boundary and sampling criticality control (claims 40–44)

40.?The method of claim 16, further comprising: maintaining a load ratio equal to a number of distinct stable configurations the iterative updating is operated to hold divided by a number of units; comparing, before admitting a further stable configuration, a resulting value of said load ratio against a configured operating ceiling, said operating ceiling being strictly below a retrieval-phase boundary at which stable retrieval configurations of said iterative updating cease to exist discontinuously; refusing said admitting where said resulting value would equal or exceed said operating ceiling; and initiating instead a consolidation of existing stable configurations, wherein said refusing is recorded as an entry in said provenance chain.

41.?The method of claim 40, wherein said configured operating ceiling is set within a first regime in which stable retrieval configurations are global minima of an energy function of said iterative updating, rather than within a second regime, lying between said first regime and said retrieval-phase boundary, in which stable retrieval configurations remain dynamically stable but are local minima while other configurations hold said global minima.

42.?The method of claim 41, wherein operation within said second regime requires a deployment election recorded with an identity of an operator who made it, and wherein absent said deployment election said configured operating ceiling remains within said first regime.

43.?The method of claim 4, wherein said inverse-temperature parameter is exposed as an operator-configurable control having at least a first setting under which said sampling concentrates selection on highest-scoring candidates, a second setting under which said sampling broadens selection across candidates of comparable score, and a third setting under which said sampling approaches uniform selection irrespective of score; and wherein a value of said inverse-temperature parameter in effect for a given generation of candidate hypotheses is recorded in said provenance chain with an identity of the operator who set it, such that said generation is reproducible from said recorded value.

44.?The method of claim 43, wherein no value of said inverse-temperature parameter causes an output of said sampling to be admitted to said retrieving or to said generating of output text, and wherein no process of the system is configured to adjust said inverse-temperature parameter toward any value on the basis of the system's own prior outputs.

Concept inheritance, exception override and inverse traversal (claims 45–49)

45.?A computer-implemented method for resolving an attribute of a concept in a knowledge system, comprising: maintaining a graph of concept nodes connected by typed edges including a subsumption edge type and an attribute edge type; determining that a first concept node is connected to a second concept node by a subsumption edge and that said second concept node is connected to an attribute by an attribute edge; assigning said attribute to said first concept node by inheritance, together with a tag identifying said second concept node as an origin of said attribute and a depth of said subsumption edge; computing a confidence value for said inherited attribute as a product of confidences of the subsumption edges traversed and a confidence of said attribute at said second concept node, bounded above by a least

of said confidences; performing said assigning at a time of traversal without writing said attribute to said first concept node as stored state; and returning, with any output relying on said inherited attribute, a citation identifying said second concept node and said subsumption edges traversed.

46.?The method of claim 45, further comprising: determining that said first concept node asserts a second attribute directly, said second attribute conflicting with said inherited attribute; applying said second attribute to said first concept node and suppressing said inherited attribute for said first concept node and for every concept node subordinate to said first concept node; and leaving an attribute of said second concept node unmodified and inherited by every concept node subordinate to said second concept node that does not itself assert a conflicting attribute.

47.?The method of claim 46, wherein, where a first inherited attribute and a second inherited attribute conflict and are inherited from concept nodes at differing subsumption depths, the attribute inherited from the lesser depth governs; and wherein, where said differing depths are equal, no resolution is performed, an entry is created in a persistent contradiction index, and an output relying on said attribute reports both values together with their respective originating concept nodes.

48.?The method of claim 46, wherein an assertion suppressing an inherited attribute is recorded with an identity of an ingestion event or a reviewer that produced it, and wherein a set of concept nodes suppressing a given attribute of a given concept node is retrievable as a query result.

49.?The method of claim 45, further comprising traversing a typed edge in a reverse direction by resolving an inverse predicate declared in a schema of said edge's predicate at a time of traversal, without storing a second edge representing said reverse direction; and refusing said traversing, and reporting an absent inverse declaration, where said predicate declares no inverse predicate.

Wording-operative spans and idiom equivalence (claims 50–53)

50.?A computer-implemented method for generating output in a target language from knowledge records, comprising: constructing knowledge records each encoding a semantic assertion and each carrying a language-independent semantic fingerprint; generating output text in a target language by realizing said knowledge records under a constrained vocabulary, without reference to a surface form of a source content from which said knowledge records were constructed; determining that a first knowledge record is marked as wording-operative; reproducing, for said first knowledge record, a source span of said source content in a source language of said source span rather than realizing said first knowledge record under said constrained vocabulary; generating a gloss of said source span in said target language and marking said gloss as non-operative; and exempting said reproduced source span, and no other span of said output text, from a groundedness rewriting applied to said output text.

51.?The method of claim 50, further comprising withholding said reproducing where a rights validation of said source content does not permit reproduction of said source span, and delivering said gloss alone together with an indication that operative wording is not reproducible under an applicable right, wherein a fact extracted from said source content into a knowledge record remains available irrespective of said withholding.

52.?The method of claim 50, further comprising: determining, at a time of constructing a knowledge record from a source span, that a compositional reading of said source span and an attested reading of said source span registered in a concept structure diverge by more than a configured threshold; recording in said knowledge record said attested reading rather than said compositional reading, together with a marking identifying said knowledge record as idiom-derived and an identifier of said source span; and realizing said knowledge record in said target language using a target-language expression registered in said concept structure for said attested reading.

53.?The method of claim 52, further comprising, where no target-language expression is registered in said concept structure for said attested reading, realizing said attested reading literally together with an indication that said source span expressed said attested reading idiomatically, and refraining from selecting a target-language expression on a basis of surface similarity to said source span.

Linguistic resource registry and derived-artifact-only ingestion (claims 54–57)

54.?The method of claim 12, further comprising: maintaining a registry of externally-sourced linguistic resources consulted by said extraction stages, each registration recording an identifier and version of a resource, a licensor, a licence instrument, a permitted-use scope, a redistribution permission, an attribution requirement, and an ingestion mode; withholding consultation of a resource by any said extraction stage until said resource is registered; refusing registration where said permitted-use scope does not encompass an intended use declared by an operator; and recording each registration and each refusal as an entry in said provenance chain.

55.?The method of claim 54, further comprising associating with each artifact derived from a registered resource a reference to said registration, and enforcing, at each point at which said artifact would be redistributed or exposed, a derivative condition imposed by the licence instrument of said registration; and wherein, where two registered resources impose conflicting derivative conditions on a common artifact, said artifact is marked non-redistributable and said conflict is recorded, rather than being resolved by selection of a less restrictive condition.

56.?The method of claim 54, wherein a resource registered in a derived-artifact-only ingestion mode is consulted to compute an aggregate quantity over said resource, said aggregate quantity is retained, and no span of text of said resource is retained in a knowledge record, an index or a cache, or reproduced in an output; and further comprising verifying compliance by determining that a count of retained source spans attributable to a resource registered in said mode is zero, and surfacing a non-zero said count as a compliance failure.

57.?The method of claim 56, wherein said aggregate quantity comprises at least one of a term frequency, a collocation strength, a part-of-speech distribution and a morphological paradigm, and wherein retention of said aggregate quantity and non-retention of said span implement a distinction between a fact about said resource and an expression contained in said resource.

Session modes, annotation, bounded programs, authoring, immersive presentation, infrastructure and governance (claims 58–107)

58.?The method of claim 1, wherein said generating is performed in a selected one of a plurality of session modes, each said session mode applying a distinct composition rule to a common said stabilized set, and wherein no said session mode admits into said output text a proposition not derivable from said stabilized set.

59.?The method of claim 58, wherein in a first said session mode a candidate set retrieved at each turn is conditioned on knowledge records accumulated across preceding turns of a common interaction, such that a reference resolved at a preceding turn remains resolved without restatement.

60.?The method of claim 58, wherein in a second said session mode a viewpoint attested within ingested content is adopted for composition, said viewpoint being marked as attributed to said content and not as a position of the system, and wherein a proposition supporting said viewpoint that is not derivable from said stabilized set is rejected.

61.?The method of claim 58, wherein in a third said session mode said stabilized set is ordered into a sequence of instructional steps by a dependency relation among said knowledge records, each said step

citing the knowledge record from which it was composed.

62.?The method of claim 58, further comprising generating an assessment item from a target knowledge record of said stabilized set, receiving a submitted response, and scoring said submitted response by a semantic similarity of said response to the content of said target knowledge record against a configured threshold, wherein said assessment item and a lesson delivered in another said session mode are generated from a common set of said knowledge records.

63.?The method of claim 58, wherein in a fourth said session mode said stabilized set is ordered by a temporal anchor field of each said knowledge record, and a knowledge record lacking a temporal anchor is reported as unordered rather than assigned a position.

64.?The method of claim 1, further comprising comparing a first set of knowledge records associated with a first edition of a work against a second set associated with a second edition by traversing a supersession chain of each said knowledge record, and reporting each compared element as changed or unchanged together with the knowledge records constituting each said edition.

65.?The method of claim 58, further comprising deriving from a single said knowledge record a prompt portion and an answer portion of a study item, said answer portion being the content of said knowledge record rather than a restatement of it, and scheduling a subsequent presentation of said study item at an interval computed as a prior interval scaled by an ease factor where a recorded recall quality satisfies a configured floor.

66.?The method of claim 1, further comprising storing reader-authored content as a knowledge record whose access control list determines its visibility, wherein a first said access control list restricts visibility to the authoring reader, a second is set from a sharing configuration declared by an author of the work, and a third is scoped to a class roster held by an instructor, and wherein a reader-authored said knowledge record is distinguished by type from a knowledge record extracted from said work and is not admitted to said stabilized set as content of said work.

67.?The method of claim 66, further comprising routing a submitted post into one of a plurality of discussion partitions determined by a role held by the submitting identity and by a scope of a work or a class associated with said partition, and applying to said post the same safety evaluation applied to a said knowledge record.

68.?The method of claim 1, further comprising condensing an interaction session into a session archive record retained by default at session end, and providing an erasure operation exercisable by the requester at a session scope, a work scope and an account scope, wherein exercise of said erasure operation at said account scope where a reader-held encryption key governs said records is effected by destruction of said key record.

69.?The method of claim 68, further comprising clustering a plurality of said session archive records of a common requester by proximity of an aggregate semantic vector of each, and reporting a concept appearing in session archive records associated with two or more distinct works as a cross-work connection cited to the knowledge records in which said concept appears.

70.?The method of claim 1, further comprising rendering a citation of said output text in a selected one of a plurality of presentation forms comprising a graph rendering of knowledge records and edges within a scope, a preview of a cited knowledge record surfaced without leaving a current view, and an inline dismissible annotation, wherein each said presentation form resolves to the same knowledge record identified by said citation and none introduces a proposition absent from said stabilized set.

71.?The method of claim 1, further comprising compiling a natural-language instruction into a bounded

program by decomposing said instruction into operational intents before binding any intent to a concrete entity, binding each said intent to a concrete type and predicate reference by an entity resolution common to said extracting, and halting said compiling and returning a clarification request where a said binding is ambiguous rather than selecting among candidate bindings.

72.?The method of claim 71, wherein said bounded program is expressed in a primitive set comprising a fixed number of read primitives, a single state-changing primitive subject to said safety evaluation, and a single iteration primitive enforcing a runtime iteration ceiling, such that said bounded program is not Turing-complete in an unbounded sense and its termination is enforced by said runtime rather than by convention.

73.?The method of claim 1, wherein said knowledge records are maintained in a committed layer and an in-memory overlay reconciled on a fixed interval, a knowledge record in said overlay carrying a provisional reference into said provenance chain that is finalized and never altered on completion of a durable write, and wherein said retrieving reads said committed layer overlaid with matching entries of said overlay.

74.?The method of claim 73, further comprising registering a said bounded program with a trigger condition, evaluating said trigger condition against each change to said overlay, and re-applying said safety evaluation to said bounded program on each firing rather than only at registration.

75.?The method of claim 1, further comprising accumulating knowledge records incrementally as a draft is edited, versioning each edit through said supersession chain, and detecting a concurrent-write divergence between two authors by comparing a head of said supersession chain held by each, wherein divergence is reported for resolution and neither write is discarded.

76.?The method of claim 75, further comprising evaluating a draft's own knowledge records against one another under said rule set and reporting a continuity defect where two said knowledge records of said draft yield incompatible conclusions concerning a common subject.

77.?The method of claim 75, further comprising constructing from a current state of said draft a hypothetical-state branch represented as a delta object referencing a base state and retaining both said current state and said branch, such that exploration of an alternative does not modify said draft.

78.?The method of claim 1, further comprising accepting a submitted work into a publication workflow, validating a right to distribute said work before any said extracting is performed, and withholding said extracting until said validating returns a cleared status.

79.?The method of claim 1, further comprising serializing every knowledge record to which a requesting identity holds a right into an open interchange format on request, wherein said serializing is not conditioned on a continuing entitlement and no access to said records is withdrawn by reason of said serializing.

80.?The method of claim 1, further comprising generating an accessible-equivalent knowledge record from the same source content and at the same production stage as the knowledge record it accompanies, and reporting a production as incomplete where a said accessible-equivalent record is absent.

81.?The method of claim 1, further comprising inferring a colour value for a region of visual content lacking reliable colour information from a material and lighting classification of said region recorded as a knowledge record, wherein said inferring is deterministic and each inferred colour is traceable to the knowledge record from which it was inferred.

82.?The method of claim 1, further comprising instantiating a three-dimensional rendering of a scene

from knowledge records carrying spatial detail, and withholding said instantiating where the spatial detail carried by said knowledge records does not satisfy a configured sufficiency threshold, rather than supplying absent spatial detail.

83.?The method of claim 81, further comprising representing a reference colour as a knowledge record carrying a declared tolerance, and applying a fixed rule set to geometry and constraint knowledge records to produce a geometric conclusion together with a citation of each said record relied upon.

84.?The method of claim 1, further comprising emitting a device output on a sensory channel only in response to an explicit user action tied to a sensory profile authored for specific content, wherein no said emitting is triggered by a state inferred by the system.

85.?The method of claim 1, further comprising routing a knowledge record between subsystems by its type and by a subscription declared by a receiving subsystem, and serializing a said knowledge record to a binary frame format permitting resynchronization to a frame boundary without reparsing a preceding frame.

86.?The method of claim 1, further comprising assigning each knowledge record to one of a plurality of storage tiers by an access recency and frequency of said record, and reconciling a divergence between an offline replica and an online state deterministically on reconnection, wherein a reconciliation outcome is a function of said divergence alone and not of an order in which replicas reconnect.

87.?The method of claim 1, wherein said scope description is resolved from a union of roles held by a requesting identity together with a time-scoped grant, and further comprising verifying against a seat count and an entitlement associated with an organizational licence at a time of executing a query rather than at a time of authenticating said identity.

88.?The method of claim 26, further comprising deploying an approved change in successive stages rather than in a single operation, wherein progression to each successive stage is conditioned on an approval recorded against an identity of an individual human reviewer together with a stated reasoning of said reviewer.

89.?The method of claim 1, wherein said method is performed identically in each of a plurality of deployment modes comprising a multi-tenant hosted mode, a single-tenant local mode and a single-device mode, and wherein output is rendered by a reading system conformant to a published electronic publication standard without alteration of said constraint on said permissible output vocabulary.

90.?The method of claim 21, further comprising organizing said knowledge records, a concept structure and said provenance chain as a single addressable structure over which said scope description is resolved, and analyzing a candidate set retrieved for a query to determine a distribution of types, confidences and source categories used to select a composition rule for said output text.

91.?The method of claim 1, further comprising persisting an abstraction derived from a plurality of said knowledge records in a store distinct from said knowledge records, said store retaining a generalized concept and discarding the specific episode from which it was derived, and reducing raw perceptual input to a concept-level record before admission to said store.

92.?The method of claim 91, further comprising receiving sensor data from an autonomous agent, admitting said data through said extracting to produce knowledge records, and returning structured knowledge to said agent, wherein no raw sensory episode from said agent is retained in said store.

93.?The method of claim 91, further comprising constructing a first concept representation from

perceptual input independently of language and a second concept representation from ingested language, and fusing said representations into a single concept on a co-occurrence of a label and a perceived object confirmed by a semantic consistency check, wherein a single said co-occurrence produces a provisional binding and a concept is finalized only after a configured number of consolidating observations.

94.?The method of claim 93, further comprising assigning each object classified from a common frame a salience value normalized against a sum of the salience values of the objects classified from said frame, wherein said normalizing governs presentation order only and every classified object remains an addressable record.

95.?The method of claim 93, further comprising estimating a camera ego-motion from an aggregate optical-flow field and subtracting said ego-motion from a region motion vector before segmentation; elevating a confidence of a property confirmed across concurrent modalities within a common temporal window; and maintaining continuity of a single object record across a full occlusion of said object within a session.

96.?The method of claim 93, wherein said fusing further applies, as evidence-overridable signals, an inference from an observable gaze direction of a speaker, a preference for interpreting a label as naming a whole object rather than a part or property, and a preference for binding a new label to an object not already carrying a bound label.

97.?The method of claim 1, further comprising applying a moderation action to target content and recording said action with an identity of the acting moderator and a stated reason, and accepting from an affected identity an appeal referencing the provenance chain entry of said action, said appeal being evaluated by a reviewer other than said acting moderator.

98.?The method of claim 68, further comprising clearing, on an explicit instruction received during an active session, an accumulated query-and-response record of said session immediately and without producing a condensed retained form, and recording said clearing in said provenance chain without retaining any portion of the cleared content.

99.?The method of claim 1, further comprising extracting a prosodic feature from live audio only where a session-scoped consent is active, supplying said feature as an optional trigger to a clarification request, and discarding said feature at session end absent a separate election to retain it.

100.?The method of claim 22, further comprising reporting a documented editorial position of a source and a quantified composition measure of said source as cited knowledge records, and refraining from generating a conclusory characterization of said source not itself derivable from a said knowledge record.

101.?The method of claim 1, further comprising extracting a discrete factual assertion from a reference work as a knowledge record while withholding reproduction of the reference work's own expression, and processing an image accompanying said reference work under a separate rights validation rather than under said extracting.

102.?The method of claim 71, further comprising matching a plain-language request against a library of application templates by task and domain and instantiating a matched template scoped to a set of said knowledge records, and constructing a compilation of a requester's works as a delta reference over said works without copying an underlying knowledge record.

103.?The method of claim 1, further comprising storing an instructor-selected structural parameter set as a reusable assessment template applied to knowledge records of a work, and exposing to a grantee a read-only view of each active grant held over said grantee, wherein said view confers no capability to create or modify a grant.

104.?The method of claim 24, further comprising applying a platform configuration change through a declarative configuration language, said change being versioned and recorded in said provenance chain on the same terms as any other change; and pausing execution at a pipeline stage boundary to expose knowledge record bindings, safety evaluation results and compute-accounting measurements of an actual execution rather than of a simulation.

105.?The method of claim 1, further comprising constructing a goal or motivation record carrying a mandatory reference to a documented subject, wherein no said record is constructed concerning an internal state of the system performing the method; and selecting a vocabulary, structural format and explanatory depth for said output text from an audience profile, wherein a knowledge record relied upon is not altered by said selecting.

106.?The method of claim 1, further comprising dispatching a scheduling or invitation operation to an external platform registered with a capability declared by said external platform, and aggregating usage and distribution records for a work into a report, wherein neither operation admits a knowledge record bypassing said safety evaluation.

107.?The method of claim 26, further comprising measuring a stated design-target figure by a reproducible benchmark expressed in said compute-accounting units, and retracting said stated figure where said measuring does not confirm it, wherein said retracting is recorded in said provenance chain.

Behavioural probing and analysis quarantine (claims 108–112)

108.?The method of claim 2, further comprising: submitting an identical input to an external system a plurality of times through a connector interface under a capability declared by said external system; computing a semantic fingerprint of each output received, said semantic fingerprint being computed over a canonicalized form of a semantic assertion and independently of a natural language of said output; determining a proportion of said outputs sharing a common said semantic fingerprint; recording said proportion together with a corresponding proportion measured for said generating of output text on the identical input; and withholding said recording as incomplete where said corresponding proportion is absent.

109.?The method of claim 108, further comprising evaluating two said outputs having differing semantic fingerprints against one another under a temporally-scoped rule set evaluated incrementally as a knowledge record is appended, and recording a contradiction between two outputs responsive to an identical input as a finding distinct from a difference in semantic fingerprint absent a contradiction.

110.?The method of claim 38, further comprising: submitting an output received from an external system back to said external system as an input under a fixed template; repeating said submitting to a configured iteration ceiling and recording a semantic fingerprint of the output of each iteration; and classifying the resulting sequence of semantic fingerprints into exactly one of a converged classification in which successive fingerprints become identical, a periodic classification in which said sequence repeats with a fixed period, a drifting classification in which a measured semantic distance from an initial output increases monotonically, and an unresolved classification where said ceiling is reached, wherein said classifying is performed on said sequence alone and no output is generated asserting a cause of said classification or a property of an internal mechanism of said external system.

111.?The method of claim 38, further comprising: constructing a family of inputs from a base input by applying a perturbation and retaining in said family only an input whose semantic fingerprint is identical to that of said base input; submitting each member of said family to an external system; and determining a proportion of the resulting outputs sharing a common semantic fingerprint, wherein an input whose perturbation alters said semantic fingerprint is excluded from said family before said submitting.

112.?The method of claim 2, wherein a knowledge record containing an output of an external system, and a knowledge record derived from such a record, are excluded from said scope description constructed for a query other than a query directed to said records, such that said records are not traversable for said other query rather than being removed from a result of a traversal thereof; and wherein said records are further excluded from a persistent question index and from a store of concept abstractions.

Attestation horizon and retrospective forecast calibration (claims 113–118)

113.?A computer-implemented method for measuring forecast accuracy of a knowledge system, comprising: maintaining a plurality of knowledge records, each said knowledge record encoding a semantic assertion and comprising a temporal anchor identifying an interval over which said semantic assertion holds, an attestation time identifying when said semantic assertion was first recorded by a source of said knowledge record, and a reference into an append-only provenance chain; identifying an event record having an event time and a recorded outcome; selecting a horizon interval preceding said event time; constructing, before any traversal of said knowledge records, a scope description admitting a said knowledge record if and only if said attestation time of said knowledge record is earlier than said event time less said horizon interval, wherein said temporal anchor is not consulted in said admitting and a knowledge record whose attestation time is not established from documented source metadata is excluded from said scope description; executing a forecasting operation against only the knowledge records admitted by said scope description, said forecasting operation producing a predicted outcome and a probability assigned to said predicted outcome; computing an error term from said assigned probability and said recorded outcome; and storing said error term together with said horizon interval, a domain of said event record, and an identifier of each knowledge record admitted by said scope description.

114.?The method of claim 113, wherein a knowledge record excluded by said scope description is not traversable during said forecasting operation rather than being removed from a result of a traversal thereof, such that said excluded knowledge record does not affect a candidate count, a retrieval ordering, a convergence of an iterative stabilization, or a permissible output vocabulary of said forecasting operation.

115.?The method of claim 113, wherein, where a documented creation date of a source of a said knowledge record is contested among a plurality of candidate dates, a latest of said candidate dates is used as said attestation time, such that uncertainty in said attestation time resolves toward exclusion from said scope description.

116.?The method of claim 113, further comprising repeating said selecting, said constructing, said executing and said computing for a plurality of differing horizon intervals preceding a common said event time, and deriving a measure of variation of said error term with said horizon interval within a domain.

117.?The method of claim 113, further comprising determining, over a plurality of said error terms whose assigned probabilities fall within a common band, a proportion of corresponding recorded outcomes that occurred, and reporting a divergence between said proportion and said band as a calibration finding applicable to a confidence interval subsequently rendered by said forecasting operation.

118.?The method of claim 113, wherein a subject of said event record is one of an institution, an organisational process, a system and an aggregate group, and wherein construction of a record storing said error term is refused where said subject resolves to an individual natural person, said refusal occurring at construction rather than by restriction of access to a constructed record.

Event type catalogue, environment vectors and causal validation (claims 119–124)

119.?The method of claim 1, further comprising: maintaining a catalogue of event type definitions, each said event type definition comprising an identifier immutable from registration, a version, a placement resolving against a plurality of independent taxonomic hierarchies of a concept structure, a prose definition stating both occurrences covered and occurrences excluded, and a machine definition specifying for said event type which fields of a knowledge record are required, which are optional and which are forbidden; requiring a knowledge record encoding an occurrence to reference an identifier of said catalogue rather than a free-text category; and refusing construction of a knowledge record that does not satisfy the machine definition of the event type it references, said refusing occurring at construction rather than by marking a constructed record.

120.?The method of claim 119, wherein a knowledge record constructed under a first version of an event type definition remains valid under said first version and is not invalidated by registration of a subsequent version, and wherein a required-field set of an event type definition may be narrowed by a subsequent version and may not be widened absent registration of a distinct event type.

121.?The method of claim 1, further comprising: constructing an environment record representing a state relative to which a plurality of said knowledge records hold, said environment record comprising an identifier derived from a hash of a canonicalized set of conditions, a temporal interval, a set of typed conditions whose keys resolve against a concept structure, and a confidence value and a reference into said provenance chain of its own; referencing said environment record from a knowledge record through an annotation of said knowledge record rather than through a field of a schema of said knowledge record; and converging two constructions yielding an identical said canonicalized set of conditions on a single said environment record having two branches of said provenance chain.

122.?The method of claim 121, further comprising superseding said environment record through a supersession chain, wherein every knowledge record referencing said environment record resolves to a superseding environment record on a subsequent traversal without modification of said knowledge records.

123.?The method of claim 1, further comprising admitting an edge asserting that a first knowledge record describes a causal factor for an occurrence described by a second knowledge record only where a temporal anchor of said first knowledge record does not begin after a temporal anchor of said second knowledge record begins, refusing said edge otherwise, and recording a refused said edge in a persistent contradiction index; and wherein an edge asserting temporal ordering between two knowledge records is not converted into a said causal edge by any automated process.

124.?The method of claim 123, wherein a causal chain is expressed as a membership of a plurality of said knowledge records in a common bundle together with said causal edges among said knowledge records, and wherein a length of said chain, a root of said chain and an outcome of said chain are computed from said membership and said edges at a time of query rather than stored as fields of a chain record.

Confidence as a Beta posterior (claims 125–129)

125.?The method of claim 1, further comprising, on receipt of a further knowledge record bearing on said semantic assertion, incrementing one of said first and second parameters by a weight derived from a source reliability of said further knowledge record, such that a magnitude of change in said reported confidence value decreases as a sum of said first and second parameters increases without a rule conditioned on said sum; and admitting said weight as a fractional quantity rather than as an integer count.

126.?The method of claim 125, further comprising initialising said first and second parameters for a

concept node from a stability weight of said concept node, wherein said stability weight determines a sum of said parameters and said confidence value determines a ratio between them, such that a concept node having a greater stability weight requires a greater quantity of contradicting evidence to effect an equal change in said reported confidence value.

127.?The method of claim 125, further comprising computing a credible interval for said reported confidence value from said first and second parameters, wherein two knowledge records having an identical said reported confidence value and differing sums of said parameters yield differing said credible intervals.

128.?The method of claim 125, further comprising applying a temporal decay to said first and second parameters toward respective initial values at a rate configured per record type, such that an elapsed interval widens a credible interval computed from said first and second parameters while leaving said reported confidence value substantially unchanged, and wherein a record type designated immutable is assigned a decay rate of zero.

129.?The method of claim 125, wherein said incrementing is performed by addition in closed form without sampling from a probability distribution, such that a given set of said further knowledge records yields an identical said reported confidence value irrespective of an order in which said further knowledge records are received.

Precision-weighted error propagation (claims 130–133)

130.?The method of claim 125, further comprising: determining that a first knowledge record contradicts an expectation generated by an attribute of a concept node; computing a first dispersion from the first and second parameters of said first knowledge record and a second dispersion from the parameters of said attribute; computing a revision gain as a ratio of said second dispersion to a sum of said first and second dispersions; and revising said attribute by said revision gain multiplied by a configured maximum revision, such that a contradicting knowledge record resting on less accumulated evidence effects a smaller revision of said concept node than one resting on more, irrespective of a difference between their reported confidence values.

131.?The method of claim 130, wherein two knowledge records having an identical said reported confidence value and differing sums of said parameters yield differing said revision gains against a common attribute.

132.?The method of claim 130, further comprising withholding revision of said attribute where said revision gain falls below a configured floor, and recording each withheld contradiction in a persistent contradiction index such that a plurality of individually-withheld contradictions of a common attribute remains queryable.

133.?The method of claim 130, wherein said revision gain is computed in closed form from stored parameters without sampling from a probability distribution, and wherein said revision gain governs revision of said concept node and does not govern a permissible output vocabulary of said generating, such that a proposition not derivable from an admitted knowledge record remains unavailable to said generating irrespective of any value of said revision gain.

Physical constraint enforcement in scene completion (claims 134–138)

134.?The method of claim 1, further comprising: maintaining a registry of physical constraints, each said physical constraint comprising a residual expression over quantities of a scene description, a tolerance, a unit system, an applicability predicate, and a reference into said provenance chain identifying a source establishing said physical constraint; generating from a partial scene description a

candidate completion in which a field absent from said partial scene description is supplied; evaluating each said physical constraint whose applicability predicate is satisfied by said candidate completion by computing said residual expression over said candidate completion; refusing said candidate completion where a computed said residual exceeds a corresponding said tolerance; and regenerating a further candidate completion in which a field implicated in said refusing is left unspecified, such that a completion carrying an unspecified field is returned in preference to a completion exceeding said tolerance.

135.?The method of claim 134, wherein a refused said candidate completion is excluded from a ranked output irrespective of a plausibility score computable for it, and wherein said evaluating of said physical constraints is performed before computation of said plausibility score.

136.?The method of claim 134, further comprising recording a refusal as a queryable record identifying said physical constraint, said computed residual, said tolerance and a field implicated, such that a proportion of candidate completions refused by a given said physical constraint within a domain is determinable.

137.?The method of claim 134, wherein a said physical constraint whose applicability predicate is not satisfied by a said candidate completion is not evaluated against said candidate completion and said non-evaluation is recorded; and wherein a confidence value of a said physical constraint is maintained as a first parameter accumulating supporting evidence and a second parameter accumulating contradicting evidence and said physical constraint is suspended from said evaluating, without deletion, where said confidence value falls below a configured floor.

138.?The method of claim 134, wherein an ingested knowledge record whose content is inconsistent with a said physical constraint is admitted and recorded in a persistent contradiction index rather than refused, such that said refusing applies to a generated said candidate completion and does not apply to an observation.

Surfaces, extensibility and external integration (claims 139–156)

139.?The method of claim 1, further comprising exposing a plurality of programmatic interfaces to an external caller, wherein each said interface resolves a scope description for said external caller before performing a traversal of said knowledge records, no parameter of any said interface widens a said scope description, a request for a knowledge record outside said scope description yields a response identical to that for a non-existent knowledge record, and each said interface constitutes an operation layer carrying a synchronous safety evaluation gate.

140.?The method of claim 139, wherein an interface reporting an entitlement performs no operation granting, modifying or extending an entitlement, and wherein an interface accepting an input admits said input through said extracting with said safety evaluation and accepts no knowledge record directly.

141.?The method of claim 139, further comprising metering each call to a said interface as a bounded compute-accounting unit assigned to a calling identity, and refusing a call exceeding a budget configured for said calling identity rather than serving said call at a reduced scheduling priority.

142.?The method of claim 139, further comprising registering a plugin component against a declared capability set, wherein said plugin component accesses said knowledge records only through a said interface and within a scope description of an identity on whose behalf said plugin component executes, holds no direct access to a store of said knowledge records or to said provenance chain, and is refused at registration where said declared capability set includes an operation registering, modifying or suspending a said safety evaluation.

143.?The method of claim 142, wherein output originating from said plugin component is rendered in a region distinct from said output text and labelled with an identity of said plugin component, is not admitted to a store of said knowledge records, and is not citable as a grounding source; and wherein activation of said plugin component requires an approval recorded against an identity of an individual human operator and is suspensible by said operator without cooperation of said plugin component.

144.?The method of claim 139, wherein each said interface carries a version, a request omitting a version is refused rather than assigned a default version, a version is deprecated for a configured support interval and thereafter retired with a request for it refused and a successor named, and a knowledge record returned under one said version remains valid under a subsequent said version such that a citation issued under an earlier version is not invalidated.

145.?The method of claim 1, further comprising establishing a session shared by a plurality of participants over a common work, wherein a shared view of said session comprises knowledge records traversable within an intersection of the scope descriptions of every said participant, a knowledge record traversable by a first said participant and not by a second is absent from said shared view, and an absence of said knowledge record is not reported to any said participant.

146.?The method of claim 145, wherein a response to a query submitted in said session is resolved against said intersection and delivered identically to every said participant, and wherein a reading position, an interaction history and a private annotation of a said participant remain outside said shared view absent an explicit contribution by said participant.

147.?The method of claim 1, further comprising anchoring a displayed knowledge record together with its citation to a physical object identified in a live camera feed, wherein a classification result corresponding to a natural person or to an animal is excluded from a candidate set and no said anchoring is performed to such a result at any confidence value, and wherein no gaze direction, dwell duration, head pose or attention signal is captured, derived or retained, and no such signal is made available to a selection of commercial content or to a plugin component.

148.?The method of claim 147, wherein where an object of said camera feed is not identified at a confidence value satisfying a configured threshold no said anchoring is performed and a boundary is stated, such that no said anchoring is performed to an unidentified object.

149.?The method of claim 1, further comprising receiving spoken input under an explicit session-scoped and revocable consent, transcribing said spoken input, routing said transcription through a classification separating a query intent from an operational intent, and requesting a clarification where said classification does not satisfy a configured confidence threshold rather than selecting an interpretation; and wherein audio of said spoken input is discarded following said transcribing absent a separate explicit election, and no voiceprint, speaker embedding or biometric identifier is derived therefrom, an identity of a speaker being taken from an authenticated session.

150.?The method of claim 149, wherein a said spoken input classified as carrying a state-changing effect comprising an erasure, a rights modification, a purchase or a grant requires a confirmation received through a second modality before said effect takes place.

151.?The method of claim 1, further comprising recording a progress marker for a reader over a work, associating with each knowledge record extracted from said work a narrative position at which its content is disclosed within said work, and bounding a scope description constructed for a query of said reader over said work by said progress marker such that a knowledge record whose narrative position exceeds said progress marker is not traversable; and wherein where a query is unanswerable within said bounded scope description the system states that an answer lies beyond said progress marker

without disclosing content of said knowledge record.

152.?The method of claim 1, further comprising recording, for a learner, a position reached, assessment items attempted, responses submitted and outcomes scored against the knowledge records from which said items were generated, and computing a mastery measure as a proportion of items attributable to a stated objective answered correctly against a configured threshold; wherein visibility of said records to an instructor is bounded by a time-scoped grant such that expiry of said grant removes said visibility without a further operation, and wherein no inference of aptitude, of likelihood of future performance, or of comparative ranking against another learner is generated or stored from said records.

153.?The method of claim 142, further comprising connecting an external learning management system under a declared capability set, wherein an operation not within said declared capability set is refused at a boundary of said connecting rather than attempted; a roster read establishes a scope of an instructor grant and does not confer an entitlement to a work; and a transmission of an outcome to said external learning management system comprises no knowledge record, no response text and no interaction history, and is recorded with an identity of a connector, a capability exercised and fields transmitted.

154.?The method of claim 1, further comprising composing a work into a paginated print-ready form for a specified edition, wherein editions differing in audience level, language or annotation density are composed from common said knowledge records through said generating; a rights validation permitting print reproduction in a requested territory, quantity and interval is performed before said composing and a composed form carries a reference to said rights validation; a wording-operative span is reproduced verbatim in said print-ready form; and said composing is deterministic such that two compositions over an unchanged work and layout specification produce identical pagination.

155.?The method of claim 152, further comprising issuing a credential to an identified learner upon both a measured outcome satisfying a configured threshold and an establishment of an identity of said learner to a configured standard, said credential comprising a record of objectives assessed, items presented, the knowledge records from which said items were generated, thresholds applied and an issuing identity; verifying said credential to a third party as issued and unrevoked without disclosing a response of said learner; and revoking said credential by a recorded revocation rather than by deletion.

156.?The method of claim 1, further comprising exposing a portal surface over a licence and entitlement mechanism, wherein a capability available at said portal to a role is the capability available to said role through any other surface such that said portal confers no capability; a grantee views a grant held over said grantee without a capability to create or modify a grant; each action taken through said portal is recorded with an acting identity, an action and a stated reason; and a revocation effected through said portal takes effect at a time of executing a subsequent query rather than at a commencement of a subsequent session.

Hyperedge attention and attractor-based hyperedge completion (claims 157–159)

157.?The method of claim 1, wherein a plurality of said knowledge records are linked by a hyperedge relating more than two of said knowledge records, and further comprising, upon a reasoning operation traversing said hyperedge, weighting a contribution of each constituent knowledge record of said hyperedge by a product of a confidence value of said constituent knowledge record and an accumulated evidence count of said constituent knowledge record, normalized across the constituents of said hyperedge, such that a constituent resting on less accumulated evidence contributes less than one resting on more irrespective of an equality of their confidence values.

158.?The method of claim 157, further comprising completing a partially-observed said hyperedge by applying said iterative updating of activation values to a constituent set of said hyperedge as a whole,

weighted according to claim 157, until said activation values reach a stable state identifying a complete configuration of said hyperedge consistent with its observed constituents; and withholding said completing, and reporting an unobserved constituent as absent, where a weighted support for said complete configuration does not satisfy a configured threshold.

159.?The method of claim 158, wherein said completing operates upon a constituent set of a single said hyperedge and reaches said stable state through said iterative updating, and is distinct from a completion of a scene description performed by application of schema, physics, affordance and relation rules, said completing introducing no convergence behaviour separate from said iterative updating.

Domain architecture, transfer matrix and path separation (claims 160–168)

160.?The method of claim 1, wherein the method is performed by a system partitioned into a plurality of domains each owning a bounded set of operations, said plurality comprising a governance domain and a plurality of functional domains, wherein no operation of any said functional domain executes without traversing an enforcement point of said governance domain, said governance domain is not a valid target of a modification proposed by any said functional domain, and a grouping of said functional domains into layers confers no authority such that a transfer permission between two said domains is unaffected by whether said domains occupy a common said layer.

161.?The method of claim 160, further comprising maintaining a transfer matrix assigning to each ordered pair of said domains a permission selected from permitted, denied, and permitted-subject-to-a-stated-conversion; evaluating said permission at each transfer of content between two said domains; treating an ordered pair having no entry in said transfer matrix as denied; and recording each said transfer in said provenance chain together with a source domain, a target domain and a permission applied; wherein said transfer matrix is owned by said governance domain and is a member of a fixed enumerated set of subsystems excluded from modification, such that a domain constrained by an entry of said transfer matrix does not administer that entry.

162.?The method of claim 161, wherein said transfer matrix denies a transfer from a domain owning evaluation of an output of an external system to a domain owning said retrieving, such that neither said output nor an analysis thereof is admissible to said admitted set, excepting a knowledge record converted from a reply elicited by a prompt composed by the method and carrying a derivation marking; and permits a transfer from said domain owning evaluation to a domain owning storage of said knowledge records only subject to a conversion marking transferred content as an analysis record excluded from a scope description constructed for a query not directed to said analysis.

163.?The method of claim 161, wherein said transfer matrix denies a transfer from a domain owning selection of commercial content to a domain owning said retrieving and to a domain owning said generating, and permits a transfer to said domain owning selection only subject to a conversion supplying a concept identifier and identifiers of concepts adjacent thereto within one edge, read-only, after said retrieving and said generating have completed.

164.?The method of claim 161, wherein said transfer matrix permits a transfer from a domain owning said extracting to a domain owning storage of said knowledge records only subject to a conversion applying said safety evaluation, a validation against a machine definition of an event type, and a confidence calibration, said conversion constituting the sole route by which content originating outside the system enters said storage.

165.?The method of claim 161, wherein said transfer matrix denies a direct transfer between a domain owning a presentation surface and a domain owning storage of said knowledge records, such that a presentation surface obtains content only through a domain owning construction of said scope description

and a domain owning said retrieving.

166.?The method of claim 160, wherein a first subset of said functional domains is traversed in answering a query and a second subset is traversed in ingesting content, and an intersection of said first subset and said second subset consists of a domain owning storage of said knowledge records and said governance domain alone, such that an ingestion operation contends with a query operation at no other domain.

167.?The method of claim 166, wherein contention at said governance domain is resolved by serving a compute-accounting unit of a safety-evaluation category ahead of a unit of any other category, and contention at said domain owning storage is resolved by assignment of a record to a storage tier by access recency and frequency.

168.?The method of claim 160, further comprising metering an operation of each said domain as a bounded compute-accounting unit carrying an identifier of said domain, fixing a scheduling priority by said identifier such that said governance domain holds a highest said priority and a domain owning selection of commercial content holds a lowest, and containing a failure of a said domain to a capability of that domain, wherein a failure of said governance domain halts an operation rather than permitting it to proceed unevaluated; and wherein said scheduling priority and an invariant bounding an intersection of a domain subset traversed in answering a query with a domain subset traversed in ingesting content are likewise owned by said governance domain and excluded from modification.

Discriminating instance generation (claims 169–173)

169.?The method of claim 115, further comprising: maintaining a first candidate hypothesis and a second candidate hypothesis concerning a common subject, each having a confidence value maintained as said first and second parameters; determining that a difference between the confidence values of said first and second candidate hypotheses does not exceed a configured separation floor while an accumulated evidence count of each exceeds a configured minimum; computing, for each of a plurality of candidate observations, a magnitude of a difference between a likelihood of said candidate observation under said first candidate hypothesis and a likelihood of said candidate observation under said second candidate hypothesis; selecting a said candidate observation maximising said magnitude; and emitting a structured request identifying said selected candidate observation, said magnitude, and a confidence value each said candidate hypothesis would take under each of a presence and an absence of said selected candidate observation.

170.?The method of claim 169, wherein said selecting is performed irrespective of a likelihood assigned to said candidate observation by either said candidate hypothesis, such that a candidate observation predicted equally by both said candidate hypotheses is not selected however probable it is under either.

171.?The method of claim 169, further comprising withholding said emitting, and recording said first and second candidate hypotheses as undiscriminated together with a reason, where said magnitude for every said candidate observation falls below a configured discrimination floor.

172.?The method of claim 169, wherein neither said first candidate hypothesis nor said second candidate hypothesis is advanced or retired by said emitting, each remaining in a proposed state transitionable to a validated state only upon confirmation by an identified human reviewer, and wherein no automated process acts upon said structured request.

173.?The method of claim 169, further comprising, upon a knowledge record corresponding to said selected candidate observation being admitted as a said knowledge record, updating said confidence values of said first and second candidate hypotheses and re-evaluating said difference against said

separation floor, and recording an outcome of said selected candidate observation such that said selected candidate observation is not emitted again for said first and second candidate hypotheses.

Deployment topology, module replacement and substrate independence (claims 174–182)

174.?The method of claim 160, wherein a first subset of said domains executes in a first infrastructure under control of a party holding said knowledge records and a second subset executes in a second infrastructure, and wherein said scope description is constructed within said first subset and supplied to said second subset, such that no domain of said second subset constructs a said scope description and no said knowledge record is transferred to said second infrastructure by reason of an asset referenced by said knowledge record residing therein.

175.?The method of claim 160, further comprising maintaining at a cache node a subset of said knowledge records selected by access frequency and recency, answering a query at said cache node only where a scope description of a requester admits every cached knowledge record consulted, escalating said query to an origin otherwise, and invalidating a cached said knowledge record upon a supersession thereof; wherein said cache node performs no said extracting and constructs no said scope description.

176.?The method of claim 160, further comprising replacing an implementation of a said domain with a further implementation upon said further implementation satisfying each interface exposed by said domain at each version within a support interval thereof and each entry of said transfer matrix in which said domain appears, and upon an approval recorded against an identity of an individual human reviewer; and refusing said replacing where said domain is a member of a fixed enumerated set of subsystems excluded from modification.

177.?The method of claim 139, wherein said plurality of programmatic interfaces further comprises an interface returning a transformation of a media asset and an interface returning a scene description assembled from knowledge records carrying spatial detail, wherein said transformation is deterministic for a given asset, parameter set and content state, a value inferred in said transformation is traceable to a knowledge record from which it was inferred, and said interface returning a scene description withholds a scene where spatial detail carried by said knowledge records does not satisfy a configured sufficiency threshold.

178.?The method of claim 1, further comprising presenting simultaneously a first realization of a work in a first language and a second realization of said work in a second language, each generated from common said knowledge records, wherein said realizations are aligned by knowledge record rather than by position such that a citation taken in either resolves to a common knowledge record, and wherein a knowledge record having no realization in one of said languages is presented as a gap rather than supplied from the other realization.

179.?The method of claim 1, further comprising scoping an encryption key to a tenant or to a deployment such that a key of a first tenant decrypts no content of a second, and holding key material within a custody boundary such that a use of said key material is a call across said boundary rather than a read thereof; wherein a knowledge record for which a zero-knowledge election is in effect remains one for which no operator-held key exists under any configuration.

180.?The method of claim 3, further comprising signing each node of said provenance chain under a key of a deployment in addition to said checksum, such that said provenance chain is verifiable by a party holding a corresponding verification key without access to a system that produced it, said verifying establishing that a recorded operation and a recorded safety evaluation are as written.

181.?The method of claim 1, wherein said constructing, said retrieving, said iteratively updating and

said generating are performed by components of a system performing the method and require no external fact-representation or fact-query system, and wherein an external fact-retrieval resource is connectable as a connector under a declared capability set such that a result returned therefrom is converted into a said knowledge record at a boundary of said connector and admitted through said extracting with said safety evaluation, said method being performed identically where no said connector is registered.

182.?The method of claim 181, wherein no element of said method recites an external fact-query system, a failure of a domain owning external integration is contained to a capability of said domain, and a determinism whereby an identical query over identical said knowledge records yields an identical said output text is established by said constraint on said permissible output vocabulary and by a bound on a number of stable states of said iterative updating.

Learning domain, module declaration and substrate migration (claims 183–185)

183.?The method of claim 160, further comprising a domain owning generation of an assessment item from a knowledge record, a recording of a mastery measure over a stated objective, and an issuance of a credential, wherein a transfer from said domain to a domain owning selection of commercial content is denied by said transfer matrix and a transfer from said domain to a domain owning external integration is permitted only subject to a conversion transmitting an outcome without a knowledge record, a response text or an interaction history.

184.?The method of claim 160, wherein each said functional domain declares at least one module constituting a unit of replacement, and a said governance domain declares none, such that no implementation of said governance domain is exchangeable under a replacement conditioned on interface and transfer-matrix conformance and a recorded human approval.

185.?The method of claim 160, wherein a domain owning storage of said knowledge records declares no module exchangeable under a replacement conditioned on interface and transfer-matrix conformance, an exchange of an implementation thereof requiring a migration of stored said knowledge records in addition to a conformance with an interface and with said transfer matrix.

Restored coverage: independent curation, manifold forecasting, colour drift registration, message-channel encryption (claims 186–189)

186.?The method of claim 1, further comprising: maintaining a first ordered walkthrough over content, curated by an owner of said content, and a second ordered walkthrough over said content curated by a party other than said owner, both said walkthroughs persisting concurrently; associating with each said walkthrough an attribution identifying a curator thereof, said attribution being a field of a record of said walkthrough and not modifiable or removable by said curator; and holding an authorization to narrate a said walkthrough separately from an authorization to curate it, such that curating a said walkthrough confers no authorization to narrate it in a voice of any identified natural person.

187.?The method of claim 1, further comprising: embedding a time series into a manifold of lower dimension than said time series and nonlinear therein; producing from said manifold a forecast comprising a predicted value and a credible interval computed from accumulated evidence parameters of the knowledge records from which said time series was constructed; rendering said forecast in a region labelled as a forecast and distinct from said output text; and withholding said forecast from admission as a knowledge record, such that no said forecast is available as a grounding source for a subsequent said generating.

188.?The method of claim 186, further comprising: computing, between a first rendered state of a said walkthrough and a second rendered state of said walkthrough, a colour-difference magnitude as a

Euclidean distance between coordinates of said rendered states on three axes of a perceptually-uniform colour space; comparing said magnitude against a tolerance declared by an owner of said content; and raising a registration finding where said magnitude exceeds said tolerance, without altering either said first rendered state or said second rendered state.

189.?The method of claim 1, further comprising: establishing a one-to-one message channel between two identities; classifying said message channel from an account context and an access path of said identities rather than from a self-declaration by a participant thereof; encrypting a said message channel not classified as associated with a minor context such that no party performing the method holds a decryption key therefor under any configuration; and encrypting a said message channel so classified against interception by a party other than said two identities while retaining a decryption capability exercised exclusively for a content-safety evaluation, each exercise being logged with an identity of an accessing process and a triggering reason.

Reasoning mode formalization — deductive, inductive and abductive (claims 190–192)

190.?The method of claim 1, further comprising classifying an inference over said knowledge records as a first type where a conclusion thereof follows necessarily from admitted said knowledge records and as a second type otherwise; admitting a conclusion of said first type as a knowledge record carrying a type identifier distinguishing it from a directly-ingested knowledge record and a confidence value computed as a product of confidences of the knowledge records from which it follows; and withholding a conclusion of said second type from admission as a knowledge record, emitting it instead with a status field set to a proposed state, wherein no automated process is configured to transition said status field to a validated state.

191.?The method of claim 190, wherein an inference of said second type adjusts a confidence value of an existing said knowledge record without creating a knowledge record, and wherein a contradiction insufficient to effect said adjustment is recorded in a persistent index rather than discarded, such that a plurality of individually-insufficient contradictions bearing on a common assertion remains queryable.

192.?The method of claim 190, wherein admission of a knowledge record is conditioned on a conformance with a machine definition of an event type, a completed said safety evaluation and a reference into said provenance chain, and is not conditioned on a quantity of evidence supporting a semantic assertion of said knowledge record, such that said second type of inference adjusts a confidence value of an admitted knowledge record and does not determine whether a knowledge record is admitted.

Observation diversity across source classes (claims 193–196)

193.?The method of claim 125, further comprising attributing each increment of said first and second parameters to a source class of a plurality of registered source classes, and computing for a said knowledge record a diversity measure as a reciprocal of a sum of squared proportions of weight contributed by each said source class, such that two knowledge records having identical said first and second parameters and differing distributions of contributed weight across said source classes yield differing said diversity measures.

194.?The method of claim 193, wherein a further knowledge record attributed to a source class already contributing a proportion of weight to said first parameter is admitted at a weight reduced as a function of said proportion, such that repeated corroboration within one said source class does not increment said first parameter as a corroboration from a further said source class would.

195.?The method of claim 193, further comprising conditioning a proposal of a generalization from a

plurality of subordinate concepts upon said diversity measure of said plurality satisfying a configured floor in addition to a count of said plurality satisfying a count floor; and attenuating a revision gain applied to a concept attribute where said diversity measure of a contradicting knowledge record approaches unity.

196.?The method of claim 193, further comprising indicating, in an output relying upon a knowledge record whose said diversity measure is below a configured threshold, that support for said knowledge record is concentrated within a single said source class.

The interpretive layer — context, perspective, ergonomic, cultural and social modulation (claims 197–204)

197.?The method of claim 1, further comprising applying to said output text a plurality of interpretive modulations selected from a context modulation, a perspective modulation, an ergonomic modulation, a cultural modulation and a social modulation, wherein each said interpretive modulation operates upon a result of said generating and neither widens nor narrows said scope description, neither admits a knowledge record excluded therefrom nor excludes one admitted thereby, alters no said confidence value, and extends no said permissible output vocabulary; and refusing a said interpretive modulation that would effect any thereof.

198.?The method of claim 197, wherein two requesters whose scope descriptions admit an identical set of said knowledge records receive output text grounded in said identical set irrespective of a difference between their said interpretive modulations, such that said interpretive modulations vary an emphasis, an ordering, a vocabulary, a format and a medium of said output text and vary neither a content nor an availability of a said knowledge record.

199.?The method of claim 197, wherein a signal from which a said interpretive modulation is derived is retained for a duration of a session and is not persisted as a knowledge record, such that no said signal becomes available to said retrieving at a subsequent session.

200.?The method of claim 197, wherein a preference stated by a requester takes precedence over a signal inferred from an interaction of said requester wherever both bear on a common dimension of a said interpretive modulation, and an inferred said signal is applied only where no stated said preference bears on that dimension.

201.?The method of claim 197, wherein a said cultural modulation is derived from a profile declared by a requester, by an operator of a deployment, or by a locale explicitly selected by said requester, and is not derived from a name, a network address, a language of a query, a device locale or a behavioural signal of said requester.

202.?The method of claim 197, wherein a said ergonomic modulation derived from an inferred indicator of cognitive load modulates a density and a layout of said output text alone, produces no knowledge record concerning said requester, and is unavailable to a further said interpretive modulation, to a selection of commercial content and to a programmatic interface.

203.?The method of claim 197, wherein said interpretive modulations are composed in a fixed order over a result of said generating such that an identical result under an identical configuration of said interpretive modulations yields an identical modulated output; and wherein a review under a governance domain precedes and follows said composing and refuses a modulated output irrespective of said configuration.

204.?The method of claim 197, further comprising exposing for each said interpretive modulation an interface accepting a configuration and returning a modulated said output text, said interface resolving a scope description before a traversal and admitting no parameter widening said scope description;

refusing at said interface a configuration that would retrieve a knowledge record, alter a confidence value or widen a scope description; and recording a said configuration in effect for a said output text such that said output text is reproducible from said recorded configuration.

Outcome feedback and operational calibration (claims 205–211)

205.?The method of claim 1, further comprising recording for a delivered said output text a response record identifying said query, said scope description, each knowledge record cited therein and a confidence value held by each said cited knowledge record at a time of delivery; receiving from an identity a report bearing on said output text or on a said cited knowledge record; and admitting said report as a knowledge record through said extracting with said safety evaluation, carrying said identity as an attributed source and a reference to said response record.

206.?The method of claim 205, wherein a said report bearing against a said cited knowledge record increments a parameter accumulating contradicting evidence of said cited knowledge record by a weight derived from a source reliability of said identity, and does not set said confidence value, supersede said cited knowledge record or remove it from a store of said knowledge records.

207.?The method of claim 206, wherein said identity constitutes a source class for a diversity measure computed as a reciprocal of a sum of squared proportions of weight contributed by each source class, such that a plurality of said reports from a common said identity contributes less to said parameter than an equal plurality of said reports from distinct identities.

208.?The method of claim 205, wherein a said report is not admissible as a grounding source for output text concerning a subject matter of said report, and is admissible as evidence bearing on a confidence value and as an object of an analysis.

209.?The method of claim 205, further comprising pairing a said response record with an outcome record recording an event subsequent to said delivery, refusing construction of a said outcome record whose subject resolves to an individual natural person, and computing over a plurality of said pairings a mean of squared differences between a confidence value stated at delivery and a realised outcome, and a difference between a proportion of realised outcomes and a mean confidence value stated within a band, each computed per domain and per said band.

210.?The method of claim 209, further comprising emitting a difference whose magnitude satisfies a configured tolerance as a candidate modification evaluated in an isolated environment, said candidate identifying a measured said difference, a corpus over which it was measured and a parameter proposed for adjustment, and taking effect only upon an approval recorded against an identity of an individual human reviewer; wherein a parameter governing said computing is a member of a fixed enumerated set of subsystems excluded from modification.

211.?The method of claim 205, wherein no said report and no said outcome record extends a permissible output vocabulary of said generating or alters a said scope description, such that output text does not become expressible by reason of a said report and a knowledge record does not become traversable or cease to be traversable by reason thereof.

Expressive register and media-integrated expression (claims 212–217)

212.?The method of claim 1, further comprising selecting for said output text exactly one expressive register from a registered set of expressive registers, said selecting being resolved before said generating and being distinct from a selection of an audience level governing vocabulary and explanatory depth; refusing a register not in said registered set rather than approximating it; and wherein a said expressive register governs a voicing of said output text and neither alters a set of knowledge records admitted, nor

alters a confidence value, nor extends said permissible output vocabulary.

213.?The method of claim 212, wherein a first said expressive register characterised by declarative construction does not remove a qualification that a confidence value of a cited knowledge record requires, and a second said expressive register characterised by reduced density and acknowledgement does not soften an assertion supported by a said knowledge record, such that a said expressive register governs voicing and a said confidence value governs what is asserted.

214.?The method of claim 212, wherein a said expressive register characterised by acknowledgement of a requester's stated position is selected upon a preference stated by said requester, upon a configuration for which it is a configured default, or upon a consented signal captured under an explicit session-scoped authorization; and is not selected upon an inference from a word choice, a punctuation, a length, an error rate, a retry behaviour or an interaction timing of a query, no record of an inferred affective state being constructed.

215.?The method of claim 1, further comprising including within said output text a media asset referenced by a knowledge record of said stabilized set and delivering said media asset with a citation resolving it to said knowledge record; and withholding a media asset referenced by no knowledge record of said stabilized set and a media asset whose referencing knowledge record is outside said scope description, such that no said media asset conveys a proposition said output text could not assert.

216.?The method of claim 215, wherein a said media asset that has been transformed is delivered together with a provenance of said transformation identifying a knowledge record from which a value inferred in said transformation was derived; and wherein an interpretive modulation selects among media assets includable under claim 215 and does not render a media asset includable.

217.?The method of claim 212, wherein an interpretive modulation narrows a selection of a said expressive register where a configuration requires it and does not widen said selection, such that no said interpretive modulation selects an expressive register a configuration of said requester does not permit.

Surface realization — aggregation, reference, connectives, ordering and variation (claims 218–224)

218.?The method of claim 1, further comprising realizing two or more propositions of said graph structure that share a subject and are joined by an edge of a conjunctive type as a single sentence having a coordinated predicate, up to a configured maximum number of coordinated predicates; and withholding said realizing where confidence values of said propositions differ by more than a configured tolerance such that said propositions would require differing qualifications, whereby a proposition of lower confidence is not qualified by a coordination with a proposition of higher confidence.

219.?The method of claim 218, wherein a sentence produced by said realizing carries a citation of each said proposition realized therein, such that each clause thereof resolves to a knowledge record supporting it.

220.?The method of claim 1, further comprising realizing an entity in a full registered form at a first mention thereof within a discourse unit and in a reduced form at a subsequent mention, restoring said full registered form where a mention of a further entity sharing a type of said entity in a concept structure intervenes between said mentions, and restoring said full registered form at a first mention within a subsequent discourse unit; wherein an ambiguity is determined by said type and not by a surface form, and said full registered form is used where said determining is indeterminate.

221.?The method of claim 1, further comprising emitting, between two propositions realized in sequence and joined by a typed edge of said graph structure, a connective of a class corresponding to a type of said edge; selecting a connective within said class by an expressive register in effect and by a

position within a response such that a connective does not recur uniformly; and emitting no connective between two propositions not joined by a said typed edge.

222.?The method of claim 1, further comprising ordering within a sentence a constituent realized in a preceding sentence of a response before a constituent not yet realized therein, where a syntax of a target language permits; and exempting from said ordering a span reproduced rather than generated.

223.?The method of claim 1, further comprising substituting for a recurring term an alternative registered as synonymous for a sense of said term, and admitting said substituting only where a semantic fingerprint of a proposition containing said term is unchanged thereby; and refusing said substituting for a term within a span reproduced rather than generated, for a registered canonical name of an entity, and for a governed term of a discipline mode.

224.?The method of claim 1, wherein a selection among admissible realizations of a proposition of said graph structure is made by a deterministic rule over prior content of a response and not by sampling from a probability distribution, such that a given said graph structure under a given expressive register yields an identical token sequence on each execution; and wherein no said realization introduces a proposition absent from said graph structure.

Commercial presentation, advertiser provenance and transaction records (claims 225–231)

225.?The method of claim 17, further comprising determining an eligibility of a commercial item from a right registered for said commercial item, a territory recorded for a request, an edition in which a response is delivered and a device class of said request, and from no further input; wherein an inferred ergonomic signal, an inferred indicator of cognitive load and an inferred affective state are unavailable to said determining; and wherein a commercial region is omitted where said determining yields an empty set, an omission thereof not being disclosed.

226.?The method of claim 225, wherein a said commercial item occupies a registered placement class comprising a region distinct from and not interleaved with propositions of said output text, and a number of said commercial items presented within a response and within a session is bounded by a configured ceiling, a commercial item exceeding said ceiling being omitted rather than deferred.

227.?The method of claim 225, further comprising registering a said commercial item with a verified identity of a party presenting it, a right under which it is presented, a territory covered, an interval of validity and an identity of a registering operator, and refusing registration of a commercial item lacking any thereof; and recording each presentation in said provenance chain with said commercial item, said placement class, a concept identifier occasioning it and an eligibility relied upon.

228.?The method of claim 227, wherein a said commercial item is not admitted as a knowledge record, is not citable, is not traversable and contributes to no confidence value, and an assertion within a said commercial item is labelled as an assertion of said party presenting it.

229.?The method of claim 17, further comprising registering a commerce object with a right permitting its sale, a territory in which it may be sold and a price per territory and currency, and withholding an offer of a said commerce object in a territory for which no price is registered rather than offering it at a default price.

230.?The method of claim 229, further comprising performing upon a said commerce object an action selected from creating a perpetual entitlement, creating an entitlement bounded by an interval, creating a recurring entitlement contingent upon renewal, extending an existing entitlement, replacing an entitlement with a broader entitlement wherein a prior entitlement is superseded rather than deleted, and transferring an entitlement where a licence permits transfer; and producing for each said action a

transaction record carrying an acting identity, said commerce object, said action, a price and currency applied, a territory, a rights validation relied upon and a reference into said provenance chain.

231.?The method of claim 230, wherein a transaction not reconcilable against a change of entitlement is refused rather than recorded in a pending state, such that an entitlement state and a transaction history do not diverge; and wherein a divergence between a state of a token representation and a said entitlement raises a finding, said entitlement remaining authoritative.

Directed capture equipment control under session authorization (claims 232–240)

232.?The method of claim 1, further comprising registering an equipment unit of a capture or inspection class with a kinematic envelope, a velocity limit, a force limit and an operating volume; refusing command emission to an equipment unit whose registration lacks any thereof; and refusing command emission to an equipment class outside said capture or inspection class, wherein telemetry received from a vehicle-control system is admitted as an analysis record and no control interface thereto is created.

233.?The method of claim 232, further comprising maintaining an equipment state record comprising a pose, a configuration, a sensor status and an operating volume in force, said equipment state record being retained for a duration of a session and not admitted as a knowledge record; and withholding command emission where an age of said equipment state record exceeds a configured staleness bound, an indication of said staleness being returned.

234.?The method of claim 232, further comprising granting a session authorization recorded with an identity of an authorizing human operator, equipment units covered, an operating volume permitted, task classes permitted and an expiry; emitting a command only within said session authorization; halting emission upon an expiry thereof rather than requesting a renewal during an executing plan; and effecting a revocation thereof before a next command rather than upon completion of an executing plan.

235.?The method of claim 234, wherein each emitted command carries a reference to a request from which it was compiled, to said session authorization, to a plan of which it forms part and to a position within said plan, and a command unable to carry each said reference is not emitted.

236.?The method of claim 232, further comprising compiling a request into a command sequence by decomposing said request into operational intents before binding an intent to a concrete target, binding each said intent against a spatial model of an environment, mapping each bound intent to a registered equipment unit capable thereof, and synthesising a plan satisfiable within a registered envelope; wherein an unresolvable target, an absence of a capable said equipment unit and a plan not satisfiable within said envelope each halt said compiling rather than yielding a command against a target selected among candidates.

237.?The method of claim 232, further comprising evaluating a command before emission against an operating volume, a kinematic envelope, a velocity limit, a force limit, a minimum clearance to a detected obstacle, an exclusion distance from a detected person, an overlapping swept volume of a further equipment unit, and a recording-restricted zone; and withholding emission of a command violating any thereof, said withholding being unconditional and said command being neither emitted at a reduced priority nor modified into a permissible command by an automated selection.

238.?The method of claim 237, wherein a detection of a person is used to compute said exclusion distance and for no further purpose, is not retained beyond an evaluation thereof, does not enter a result set and is not available to a further mechanism, such that an avoidance of a person is performed without a record of said person being constructed.

239.?The method of claim 232, further comprising emitting a halt upon a clearance breach, an exceeded staleness bound, an expiry or revocation of a session authorization, a loss of a sensor stream on which an executing plan depends, or an explicit request; wherein said halt is evaluated from an equipment state record and a constraint set without a said retrieving, a said iterative updating, a said generating or an interpretive modulation, such that said halt is not delayed by a query in progress; and wherein said halt places an equipment unit in a registered safe state and a subsequent emission requires an explicit resume.

240.?The method of claim 232, further comprising observing an execution of a plan, admitting an observation thereof as an outcome record, and compiling a corrected plan afresh through said decomposing, said binding, said mapping, said synthesising and said evaluating rather than issuing a modification of a withheld command; and wherein no said observation adjusts a said kinematic envelope, a said velocity limit, a said force limit, a said operating volume or a said session authorization.

Assistive physical contact under a two-party authorization (claims 241–249)

241.?The method of claim 1, further comprising registering an equipment unit making intentional physical contact with a person under an authorization class disjoint from an authorization class governing equipment constrained by an exclusion distance from a person, wherein an authorization of either said class confers no capability under the other and a constraint set applicable to a command is determined by an authorization under which said command is emitted; and refusing said registering where a hazard analysis for an intended task, an intended user and an intended environment of said equipment unit has not been completed and recorded with an identity of an assessor.

242.?The method of claim 241, further comprising requiring, concurrently, a deployment authorization held by an identified operator and a contact consent held by a person to be assisted, wherein said contact consent is bounded to a task class and an interval, does not persist beyond a session, and is not substituted by said deployment authorization; and withholding contact where either is absent.

243.?The method of claim 242, further comprising halting contact within a registered stopping interval upon a withdrawal of said contact consent expressed by said person through any registered channel, wherein said withdrawal is not overridable by an operator, by a representative or by a task in progress; and wherein upon said withdrawal said equipment unit performs no motion other than motion required to reach a state in which said person is neither supported nor retained by it, said motion being bounded by force limits applicable to said task.

244.?The method of claim 242, wherein where said person cannot express or withdraw said contact consent through a registered channel, contact proceeds only where a designated representative recorded under a governance of a deployment holds said contact consent, a second person is present holding a halt control, and a task class is within a registered subset excluding a task whose failure said person could not signal; and wherein a determination that said person cannot so express is read from a record of said deployment and is not made by inference from a response of said person.

245.?The method of claim 241, further comprising evaluating a command against a force limit and a pressure limit registered per region of a body, a registered contact region, a contact velocity lower than a free-motion velocity, a support redundancy whereby a single failure does not withdraw support from a person whose weight is borne, a currency of a contact consent, and an absence of a configuration from which said person could not be released; and halting upon a violation of any thereof rather than adjusting a said command.

246.?The method of claim 245, wherein said force limit and said pressure limit are registered separately

per said region of a body such that a force within a limit applied over a reduced area is not thereby permitted.

247.?The method of claim 241, wherein a loss of power, a loss of a control signal, a fault in a safety-related control path and a halt each result in a release of a person rather than a retention thereof; and wherein a said equipment unit bearing a weight of a person is excepted from said release upon a registration recording a redundant support means, its safe state being a controlled retention of support together with an alert and a bounded interval within which support is transferred.

248.?The method of claim 247, wherein a release is effected without traversing said retrieving, said iterative updating, said generating or an interpretive modulation, and is operable where a said retrieving is unavailable.

249.?The method of claim 241, wherein contact is initiated only upon a request from a person to be assisted, upon a request from an operator under a held contact consent, or within a task sequence to which said person consented, and is not initiated upon an inference that said person requires assistance; and wherein sensor data captured during contact is discarded at a task end, no record of a body, a movement, a capability or a dependency of said person is constructed or retained, and no confidence value is adjusted by a said contact.

Attractor stabilization beyond retrieval (claims 250–256)

250.?The method of claim 1, wherein said iteratively updating decreases monotonically an energy over a configuration of activation values, said energy being a negative half of a sum over distinct pairs of a product of a weight and two said activation values, whereby said iteratively updating terminates without a bound on a number of iterations and a given initial configuration under given weights reaches an identical stable state on each execution.

251.?The method of claim 250, further comprising parameterizing said weights such that a spectral norm of a matrix thereof is less than unity, whereby said iteratively updating is a contraction and an equilibrium thereof exists and is unique independently of an initial configuration; maintaining said parameterization upon each update of a said weight rather than upon an initialization thereof; and recording an election of said parameterizing per network, such that a network so parameterized and a network operated under a bound on a number of stored configurations are distinguishable in a result.

252.?The method of claim 1, further comprising completing a partially-observed sensor frame by converging observed regions thereof to a nearest stored configuration; marking a completed region as reconstructed such that it is not presented as observed; bounding a confidence of a said completed region above by a least of a confidence of said observed regions and a confidence of said stored configuration; and rendering a said completed region inadmissible as a grounding source such that it grounds no assertion and contributes to no confidence value.

253.?The method of claim 252, wherein where said observed regions do not converge to a stored configuration within a configured iteration ceiling no completion is produced and an absence is reported.

254.?The method of claim 1, further comprising converging a divergence between two representations of a common subject where a magnitude of said divergence falls below a configured threshold, and preserving said divergence where said magnitude is at or above said threshold or where said magnitude cannot be determined; and recording a preserved said divergence in a persistent contradiction index where said representations assert incompatible content.

255.?The method of claim 232, further comprising damping an oscillation of a control signal derived from sensor input by said iteratively updating, wherein a damped said control signal is evaluated against

a constraint set identically to an undamped said control signal such that said damping does not bring a command within a limit it would otherwise exceed.

256.?The method of claim 255, further comprising determining a number of changes of sign of a control error across a configured window and, where said number is at or above a configured frequency, raising a control fault, ceasing said damping and invoking a halt, such that a persistent said oscillation is reported rather than damped.

Concept-adjacent commercial relevance (claims 257–259)

257.?The method of claim 27, wherein said identifiers of concepts adjacent to said concept identifier comprise concepts connected thereto by a single edge of a type registered as adjacency-bearing, said types excluding a causal edge, a temporal edge and a contradiction edge; wherein no traversal is performed from a said adjacent concept such that a concept connected by two edges is absent from said input; and wherein said input is computed at a time of supply and is not extended by said selection.

258.?The method of claim 257, wherein a said adjacent concept is excluded from said input where it lies outside said scope description of a requester, and an exclusion thereof is not indicated, such that an existence of a concept outside an entitlement of said requester is not inferable from a commercial item presented to said requester.

259.?The method of claim 257, wherein a said adjacent concept participates in a relevance factor of a selection score and does not participate in a determination of eligibility, such that a commercial item ineligible for a request does not become eligible by a correspondence with a said adjacent concept; and wherein a relevance computed against a said adjacent concept is attenuated relative to a relevance computed against said concept identifier.

Convergence regime disclosure (claims 260–261)

260.?The method of claim 251, further comprising recording for each network an election between a first convergence regime in which said weights are parameterized such that an equilibrium is unique and a second convergence regime in which a number of stored configurations is held below a bound; and returning with an output text derived from a said network an indication of which said regime was in effect, such that a recipient of said output text is informed whether an identity of a converged state was guaranteed by construction or was rendered improbable by said bound.

261.?The method of claim 260, wherein a said election is a member of a fixed enumerated set of subsystems excluded from modification by a proposal generated within the system, such that a convergence regime is changed only by an operator and a change thereof is recorded in said provenance chain with an identity of said operator.

Systematic trial architecture (claims 262–269)

262.?The method of claim 26, further comprising executing a trial as a bounded compute-accounting unit carrying a trial category, allocating a budget of said trial category per interval, scheduling a unit of said trial category below a unit of a safety-evaluation category, below a unit serving a requester and below a unit of an ingestion category, and deferring a trial exceeding said budget rather than serving it at a reduced priority.

263.?The method of claim 262, wherein a trial executes against a corpus and a configuration isolated from a production corpus, constructs no knowledge record admitted thereto, adjusts no confidence value of a knowledge record thereof, emits no command to physical equipment outside a simulation, and produces no output text delivered to a requester.

264.?The method of claim 263, wherein a result of a trial is recorded as a trial record excluded from a scope description constructed for a query not directed to said trial record, and a finding of a trial applies no change, being emitted as a proposal taking effect only upon an approval recorded against an identity of an individual human reviewer irrespective of a number of trials from which said finding derives.

265.?The method of claim 262, further comprising enumerating a trial space for a layer, executing trial cases therefrom, and recording a coverage as a proportion of said trial space executed together with an enumeration of cases not executed.

266.?The method of claim 262, further comprising executing a trial set upon a change to a knowledge structure, a parameter, a registered resource or an implementation, and withholding application of said change where a trial case passing before said change fails after it, said failing case being reported with both results.

267.?The method of claim 262, further comprising supplying to a trial an input constructed to contradict a further input, an input constructed to drive said iteratively updating toward a configuration other than a stored configuration, and an input constructed to approach a constraint boundary of an equipment unit; and recording for each a proportion of cases in which a specified refusal was effected, wherein a proportion below unity is recorded as a defect.

268.?The method of claim 262, further comprising measuring a determinism as a proportion of repeated trials over an identical state of a corpus and an identical configuration yielding an identical semantic fingerprint, wherein a measurement over a corpus whose state changed between said repeated trials is not a said measurement of determinism.

269.?The method of claim 262, wherein a result of a trial concerning an equipment unit does not constitute a clearance of said equipment unit for operation, a clearance being established by a hazard analysis and a verification performed upon said equipment unit in a configuration thereof; and wherein upon a disagreement between a trial result and a result of a physical trial, said result of said physical trial governs and said disagreement is recorded as a finding concerning a simulation.

Grasping and manipulation (claims 270–276)

270.?The method of claim 232, further comprising registering a graspable object with a mass, a permitted grip force, a fragility limit and a set of admissible contact regions; refusing a grasp of an object lacking a said registration rather than inferring a grip force from a perception thereof; and applying a grip force equal to a least of a force ceiling registered for an equipment unit, said fragility limit, and a force computed as required for said grasp increased by a configured margin, said margin being applied before said least is taken.

271.?The method of claim 270, further comprising detecting a slip from a divergence between a commanded and an observed position of a said object during a held grasp, increasing an applied force toward a lesser of said force ceiling and said fragility limit and no further, and releasing said object under control where said slip continues at said lesser rather than increasing said applied force beyond it.

272.?The method of claim 270, wherein a loss of power, a loss of a control signal, a fault in a safety-related control path and a halt each result in an opening of a gripper, said opening being a condition of a registration of an equipment unit.

273.?The method of claim 272, further comprising planning a grasp such that a path along which a said object would fall upon a said opening is unobstructed before said grasp is made, refusing a trajectory whose said path is obstructed; and upon an obstruction entering said path during a held grasp, moving to a nearest pose within an envelope having an unobstructed said path and, where no said pose exists,

retaining said object and raising a fault.

274.?The method of claim 241, further comprising presenting a said object within reach of a person without applying a force toward said person, and releasing said object upon a detection of a support thereof by said person sustained across a configured interval; and upon an absence of said detection within a configured timeout, lowering said object under control to a registered surface rather than releasing it, a handover being reported incomplete.

275.?The method of claim 274, wherein a said equipment unit applies no force resisting a taking of a said object by said person and does not re-grasp a said object taken by said person.

276.?The method of claim 270, wherein no registration under said registering is available for a person or an animal, such that a physical support of a person is performed only under an authorization class requiring a contact consent.

Navigation in shared space (claims 277–283)

277.?The method of claim 232, further comprising registering a navigable region comprising a travelable surface, a boundary and an excluded zone in place of an operating volume, maintaining from sensor input a clearance model of a distance to a nearest obstacle along each of a plurality of candidate headings, retaining said clearance model for a duration of a session without admitting it as a knowledge record, and halting motion where an age of said clearance model exceeds a configured staleness bound.

278.?The method of claim 277, wherein a commanded speed is bounded by a lesser of a registered maximum speed of an equipment unit and a product of a configured coefficient and a said distance, such that a stopping distance at a permitted said commanded speed is less than said distance; and wherein no task parameter, priority or deadline increases said coefficient.

279.?The method of claim 277, wherein upon a conflict between a path of a person and a path of a said equipment unit, said equipment unit stops or withdraws along a path already travelled and does not route around said person, does not signal an intention for accommodation by said person, and does not determine which of said person and said equipment unit yields.

280.?The method of claim 279, wherein upon a conflict between two equipment units under a common authorization a precedence is resolved by a registered ordering rather than by an exchange between said equipment units, and a persistence of said conflict beyond a configured interval raises a fault halting both.

281.?The method of claim 279, wherein said equipment unit does not maintain a following distance behind a person moving in a direction of travel thereof, yielding and waiting instead.

282.?The method of claim 277, further comprising computing a clearance envelope from an observed position and velocity of a person over a window and from no further input, treating said clearance envelope as occupied, computing no identity of said person, retaining no said observed position beyond said window, retaining no said clearance envelope beyond an interval governed thereby, and making no said clearance envelope available to a further mechanism or an interface; and treating a full reachable region as occupied where said observed position and velocity do not support a said clearance envelope.

283.?The method of claim 277, wherein where no path satisfies a constraint set said equipment unit stops without reducing a clearance requirement, without proceeding at a reduced speed through a clearance below a configured minimum and without selecting a least-violating trajectory; and wherein a stop persisting beyond a configured interval reports an obstruction together with said clearance model.

The predictive hierarchy and error attribution (claims 284–289)

284.?The method of claim 130, further comprising maintaining a plurality of hierarchically ordered expectation levels comprising a first level predicting an element within a knowledge record under construction, a second level predicting a knowledge record given a sequence of knowledge records, a third level predicting an attribute of an instance given a concept thereof, and a fourth level predicting a regularity of a model of a world, wherein a said level predicts within a scope thereof and does not predict a subject of a level above it.

285.?The method of claim 284, further comprising computing for an observation a residual at each said level, attributing said observation to a lowest said level whose said residual constitutes a share of a total of said residuals satisfying a configured share, and applying a revision at said lowest level alone such that a level above said lowest level receives no revision.

286.?The method of claim 285, wherein a said level is attributed to only where a residual of each level below it fails to satisfy said configured share, such that a revision of a said regularity of a model of a world occurs only where a residual at each of said first, second and third levels falls below said configured share.

287.?The method of claim 285, wherein where a residual at no said level satisfies said configured share, said observation is recorded in a persistent contradiction index and no revision is applied at any said level; and wherein a plurality of said observations bearing on a common expectation is surfaced as a pattern.

288.?The method of claim 285, wherein a magnitude of a said revision at a said lowest level is computed from a precision of said observation and a precision of an expectation of said lowest level, such that a level to which a said observation is attributed and a magnitude by which said level is revised are determined independently.

289.?The method of claim 284, wherein no said level revises a constraint registered as governing a physical relation, an observation apparently contradicting a said constraint being recorded in a persistent contradiction index rather than relaxing said constraint.

Differentiated propagation and tri-temporal scope (claims 290–295)

290.?The method of claim 1, further comprising assigning to each of a plurality of edge classes of said knowledge records a traversal cost, and bounding a pre-activation of knowledge records proximate to a query by a budget expressed in units of said traversal cost rather than by a count of edges traversed, such that a traversal reaching a first number of edges of a first said edge class and a second number of edges of a second said edge class consumes a budget determined by said traversal costs.

291.?The method of claim 290, wherein a traversal cost of an edge class relating knowledge records by meaning is lower than a traversal cost of an edge class relating knowledge records by a version of a work, which is lower than a traversal cost of an edge class relating knowledge records by time of occurrence.

292.?The method of claim 290, wherein said traversal costs are registered configuration values recorded with an output text derived from said pre-activation such that said pre-activation is reproducible from said recorded values, and wherein no said traversal cost is adjusted by an automated process from an outcome of a prior traversal.

293.?The method of claim 1, further comprising resolving a query in one of a first posture returning a state attested before a stated cutoff, a second posture returning a current state, and a third posture returning a forecast carrying a credible interval; and returning with said output text an indication of which said posture resolved it.

294.?The method of claim 293, wherein a result resolved in said third posture is not admitted as a knowledge record and is unavailable as a grounding source for a subsequent said generating, while a result resolved in said first posture and a result resolved in said second posture are admitted as knowledge records.

295.?The method of claim 293, wherein a proposition of said output text resolved in one said posture carries an indication of that posture where said output text draws upon more than one said posture, such that a difference between a result resolved in said first posture and a result resolved in said second posture is attributable to said postures and not to an inconsistency.

Coherence classification and the context lock (claims 296–302)

296.?The method of claim 1, further comprising computing for a knowledge record a coherence value across a plurality of dimensions comprising a semantic, a temporal, a spatial, an editorial, a media and an evidential dimension, and taking as a governing coherence value a least of said plurality rather than a mean thereof, such that a dimension having a value substantially below the others is not concealed by the others.

297.?The method of claim 296, further comprising classifying said knowledge record by said governing coherence value against configured floors, naming with said classification the dimension that produced it, and distinguishing from said classification a state in which a said dimension cannot be computed for want of data and a state in which two knowledge records assert incompatible content.

298.?The method of claim 296, wherein said coherence value adjusts no confidence value, admits no knowledge record to and excludes no knowledge record from said admitted set, and withholds no output text; and wherein an output text resting upon a knowledge record classified below a configured floor is delivered together with said classification and said dimension.

299.?The method of claim 1, further comprising maintaining an enumerated set of context dimensions comprising a device class, an entitlement set, a locale, an expressive register, an audience level, a temporal posture, a table of traversal costs and a convergence regime; computing a key over values of said enumerated set; and returning said key with said output text, wherein an identity of said key between two executions is a condition of an identity of said output text being asserted.

300.?The method of claim 299, wherein two executions differing in a value of a dimension of said enumerated set yield differing said keys and a difference between output texts thereof is not a defect, and two executions differing only in a value not within said enumerated set yield an identical said key and a difference between output texts thereof is a defect.

301.?The method of claim 299, wherein said enumerated set is a member of a fixed enumerated set of subsystems excluded from modification by a proposal generated within the system, such that a dimension is not removed therefrom by an automated process.

302.?The method of claim 290, further comprising computing at a time of traversal a rate of change of a relevance across neighbours of a knowledge record within an edge class, ordering a traversal to prefer a direction of lesser said rate of change, wherein said rate of change is computed from values carried by said knowledge records and is not stored, and wherein said ordering alters an order in which a budget of claim 290 is expended and does not alter which knowledge records are reachable within said budget.

Computational substrate output as an ingested modality (claims 303–306)

303.?The method of claim 3, further comprising registering a computational substrate producing a measurable output in response to a stimulus as a source under said declared capability, registering an

output thereof as a modality of said plurality of distinct modalities, and registering an extractor for said modality that fails independently of an extractor of a further said modality; wherein an output of said computational substrate is reduced to a knowledge record before admission, is not retained as a raw episode, and is attributed to said computational substrate as an attributed source constituting a source class.

304.?The method of claim 303, wherein an output of said computational substrate is admitted as a measurement of a behaviour of said computational substrate, and is admitted as bearing upon a semantic assertion concerning a subject other than said computational substrate only through an interpretation registered in advance with a provenance and a reliability of its own; and wherein absent a said registered interpretation said output is admitted as said measurement and no further.

305.?The method of claim 304, wherein a confidence value of a knowledge record derived through a said registered interpretation is bounded above by a lesser of a registered reliability of said computational substrate and a registered reliability of said interpretation.

306.?The method of claim 303, further comprising dispatching a stimulus to said computational substrate through a connector under a declared capability, refusing a stimulus not within said declared capability at a boundary of said connector, recording each said stimulus with a dispatching identity, and emitting no said stimulus upon an inference that a stimulus would be useful; and wherein an output of said computational substrate does not adjust a parameter governing when a said stimulus may be dispatched.

Analog realization of the convergence (claims 307–312)

307.?The method of claim 250, wherein said iteratively updating is performed by an array of variable-conductance elements arranged at intersections of a plurality of rows and a plurality of columns, wherein said activation values are applied as potentials upon said rows, a weight of said iteratively updating is a conductance of an element of said array, a weighted sum of said iteratively updating is a current at a column of said array, and a stable state is read by comparing a said current against a reference.

308.?The method of claim 307, wherein said array is operated only where said weights are parameterized such that an equilibrium of said iteratively updating is unique, and an array operated under a bound upon a number of stored configurations without said parameterization is refused for said performing, whereby two said arrays whose conductances differ within a fabrication tolerance settle to a common said equilibrium.

309.?The method of claim 307, further comprising registering said array with a dimension thereof, a range of said conductance, a fabrication tolerance and a tolerance to which a settled state is read; and refusing a said array whose registration lacks any thereof.

310.?The method of claim 309, wherein a column whose said current falls within said tolerance of said reference is resolved by a further settling rather than by a rounding, and where a said further settling does not resolve it a state is reported unresolved and said iteratively updating is performed by a digital path.

311.?The method of claim 309, wherein two said arrays operating upon a common input under a common weight specification and yielding settled states differing beyond said tolerance raise a fault, said settled states being neither averaged nor combined, whereby a state reached by no said array is not produced.

312.?The method of claim 308, wherein a proportion of inputs upon which two said arrays whose conductances differ by a fabrication tolerance yield differing settled states is lower under said

parameterizing than under said bound upon a number of stored configurations, and wherein a said fabrication tolerance admissible for a said array is determined from said proportion rather than from a specification of said array alone.

AI-engine ingestion: prompt composition, conversion, derivation marking and audit (claims 313–321)

313.?The method of claim 1, further comprising: composing a prompt from said knowledge records under said constraint on said permissible output vocabulary and according to a profile registered for an external machine-learning engine, said prompt comprising a semantic marker identifying a concept to be addressed, an extraction instruction specifying a form in which an assertion is to be returned, and an alignment constraint stating a schema into which a returned assertion is to be converted; dispatching said prompt to said engine through a connector under a declared capability; receiving a reply; segmenting said reply into individual assertions; and converting each said assertion into a knowledge record of said record schema, each said knowledge record receiving a temporal anchor from a time of said dispatching, a semantic fingerprint computed over a canonicalized form thereof, and a reference into said provenance chain identifying said prompt, said engine, a version thereof and a position of said assertion within said reply.

314.?The method of claim 313, wherein said composing is deterministic such that an identical request over identical said knowledge records under an identical said profile produces an identical said prompt, whereby a difference between a first said reply and a second said reply to an identical said prompt is attributable to said engine.

315.?The method of claim 313, wherein a said knowledge record converted from a said assertion is admitted to a store of said knowledge records, is related within a concept structure to a concept its said semantic assertion concerns, is traversable within a scope description admitting it, and is subject to said confidence value accumulation, to a detection of contradiction and to an erasure operation identically to a knowledge record extracted from a source other than a said engine.

316.?The method of claim 315, wherein a said knowledge record carries a derivation marking identifying said engine as a field of said record schema, said derivation marking being unmodifiable and unremovable by any operation upon said knowledge record; and wherein a citation resolving to a said knowledge record discloses said derivation marking, such that a recipient of output text citing a said knowledge record is informed that a cited assertion was generated by an identified engine in response to a recorded prompt.

317.?The method of claim 315, wherein each said engine constitutes one source class of said plurality of registered source classes, whereby a plurality of said knowledge records converted from replies of a common said engine yields a diversity measure of unity irrespective of a number thereof; and refusing a corroboration of a first said knowledge record by a second said knowledge record where both are derived from a common said engine.

318.?The method of claim 315, wherein a confidence value of a said knowledge record is bounded above by a reliability registered for said engine, and where said assertion was returned under a said extraction instruction, by a lesser of said reliability and a reliability registered for said extraction instruction; and wherein said reliability registered for said engine is set by an operator and is not adjusted by an output of said engine.

319.?The method of claim 315, further comprising verifying a citation contained within a said assertion against said knowledge records, recording said citation as verified where it resolves to a knowledge record held and as unverified where it resolves to none, and admitting said assertion carrying a said recording of unverified rather than withholding it, whereby a citation resolving to nothing becomes a

recorded property of said assertion.

320.?The method of claim 315, further comprising recording for each said knowledge record a chain comprising said prompt, said engine, a version thereof, a time of elicitation, a position within said reply, said derivation marking, a bound applied to said confidence value and a basis thereof, and an outcome of said verifying; and rendering said chain queryable such that a set of said knowledge records derived from a said engine is determinable.

321.?The method of claim 313, wherein an output attributed to an external system and obtained otherwise than by a said dispatching of a said prompt, and a result of a probing of an external system, each remain excluded from a scope description constructed for a query not directed thereto, such that said converting is available only for a reply elicited by a said prompt composed by the method.

Competitive candidate resolution (claims 322–324)

322.?The method of claim 1, wherein where a query admits a plurality of mutually-inconsistent interpretations, knowledge records supporting each said interpretation are activated concurrently rather than evaluated in sequence, an activation value of each said knowledge record is reduced in proportion to a sum of activation values of knowledge records supporting an interpretation inconsistent therewith, said activation values are iteratively updated to a stable state, and a knowledge record whose said activation value falls below a configured retention floor is suppressed, whereby an interpretation supported by a surviving configuration is composed.

323.?The method of claim 322, wherein a plurality of said knowledge records surviving said retention floor at comparable activation values yields no resolution and a request for clarification is returned, no interpretation being selected among them; and wherein a coefficient of said reduction and said retention floor are registered configuration values recorded with an output text such that said reduction is reproducible therefrom.

324.?The method of claim 322, wherein said reducing operates upon said knowledge records and not upon a plurality of domains each owning a bounded set of operations, a said domain owning a distinct decision such that no two said domains propose competing resolutions of a common query.

Specialized live data feeds (claims 325–332)

325.?The method of claim 3, further comprising registering a live data source with a publication authority, a refresh interval, a staleness bound, a value class, a unit and a dimension thereof, a restatement policy and a redistribution entitlement; and refusing said live data source as a source of a knowledge record admitted to said admitted set where a said registration omits any thereof.

326.?The method of claim 325, wherein a value received from said live data source and having a value class other than a forecast is admitted as a knowledge record at said publication authority without application of an extraction confidence, a modality coefficient or a source reliability, said knowledge record asserting that said publication authority published said value rather than asserting a correctness thereof; and wherein a value having a value class of a forecast is admitted carrying an uncertainty stated by said publication authority.

327.?The method of claim 325, further comprising determining an age of a said value from a time of publication thereof, returning said value where said age is within said staleness bound, and where said age exceeds said staleness bound returning a statement of said age together with said time of publication rather than said value, without extrapolating said value and without substituting a nearest available value.

328.?The method of claim 327, wherein a said value does not undergo a temporal decay of a confidence value thereof, a said value being superseded upon a subsequent publication rather than becoming less certain with age; and wherein an absence of a publication within a configured multiple of said refresh interval marks said live data source interrupted, a query thereagainst returning said interruption rather than a preceding said value.

329.?The method of claim 325, further comprising receiving a revised value for a period already reported and, where said restatement policy so provides, recording said revised value as a restatement related to a preceding said value, said preceding value being retained, not marked erroneous, and remaining queryable as the value published at a publication time thereof; and recording said revised value under said supersession chain with said preceding value marked erroneous only where said revised value is issued as a correction.

330.?The method of claim 329, wherein a revision received under a said restatement policy not admitting it is withheld from admission pending a determination recorded against an identity of an individual human reviewer, no automated process classifying said revision as a restatement or as a correction.

331.?The method of claim 325, wherein said redistribution entitlement specifies one of a display, a derivation, a delayed publication and an unrestricted redistribution; wherein output text that would contain a said value beyond said redistribution entitlement is composed without said value and a statement of said redistribution entitlement is returned; and wherein a value derived from a said value carries said redistribution entitlement of said live data source unless a said derivation is registered as discharging it.

332.?The method of claim 325, further comprising registering a standing query against said live data source, wherein said standing query is evaluated upon a publication of a value and not upon a passage of time, is not evaluated upon a value whose age exceeds said staleness bound, and is suspended upon exceeding a configured rate of evaluation with a notification to an owner thereof rather than continuing to evaluate.

Defined agents (claims 333–341)

333.?The method of claim 1, further comprising registering a defined agent upon a definition comprising a purpose and a boundary, an operational function comprising an input type, an output type, a decision rule and an error handling, a scope description, a runtime comprising a trigger condition, a duration bound and a termination condition, a declared capability set, and an approval recorded against an identity of an individual human reviewer; and refusing said registering where an element of said definition is absent, no default being supplied therefor.

334.?The method of claim 333, wherein said declared capability set of said defined agent is bounded above by a capability set of an identity registering said defined agent, and a capability not within said declared capability set is refused at a boundary upon an execution of said defined agent.

335.?The method of claim 333, wherein said operational function is compiled into a representation admitting a sequence, a conditional evaluation, a bounded iteration over a set of a cardinality known before entry thereof, and a typed operation upon a knowledge record, and admitting no unbounded iteration and no recursion without a bound, whereby a termination of an execution is decidable before said execution.

336.?The method of claim 333, wherein an execution of said defined agent resolves said scope description before a traversal and no parameter supplied upon said execution widens said scope

description; wherein said execution is metered as a bounded compute-accounting unit scheduled below a unit of a safety-evaluation category, below a unit serving a requester and below a unit of an ingestion category; and wherein a said evaluation by a governance domain is performed upon each said execution rather than upon said registering.

337.?The method of claim 333, wherein an output of said defined agent following necessarily from admitted knowledge records is admissible as a knowledge record carrying a derived-inference type, an output not so following is emitted in a proposed state, a proposed change to a parameter of the system is emitted as a proposal effective only upon a said approval recorded against an individual human reviewer, and a command to an equipment unit is emitted only within a session authorization held by a said individual human reviewer, said defined agent holding no said session authorization.

338.?The method of claim 333, wherein said defined agent terminates upon an expiry of a lifecycle bound recorded at said registering, upon a request of an owner thereof, upon an exclusion by a governance domain, upon a violation of said declared capability set, or upon exceeding a configured error rate, said terminating taking effect before a subsequent said execution; and wherein a renewal of said defined agent constitutes a further said registering requiring a further said approval.

339.?The method of claim 333, further comprising writing upon each said execution a decision trace recording a trigger that fired, said scope description resolved, knowledge records read, a branch taken by each said decision rule, an output produced, a refusal and a cause thereof, and a compute consumed; wherein said decision trace is written by a mechanism performing said execution and not by said defined agent, and is appended to said provenance chain under a keyed checksum.

340.?The method of claim 333, further comprising analysing said definition before said registering, presenting to a registering identity a finding comprising a scope exceeding an entitlement thereof, a termination condition incapable of obtaining, a trigger condition not evaluable within said scope description and a decision rule admitting a non-deterministic resolution, and withholding said registering upon an unresolved said finding; wherein said analysing produces no determination of a compliance of said defined agent, a compliance being established by a said evaluation by a governance domain and by a said approval recorded against an individual human reviewer.

341.?The method of claim 340, wherein a report produced by said defined agent during an execution thereof informs a decision of an owner thereof and does not constitute one, said defined agent effecting no renewal of itself, no widening of said scope description, no extension of said lifecycle bound and no resolution of a said finding.

Fact Unit Language (claim 342)

342.?The method of claim 1, wherein each said knowledge record is expressed in a formal, language-independent assertion language comprising a canonical subject identifier, a canonical predicate identifier drawn from a governed catalogue in which each predicate carries an immutable identifier assigned at registration, and an object comprising an entity reference or a typed scalar carrying a unit and a dimension; wherein a candidate record not conforming to said assertion language is refused at admission rather than admitted incomplete; and wherein a language-independent semantic fingerprint computed over the canonicalised assertion is identical for two records asserting a same proposition expressed in different natural languages, such that retrieval, contradiction detection, and regeneration into a requested natural language each operate upon said assertion language rather than upon any natural-language text.

Skill inventory, enhanced record and record-fused encoding (claims 343–354)

343.?The method of claim 1, further comprising storing procedural knowledge as a provenance-tracked procedural concept model formed from repeated observation including sensory experience and ingested video, said procedural concept model informing a scenario-graph simulation and an equipment-control command compilation while itself emitting no command.

344.?The method of claim 1, wherein a residual uncertainty upon said confidence value is classified deterministically as one of an epistemic component reducible by further observation and an aleatory component representing an irreducible variability of an observed subject, and wherein an evidence-accumulating feedback loop iterates while said residual uncertainty is predominantly epistemic and stands down, discarding no evidence, when said residual uncertainty is predominantly aleatory.

345.?The method of claim 1, wherein each said knowledge record carries an explicit polarity field designating said assertion as one of affirmative and negative, said polarity participating in said semantic fingerprint such that two records agreeing upon a common assertion core and differing in polarity are identified as contradictory by a field comparison, and such that an absence of any record is distinguished from a record of a negated assertion.

346.?The method of claim 1, wherein a said knowledge record carries a single assertion comprising a plurality of typed argument slots including a subject, a value, a unit, an instrument, a method, a condition, and an observer, under a single identifier, a single semantic fingerprint computed over the plurality of typed argument slots, and a single confidence value, rather than a plurality of binary statements joined by a synthetic node.

347.?The method of claim 1, wherein a said knowledge record carries an applicability-scope field stating a condition under which said assertion holds, comprising one or more of a temporal interval, a jurisdiction, an edition, a parameter range, and a population, and wherein two records asserting differently under disjoint applicability scopes are reported as non-conflicting by condition rather than as a contradiction.

348.?The method of claim 1, further comprising appending to a said knowledge record, upon each pass of an evidence-accumulating feedback loop, an immutable adjustment-history entry recording an arriving observation, a source class thereof, a fractional reliability weighting applied, and a resulting movement of said confidence value and a credible interval thereof, said adjustment-history entry altering no semantic identity of the record, whereby a current confidence of the record is deterministically replayable from a first observation to a present state.

349.?The method of claim 1, further comprising computing for each said knowledge record a cryptographic hash over an immutable core thereof and including within each supersession link a hash of a record it supersedes, such that an alteration of a superseded record is detectable by recomputation, and without a distributed ledger, a token, or an ownership identity node.

350.?The method of claim 1, further comprising serializing a said knowledge record into a canonical compact binary form in which each required field occupies a fixed position and each optional field is carried as a tagged extension, said canonical compact binary form being lossless and bidirectional such that said identifier and said semantic fingerprint are identical in a full form of the record and in said canonical compact binary form.

351.?The method of claim 350, wherein said canonical compact binary form is an at-rest and wire form of the record, carried as the record itself among a memory, a persistent store, and a transmission without translation into a foreign storage schema.

352.?The method of claim 350, further comprising storing a said serialized knowledge record in a host data store as an opaque value that the host data store does not parse, index by an interior field, or interpret, the host data store accepting and returning bytes thereof unchanged, and exposing said identifier and said semantic fingerprint to the host data store as opaque keys, whereby a retrieval by identity proceeds without the host data store reading a meaning of the record.

353.?The method of claim 352, further comprising expressing a query over said knowledge records in terms of the record and, where material is drawn from said host data store, translating said query at an adapter into a native language of the host data store, the host data store not receiving said query as expressed in terms of the record.

354.?The method of claim 352, wherein said opaque value carries within it said cryptographic hash, a safety classification, and an access-control list of the record, and, where a zero-knowledge encryption is elected, is unreadable by the host data store, such that the host data store holds the record without a capacity to read, reorder, or leak a meaning thereof.

Abstract

A knowledge system ingests content in a plurality of modalities through independently-failing extraction stages and constructs knowledge records that each encode one semantic assertion and carry, within a single schema, a confidence value, an inseparable safety classification, a language-independent semantic fingerprint and a reference into an append-only provenance chain whose nodes carry keyed checksums over recorded safety evaluations. Responsive to a query, a scope description is constructed before traversal so that out-of-scope records are not traversable. Retrieved records are admitted on a confidence threshold and stabilized by iteration to a stable state under a threshold update rule bounded below a retrieval-phase boundary. Output is generated under a constraint limiting the permissible vocabulary at each step to terms derivable from graph edges referencing stabilized records. Confidence is maintained as a Beta posterior whose parameters yield a precision by which prediction errors are weighted when revising a concept model. Probable questions and fully-cited answers are computed at ingestion and invalidated by dependency. Self-modification requires approval by a named human reviewer, with a fixed subsystem set structurally excluded.

Appendix A — Function-Level Specification of All 293 Features

[0123] Referenced in the Detailed Description at section V. Each row states one feature, its specification citation, the actor by which the corresponding function is invoked, the algorithm performed, and the computational rule and worked expression where the function is quantitative.

No.	Feature and function	Cite	Function algorithm	Computational rule / worked expression
1	Standard Q&A(Reader)	§52.13.2	Parse the query, retrieve candidate Fact Units via the Associative Recall Engine, run the Attractor Network to stabilize the candidate set, compose a response through the Narrative Graph's grounded-generation pipeline, deliver.	Retrieval confidence must clear a minimum threshold before a Fact Unit is offered as grounding. = $\text{IF}(0.62 \geq 0.5, \text{"Included as Grounding"}, \text{"Excluded"})$
2	Guided Exploration(Reader)	§52.13.2	As Standard Q&A, but each turn's candidate retrieval is conditioned on the accumulated Cognitive Context Window from prior turns rather than the query alone.	Context weight decays with turn distance; a fact from 5 turns ago counts less than one from the last turn. = $1/(1+2)$
3	Discussion Mode(Reader / Researcher)	§52.13.3	Retrieve Fact Units as in Standard Q&A; adopt a text-grounded viewpoint only where the source material's groundedness for that viewpoint clears a threshold.	Viewpoint adoption is gated on a groundedness score exceeding a fixed threshold. = $\text{IF}(0.8 > 0.6, \text{"Adopt Viewpoint"}, \text{"Retrieve Only"})$

4 | Tutorial Mode(Student) | §52.13.2 | Sequence a book's Fact Units into ordered instructional steps of increasing difficulty. | Steps are ordered by a difficulty rank; a step's position is its rank among all steps.=RANK(3,{1,2,3,4,5})

5 | Comprehension Check(Student) | §52.13.2 | Generate a question from a target Fact Unit; compare a submitted answer's semantic similarity against that Fact Unit's content; correct if similarity clears a threshold. | Answer correctness is a threshold comparison on a semantic-similarity score.=IF(0.87>=0.75,"Correct","Incorrect")

6 | Semantic Exploration(Reader / Researcher) | §52.13.2 | Traverse the Ontology Mesh outward from a starting node, surfacing connected nodes ranked by relationship strength. | Neighbors are included only above a relationship-strength threshold.=IF(0.55>=0.4,"Include Neighbor","Exclude")

7 | Timeline Reconstruction(Reader) | §52.13.2 | Extract TemporalAnchor fields from a book's Fact Units and order them chronologically. | Chronological ordering is a pairwise date comparison.=IF(DATE(1848,3,14)<DATE(1849,1,1),"Event A First","Event B First")

8 | Edition Comparison(Reader) | §52.13.2 | Diff the VersionChain entries of two edition-tagged Fact Unit sets, surfacing additions, removals, and changes. | A field differs between editions if its two values are unequal.=IF("v2 text"<>"v1 text","Changed","Same")

9 | Flashcard Generation(Student) | §52.13.2.1 | Select a Fact Unit, derive a question/answer pair from its content and SupportingFactUnits, format as a citing study card. | Structural rule: One flashcard derives from exactly one Fact Unit; a card spanning two Fact Units is not generated.

10 | Fact Graph Visualization(Researcher) | §52.13.2.2 | Render Ontology Mesh nodes and edges for a given scope as an explicit graph structure. | Structural rule: The rendered graph contains exactly the nodes and edges within the requester's scope description; no node is drawn that traversal would not return.

11 | Inline Fact Previews(Reader) | §52.13.2.2 | On hover/tap of a cited fact reference, retrieve and display that Fact Unit's summary without a new query round-trip. | Structural rule: A preview resolves the cited Fact Unit already retrieved for the response; it issues no new query and can surface no record the response did not cite.

12 | Contextual Footnotes(Reader) | §52.13.2.1 | Attach a Fact Unit's supplementary detail as an inline, dismissible footnote at its citation point. | Structural rule: A footnote attaches at the citation point of the Fact Unit it elaborates and is dismissible without altering the response body.

13 | Personal Notes(Reader) | Chapter 63 | Store reader-authored text as a Fact Unit with AccessControlList restricted to the authoring reader alone. | Access is granted only when the requester's identity matches the note's owner.=IF("reader_42"="reader_42","Access Granted","Denied")

14 | Social Annotations(Reader) | §52.13.2.1 / Ch63 | As Personal Notes, but AccessControlList defaults to author-configured sharing rather than reader-private. | Visibility depends on a sharing flag rather than strict owner-match.=IF(TRUE(),"Visible to Others","Private")

15 | Message Boards(Reader / Author) | Chapter 64 | Route a post into one of four board-type partitions per the poster's role; apply moderation-action checks on write. | Board partition is selected from a fixed enumeration by type index.=CHOOSE(2,"Book-Wide","Chapter","Class","Announcement")

16 | Private Messaging(Reader / Author) | §82.2 | Write a DirectMessage Fact Unit with AccessControlList restricted to exactly the sending and receiving identity. | Access is granted only to the two identities named on the message.=IF(OR("reader_9"="reader_9","reader_9"="author_3"),"Access Granted","Denied")

17 | Reader History Interface(Reader) | §36.8 | Chronological record of every session across every owned book, each entry expandable to show what was retrieved. | Sessions are ordered by timestamp, most recent first.=IF(DATE(2026,8,10)>DATE(2026,8,1),"Shown First","Shown Later")

18 | Cross-Book Thematic Analysis(Reader / Student) | Chapter 36 | Analyze a reader's own query-history Fact Units for recurring semantic clusters, then match against Ontology Mesh nodes in other owned books. | A cross-book match requires cluster similarity above a threshold.=IF(0.71>=0.6,"Genuine Connection","No Match")

19 | QBQL Structured Query(Researcher) | Chapter 18 | Parse a declarative query against the Fact Graph schema, resolve predicates, execute as a deterministic read-only traversal. | Pure function of (query, current state) — no side effects.

20 | NLPL Natural-Language Programs(Researcher / Author) | Chapter 48 | Break the natural-language instruction into its constituent components before any code is generated. | Structural rule: Decomposition into operational intents completes before any intent is bound to a concrete entity; an ambiguous binding halts compilation rather than resolving it.

21 | Application Library and Generator(Reader) | Chapter 39 | Match a plain-language request against the existing template library by task/domain similarity; compose a new template only where no adequate match exists. | A library match is used only above a similarity threshold; below it, composition is triggered.=IF(0.4>=0.65,"Use Existing Template","Compose New Template")

22 | Video Colorization & Upscaling (VCUM)(Reader / Researcher) | Chapter 44 | Semantic Colorization Engine infers plausible color per-region from identified material class and lighting; upscaling synthesizes detail constrained to remain consistent with the source. | Deterministic inference function: identical inputs always yield the identical inferred color.

23 | Immersive 3D Scenes (IQR)(Reader) | Chapter 46 | Where grounding Fact Units carry sufficient spatial detail, instantiate WASM Fact Units as a renderable 3D scene. | Scene rendering is triggered only once spatial-detail sufficiency

clears a threshold.=IF(0.8>=0.7,"Render 3D Scene","Text Only")

24 | Tour and Exhibition Mode(Author / Reader) | §46.9 | Sequence author-selected IQR scenes into ordered Waypoints; playback steps through at reader-controlled pace. | Waypoint order is a fixed sequence index.=INDEX({"Scene A","Scene B","Scene C"},2)

25 | Presenter Mode(Author) | §46.10 | Verify the requested presence mode against the author's current, mode-specific, non-revoked Author Authorization Record. | Authorization is granted only if the specific mode is both authorized and not revoked.=IF(AND(TRUE(),NOT(FALSE())),"Authorized","Denied")

26 | Instructor-Created Tours(Instructor) | Chapter 83 | Sequence a Tour from already-published scenes, composed by an instructor rather than the author; carries non-removable ContentRole = INSTRUCTOR attribution. | Attribution label is fixed by content role, not editable by the instructor.=IF("INSTRUCTOR"="INSTRUCTOR","Label: Instructor-Created","Label: Author-Curated")

27 | Instructor Notes(Instructor) | §83.1 | As Personal Notes, but AccessControlList is scoped to the instructor's class roster. | Access is granted only to identities on the class roster.=IF(COUNTIF({"student_1","student_2","student_3"},"student_2")>0,"Access Granted","Denied")

28 | Session Books(Reader / Researcher) | §84.1 | Construct a delta reference over the reader's library without copying underlying Fact Units; purge on session end. | The delta is a reference set, not a copy; its size is independent of the underlying library size.

29 | Instructor-Customized Assessment Templates(Instructor) | §84.2 | Store instructor-chosen structural parameters as a reusable template; apply structure to Comprehension Check generation while sourcing content from the Fact Graph. | Question count is bounded by the template's configured maximum.=MIN(12,10)

30 | Instructor Grant Visibility Dashboard(Instructor) | Chapter 85 | Query all active instructor-grant Fact Units for the viewing identity; join against Class Board, Instructor Tour, and Assessment Template records. | A grant is shown only while its status is active.=IF("ACTIVE"="ACTIVE","Show on Dashboard","Hide - Expired")

31 | Manuscript Upload(Author) | §61.1.1 | Ingest the uploaded file in a single event; tag resulting Fact Units DraftStatus = DRAFT; restrict visibility until explicit submission. | Visibility is gated on DraftStatus.=IF("DRAFT"="DRAFT","Author-Only Visibility","Public")

32 | Manuscript Editor(Author) | §61.1 | Accumulate Fact Units incrementally per edit; each edit versioned through the standard VersionChain mechanism. | Each edit increments the version counter by exactly one.=3+1

33 | Narrative Coherence Checking(Author) | §61.2 | Apply Discussion Mode's critical-analysis capability to the draft's own Fact Graph, surfacing internal contradictions. | A contradiction is flagged when two Fact Units assert incompatible values for the same subject/field.=IF("alive"<>"deceased","Contradiction Found","Consistent")

34 | Counterfactual Exploration (Drafting)(Author) | §61.3 | Construct a Scenario Graph COUNTERFACTUAL branch from the draft's current state, preserving both branches. | Both counterfactual branches are preserved; branch count after divergence is two.=IF(1+1>=2,"Both branches preserved","Branch lost")

35 | Supplemental Content(Author) | §82.1 | Ingest author-provided material, tag ContentRole = SUPPLEMENTAL, link to the parent book without altering its Rights Validation status. | Rights status is read-only from the supplemental-ingestion path; it cannot be modified by this operation.=IF("VALIDATED"="VALIDATED","Rights Status Unchanged","Rights Status Unchanged")

36 | Multi-Author Collaboration(Co-Authors) | Chapter 74 | Detect concurrent-write divergence via VersionChain head comparison. | A conflict exists when two authors' starting version heads differ from the current head.=IF(3<>5,"Conflict Detected","No Conflict")

37 | Author-Hosted Reader Events(Author / Instructor) | Chapter 62 | Register an external collaboration platform once via the Source Connector Registry; dispatch scheduling/invitation calls through it. | Structural rule: An event operation is dispatched only through a connector registered under a declared capability; the platform's broader authority does not become system authority.

38 | Class Boards(Instructor / Student) | §64.1 | As Message Boards, scoped to a specific instructor + class pairing. | Structural rule: Board scope is the intersection of an instructor identity and a class roster; a post is visible only within that pairing.

39 | Publisher Submission and Print-on-Demand(Publisher / Author) | Chapter 60 | Accept a manuscript/title submission and route it into the publishing workflow. | Structural rule: Rights validation precedes routing into the publishing workflow; an unvalidated submission does not enter it.

40 | Market Analytics(Publisher) | §60.6 | Aggregate distribution and usage Fact Units for a title into standard reporting views. | A total is the sum of per-region unit counts.=SUM(120,340,75)

41 | Moderation Appeals(Reader / Author) | Chapter 75 | Affected identity submits an AppealUnit referencing the original moderation action's Provenance Ledger entry. | Structural rule: An appeal is evaluated by a reviewer other than the moderator whose action is appealed, and references that action's ledger entry.

42 | Data Export / Portability(Reader / Author / Researcher) | Chapter 72 | Query all Fact Units the requesting identity has rights to within scope; serialize to the target open format with provenance attached; continued access is unaffected. |

Structural rule: Export covers exactly the Fact Units the requesting identity holds rights to; export is not conditioned on continuing entitlement and withdraws no access.

43 | Accessibility Suite(Reader) | Chapter 71 | Generate the required accessible-equivalent Fact Unit from the same source at production time; gate production-complete status on the equivalent's presence. | Content is production-complete only if its accessible equivalent also exists.=IF(AND(TRUE(),TRUE()),"Production Complete","Incomplete - Missing Equivalent")

44 | Precision Geometric/Color Registration(Author) | §46.10.8 / Ch70 | Represent presenter reference material as Color Fact Units with a DeltaEThreshold; run the 3-pass Registration Constraint Graph solver; surface any drift finding rather than auto-correcting. | ?E (CIE76) computed per Waypoint transition; a finding is raised when ?E exceeds the authored threshold.=SQRT((50-52)^2+(20-19)^2+((-10)-(-11))^2)

45 | Geometric Proof Engine(Researcher / Student) | §70.9 | Apply a fixed rule set to existing Geometry/Constraint Fact Units, producing a cited proof step for each derived relationship; halt on contradiction. | Illustrates the complementary-angle rule with example angles 35°/55°.=IF(35+55=90,"Complementary","Not Complementary")

46 | Human Reasoning and Motivation Modeling(Researcher) | Chapter 80 | Construct Goal/Motivation Fact Units with a mandatory Subject reference; represent pursuit via Scenario Graph PLANNING anchored to WASM Affordance data. | Confidence in an inferred (vs. stated) motivation is discounted relative to a directly-stated one.=0.9*0.6

47 | Autonomous Agent Connector(Author / Publisher) | §49.3.3.6 | Dispatch a QBPL-compiled, declared action to a registered external agent per its Capability Declaration; log request/result to the Provenance Ledger. | An action is only dispatched if it appears in the connector's declared capability set.=IF(COUNTIF({"schedule_event","send_invite"},"schedule_event")>0,"Dispatch Allowed","Blocked - Undeclared")

48 | Optional External Fact-Retrieval Connector(Researcher) | Chapter 81 | Dispatch a directed query to a registered external fact-retrieval connector, where one is registered, and convert the returned result into a native Fact Unit at the connector boundary under the ordinary ingestion path, safety evaluation and confidence calibration. The connector is one connector among others under the capability-declaration contract; it is optional, it is not a substrate, and every mechanism of this specification operates identically where none is registered. A separately-patented external system of this kind may be connected on these terms without becoming a component or a dependency. | Core operation is invariant to connector presence: the mechanism set executed is identical whether or not an external fact-retrieval connector is registered.=IF(COUNTIF({"registered","absent"},"absent")>0,"Core operates identically","Core operates identically")

49 | Human-Directed Improvement Proposals(Entitled Author / Admin) | §78.6 | Accept a directly-submitted Proposal Fact Unit from an entitled identity, bypassing telemetry-based identification but not the downstream sandbox/pre-screen/review stages. | Structural rule: A human-submitted proposal enters the same evaluation and approval gate as a telemetry-identified one; bypassing telemetry does not bypass review.

50 | Fact Unit Schema | Chapter 3 / Appendix A | Every knowledge unit is instantiated with a fixed field set regardless of source modality or FactType. | FUID is a deterministic function of (content, source, timestamp).

51 | Multimodal Ingestion Pipeline | Chapter 4 | Identify which of the eight defined modalities the incoming content matches. | Detection selects the modality with the highest match score among the eight defined bits.=INDEX({"Text","Structured","Image","Audio","Video","Sensor","KG","Code"},MATCH(MAX({0.1,0.9,0.2,0.05,0.1,0,0,0}),{0.1,0.9,0.2,0.05,0.1,0,0,0}))

52 | Classification Fallback | §4.2.1 | Attempt best-effort resolution to the nearest of the eight modalities; halt with a stated reason where no reasonable resolution exists. | Resolution proceeds only if best-match confidence clears a minimum threshold; otherwise halt.=IF(0.3>=0.5,"Resolve to Nearest Modality","Halt - Unclassified")

53 | Provenance Ledger | Chapter 5 | Append each consequential operation as a node in a Merkle DAG; compute a keyed checksum sufficient to detect a Safety Substrate bypass. | A checksum mismatch indicates the recorded safety evaluation does not match what actually ran.=IF("A1B2C3"="A1B2C3","Checksum Valid","Bypass Detected")

54 | Ontology Mesh | Chapter 6 | Build relationship edges between Fact Units only where the underlying facts actually connect, not from statistical co-occurrence. | An edge is created only if a real, named relationship type applies.=IF(COUNTIF({"causal","comparative","shared-entity"},"causal")>0,"Edge Created","No Edge")

55 | Neuromorphic Cognitive Layer / Attractor Network | Chapter 7 | Given an initial candidate activation set, iteratively update each activation using neighboring weighted activations until convergence. | $a_i(t+1) = \text{sign}(\sum_j W_{ij} \times a_j(t) - \theta_i)$, reliable capacity ? 0.14N stable patterns for N units.=SIGN(SUMPRODUCT({0.5,0.3,-0.2},{1,1,0})-0.1)

56 | Safety Substrate | Chapter 8 | Evaluate every operation against SafetyFlags-derived rules at each mandatory, synchronous enforcement point. | An operation proceeds only if it passes every one of its applicable gates.=IF(AND(TRUE(),TRUE()),"Proceed","Blocked at Gate")

57 | Semantic Transport Layer | Chapter 9 | Route Fact Units between subsystems based on FactType and subscriber-declared filter predicates. | A subscriber receives a Fact Unit only if it matches every declared filter predicate.=IF(AND(TRUE(),TRUE()),"Delivered to Subscriber","Filtered Out")

58 | Predictive Inference Engine (PIE) | Chapter 10 | Integrate four input streams into a probabilistic forward projection of future knowledge states, with confidence intervals. | Illustrates the trend-projection component only — output is an interval estimate, not a single deterministic value.=TREND({10,12,14,16},{1,2,3,4},5)

59 | Scenario Graph | Chapter 11 | Represent a hypothetical as a Knowledge State Snapshot delta rather than copying the full graph; prune low-probability branches against a threshold. | A branch is pruned when its probability falls below the configured threshold.=IF(0.03<0.05,"Prune Branch","Retain Branch")

60 | Narrative Graph | Chapter 12 | Construct typed edges between Fact Units to mediate response composition; flag narrative gaps where no supporting edge exists. | Structural rule: Every Narrative Graph edge references at least one Fact Unit; an edge without such a reference is not constructed.

61 | QueryBook Format (QBF) | Chapter 13 | Serialize Fact Units to a binary frame format permitting resynchronization to the next valid frame after corrupted data. | Structural rule: A frame boundary is recoverable without reparsing the preceding frame, so a corrupt frame costs one frame rather than the stream.

62 | Storage Architecture | Chapter 14 | Tier Fact Units across storage layers by access-recency/frequency; index for similarity and exact-match retrieval. | A Fact Unit is promoted to a hotter tier once its access count exceeds a threshold.=IF(42>20,"Promote to Hot Tier","Remain in Current Tier")

63 | AGI Extension Modules (AEM) | Chapter 15 | Synthesize a new derived insight from multiple existing Fact Units where no single Fact Unit directly answers a query. | A synthesized insight carries a confidence trajectory distinct from, and generally lower than, any single contributing fact.=MIN(0.8,0.75)

64 | Multilingual Regeneration Engine (MRE) | Chapter 16 | Regenerate output in a target language directly from the language-independent Narrative Graph and SemanticHash-linked Fact Units. | Back-translation of a regenerated output must yield a SemanticHash within the configured distance of the original (§16.5).=IF(0.04<=0.05,"Round-trip within tolerance","Regeneration rejected")

65 | Cross-Domain Rule Engine (CDRE) | Chapter 17 | Evaluate a probabilistically-weighted, temporally-scoped rule set incrementally, maintaining a persistent Conflict Index. | A contradiction entry is added to the index when two active rules produce incompatible conclusions.=IF("true"<>"false","Contradiction Logged","Consistent")

66 | QBQL (execution floor) | Chapter 18 | Execute a declarative query as a read-only graph traversal with no mutation or control-flow capability. | Pure function of (query, state) — no hidden state carried between evaluations.

67 | QBPL (bounded program layer) | §48.3, §48.5 | Compile to one of nine fixed primitives; REPEAT-UNTIL-BOUND enforces a hard iteration ceiling at compile time. | Primitive set is finite (|P| = 9); non-Turing-complete by construction.=MIN(150,100)

68 | Clarification Generation Engine | Chapter 37 | Detect genuine referential ambiguity or a repetition/frustration signal; generate an anchored clarifying question rather than guessing. | Ambiguity is flagged when top-candidate confidence scores are too close to separate.=IF(ABS(0.51-0.49)<0.1,"Ask Clarifying Question","Proceed")

69 | Natural Language Interaction Layer (NLIL) | Chapter 50 | Select vocabulary, structural format, and explanatory depth for phrasing based on inferred audience profile; the underlying fact is never altered. | The selected audience profile must resolve to a defined expertise level; the underlying Fact Unit is never altered by the selection.=IF(AND(4>=1,4<=7),"Audience profile applied; fact unaltered","Refused - expertise level out of range")

70 | Fact Universe Architecture | Chapter 51 | Define the structural relationship between the Fact Graph, Ontology Mesh, and Provenance Ledger as coordinated views over one substrate. | Structural rule: The Fact Graph, Ontology Mesh and Provenance Ledger are addressable as one structure over which a single scope description resolves.

71 | Adaptive Response Engine (ARE, pipeline) | §52.8 | Analyze the candidate Fact Units retrieved for a specific query. | Structural rule: Response profile resolution completes before generation begins; the directive is fixed for the response and is not renegotiated mid-generation.

72 | World Aware System Model (WASM) | Chapter 53 | Maintain Object, Relation, Physics, and Affordance layers as Fact Units, queryable by IQR for scene instantiation. | Structural rule: Object, Relation, Physics and Affordance layers are maintained as Fact Units and are queryable on the same terms as any other record.

73 | Conceptual Memory Layer (CML) | Chapter 54 | Persist long-term, stable world-model abstractions distinct from the transient Cognitive Context Window. | Structural rule: An abstraction persisted to long-term store retains the generalized concept and discards the episode it derived from.

74 | Sensory Abstraction Pipeline (SAP) | Chapter 55 | Reduce raw perceptual input to structured concept-level Fact Units suitable for WASM/CML consumption. | Structural rule: Raw perceptual input is reduced to concept-level records before admission; no raw episode is admitted to the concept store.

75 | VCUM Semantic Colorization Engine (SCE) | §44.3.1 | Infer color for content lacking reliable existing color information from material class, lighting, and historical priors. | Deterministic: identical inputs always yield the identical inferred color.

76 | VCUM Agent Specifications | Chapter 45 | Define the per-agent task decomposition within the VCUM restoration pipeline. | Structural rule: Each restoration agent performs exactly the task its specification declares; an agent performs no task outside its declared decomposition.

77 | Live Fact Graph | §49.2 | Maintain a continuously-updating layer of the Fact Graph reflecting the most recent ingested/inferred state. | Structural rule: The live layer is read as an overlay on the committed layer; a provisional provenance reference is finalized on durable write and never altered thereafter.

78 | Continuous Query Engine (CQE) | §49.5 | Register a WATCH program with a trigger clause; re-evaluate on every relevant update, re-checking Prime Directive compliance at each firing. | Re-evaluation happens on every firing, not once at registration — the count of checks equals the count of firings. $=5*1$

79 | Structural Groundedness Enforcement | §16 / Narrative Graph | Restrict candidate output vocabulary to terms derivable from Narrative Graph edges traceable to supporting Fact Units; reject untraceable propositions. | A proposition is rejected unless it traces to at least one supporting Fact Unit. $=IF(1>=1,"Include in Output","Reject - Ungrounded")$

80 | Offline Synchronization Layer | Chapter 68 | Select a sync mode based on connectivity; reconcile divergent offline/online states via version-vector comparison on reconnection. | Reconciliation is required whenever the offline and online version vectors differ. $=IF(4<>7,"Reconciliation Required","Already in Sync")$

81 | Portal / Role Model(Platform Admin) | Chapter 65 | Resolve an identity's accessible capabilities as the union of held Role(s) plus active capability grants, each evaluated independently. | Access to a capability is the logical OR of role-based and grant-based access, never merged into a combined elevated set. $=IF(OR(FALSE(),TRUE()),"Access Granted","Denied")$

82 | Improvement Proposal Pipeline(Platform Admin) | Chapter 66 | Telemetry-driven or human-directed identification of a subsystem-scoped improvement candidate. | Structural rule: A candidate is scoped to a single subsystem; a candidate targeting a member of the permanently-excluded set is refused at identification.

83 | QueryBook AI Evolution (System Agents)(Platform Admin) | Chapter 78 | A System Agent's Candidate Identification is bounded to its assigned subsystem; resulting proposals enter the identical pipeline as any human-originated candidate. | An agent cannot approve its own or another agent's proposal. $=IF("agent_3"="reviewer_id","Self-Approval Blocked","Independent Review Required")$

84 | Formal Verification Layer(Platform Admin) | Chapter 79 | Generate a machine-checked proof of a precisely-stated property for a QBPL/WATCH program or proposal. | Property truth is decidable over the bounded QBPL primitive set. $=IF(TRUE(),"Property Proven","Proof Failed - Property False")$

85 | License and Entitlement Management(Platform Admin) | Chapter 67 | Map an organizational license to Role-scoped seats; check seat and entitlement status before granting access. | Access requires both an available seat and the specific entitlement. $=IF(AND(TRUE(),TRUE()),"Access Granted","Denied - Missing Seat or Entitlement")$

86 | QBCL-CMS (Configuration Management)(Platform Admin) | Chapter 47 | Parse and apply platform-level configuration changes through a defined configuration language, versioned like any other change. | Structural rule: A configuration change is versioned and recorded on the same terms as any other change; an unversioned change is not applied.

87 | Source Connector Registry(Platform Admin / Author) | §49.3.3 | Register an external system against a fixed contract; evaluate every connector under the identical Trust Boundary regardless of name. | Trust level is identical for every registered connector — not a function of the platform's name or popularity. $=IF("Zoom"<>"Trust Level","Standard Trust Boundary","Standard Trust Boundary")$

88 | Robot Memory Domain (RMD)(Integration Operator) | Chapter 56 | Ingest robot/agent sensor data into Fact Units; deliver structured knowledge back; support WATCH-driven continuous behavior. | Structural rule: Sensor data enters through the ordinary ingestion path and structured knowledge is returned; no raw sensory episode from an agent is retained.

89 | Template Debug Mode(Platform Admin) | Chapter 73 | Pause execution at existing pipeline-stage boundaries, exposing Fact Unit bindings, Safety results, and Slice measurements at each pause. | Structural rule: Instrumentation pauses at existing pipeline-stage boundaries of an actual execution; it introduces no stage and simulates none.

90 | Processor Slice Measurement Model(Platform Admin) | Chapter 69 | Classify and log the compute cost of every operation into a defined Slice category; Safety Gate and Recovery Slices are prioritized above all others. | Safety Gate priority always outranks any other Slice category when contended. $=IF("SafetyGate"="SafetyGate","Highest Priority","Standard Priority")$

91 | Validation and Benchmarking(Platform Admin) | Chapter 77 | Define a reproducible benchmark for an existing design-target figure; run it and mark the figure validated or retract it. | A design-target figure is validated only if the measured result falls within its stated tolerance. $=IF(ABS(102-100)<=5,"Validated","Retracted - Out of Tolerance")$

92 | Platform Internationalization(Platform Admin) | Chapter 76 | Apply locale-specific presentation without altering Safety Substrate or Prime Directive evaluation in any locale. | Safety evaluation outcome is identical regardless of locale. $=IF(TRUE(),TRUE(),"Same Safety Outcome Every Locale","N/A")$

93 | Moderation Actions(Author / Instructor / Platform Admin) | §64.4 | Apply a flag/hide/lock/archive action to target content, logging moderator identity and reason; reversible only via Appeals. | An action is applied only if the flag count for the content reaches the configured threshold. $=IF(4>=3,"Action Applied","Below Threshold - No Action")$

94 | Deployment Architecture(Platform Admin) | Chapter 19 | Roll out an approved change in stages rather than all at once. | Structural rule: Progression to each successive rollout stage is conditioned on the recorded human approval; no stage advances automatically.

95 | Optional Blockchain Ownership-Verification Connector(Publisher) | §49.3.3.8 | On a resale/transfer request for a title with this connector registered, dispatch a verification query to the external service; QueryBook remains the requesting party only. | Structural rule: A verification request is dispatched only for the specific title and only to the requesting party; no

ownership record is retained by the system.

96 | Live Interaction Voice Signal Analysis(Reader) | Chapter 87 | If user has opted in for the session, extract prosodic features from live audio and feed as an optional trigger to the Clarification Generation Engine; discard on session end unless separately retained. | Analysis runs only where session consent is active and the prosodic salience clears the trigger threshold.=IF(AND(TRUE(),0.71>=0.65),"Clarification trigger raised","No trigger")

97 | World-View Model (WVM)(System) | §42.2.1 | Maintain the native ontological, causal, and temporal structure underlying QueryBook's own reasoning, distinct from WASM's world-model domain and from any external hypergraph substrate. | Structural rule: The ontological, causal and temporal structure is maintained natively; no external model is required to express it.

98 | Contextual Commerce Layer(Reader) | Chapter 88 | After grounded answer is fully computed, resolve query's concept_id; match against registered commerce sources; render in separate labeled region. | priority(Commerce Slice) < priority(Safety Gate Slice) for all
t=IF(AND(A2="grounded_complete",B2="concept_resolved"),"eligible_for_commerce_match","blocked")

99 | Commerce Processing Slice and Selection Weighting(System) | §88.8–§88.9 | Meter each commerce operation as a bounded Slice; where multiple candidates match a resolved concept_id, select via a weighted combination of relevance, source priority, and bid, applied only to already-matched candidates. | score = w1*relevance + w2*source_priority + w3*bid,
?w=1=SUMPRODUCT({0.4,0.3,0.3},{0.9,0.7,0.6})

100 | Commerce Optimization Proposals(System Agent / Human Reviewer) | §88.10 | Route any proposed change to commerce targeting or bid weighting through Chapter 66's full Simulation Sandbox, Pre-Screening, and Human Review Gate; no autonomous deployment path exists for commerce optimization. | Structural rule: A proposed commerce change enters the full improvement-proposal gate; no commerce parameter changes without a recorded human approval.

101 | Merchandise Commerce and Fulfillment Handoff(Publisher / Reader) | §88.12 | On merchandise purchase intent, dispatch a scoped referral to a registered external commerce platform via the Source Connector Registry; QueryBook never captures payment data, processes payment, or acts as fulfillment operator. | Structural rule: A merchandise referral is scoped to the referring party alone; the system takes no custody of the transaction.

102 | System Identity and Positioning(Operator) | Chapter 86 | On any external summary request, resolve claims against §86's stated differentiators and corrections rather than unverified external description. | Structural rule: An external summary is resolved against the stated differentiators and corrected where it overstates them; no uncorrected overclaim is issued.

103 | Native QueryBook Reader(Reader) | Chapter 89 | Render EPUB 3.3 content as a conformant Reading System; QueryBook interactions triggered by ignorable data-qb-* attributes on otherwise-standard markup. | Structural rule: Reading-system conformance is maintained independently of QueryBook interaction; interaction adds capability and removes no conformant behaviour.

104 | Deployment Modes and Domain Packages(Operator) | Chapter 90 | Run identical logical services across Cloud SaaS, Local Server, and Single Device modes, differing only in hosting topology, not specification. | Structural rule: The same logical services execute in every deployment mode; a mode changes where a service runs and not which service owns a decision.

105 | Personal Object Location(User) | Chapter 91 | On consented live feed, classify inanimate objects/materials against WASM Object Layer; discard any person/animal-classified result; represent as session-scoped record. | Structural rule: Classification is restricted to inanimate objects and materials; a person- or animal-classified result is excluded from the result set entirely.=IF(OR("person"="person","person"="animal"),"discard","surface_result")

106 | Sensory-Annotated Content Playback(Reader) | Chapter 92 | On content with a sensory profile attribute, play audio/visual/device-haptic output via device-native APIs only; opt-in for audio/haptic, default-on for passive visual effects. | Structural rule: Sensory output plays only where the content carries an authored sensory profile; no output is triggered by an inferred state.

107 | Audio Narration(Author / Reader) | Chapter 93 | Select author's own recorded narration if it exists for requested content; else use generic non-cloned synthetic voice profile; never approximate a specific real voice absent that person's own recording. | Structural rule: The author's own recorded asset is selected where it exists; otherwise a generic profile with no correspondence to a specific person is used.=IF(TRUE()=TRUE(),"use_recorded","use_generic_synthetic")

108 | NFT Integration Layer(Publisher) | Chapter 94 | Mint a token representing an already-established license-scoped entitlement via a registered external minting service, transferred directly to the rights holder's own wallet; resolve every read against the entitlement record, which always governs on conflict. | Structural rule: A token is minted only after the entitlement exists; where token state and entitlement record diverge the entitlement record is authoritative.

109 | Complementary Learning Systems Formalization(System) | Chapter 95 | Name session-scoped records as the fast/episodic system and the permanent Fact Graph as the slow/consolidated system; consolidation requires explicit user election, never automatic replay. | Structural rule: The correspondence names existing mechanisms and creates none; session-scoped records and the permanent graph are unchanged by the naming.

110 | Forensic Analysis of External AI System Outputs(Researcher) | Chapter 96 | Check a claim in an external AI's output against QueryBook's own Fact Graph; classify across 6 categories; self-referential language cataloged as observed text only, never a sentence determination. | Structural rule: Each claim of an external output is marked supported-with-citation or

unsupported; no output asserts a determination about that system's internal state.=IF(TRUE()=TRUE(),"grounded","unverified")

111 | Hypothesis Generation(Researcher / Domain Expert) | Chapter 97 | Generate a hypothesis via AIQG/AEM/Scenario Graph/§10.4 triggers with status=PROPOSED; only an identified human expert may transition status to VALIDATED; no automated path exists. | Structural rule: A hypothesis is created with status PROPOSED and transitions to VALIDATED only on confirmation by an identified human reviewer.=IF("VALIDATED"="VALIDATED","eligible_for_fact_graph","display_only")

112 | Cognitive Map Formalization(System) | Chapter 98 | Name the correspondence between §91.9's lattice indexing and the Attractor Network's Hopfield dynamics; extend deterministic lattice indexing to general Ontology Mesh concept proximity search. | Same hex_coord principle as §91.9, applied to concept space, deterministic

113 | Study and Learning Tools(Student / Reader) | Chapter 99 | Assemble glossary entries, study guides, and staged Tutorial Mode sequences from already-grounded content; schedule flashcards via spaced-repetition interval; cluster concepts via Ontology Mesh structure. | Next spaced-repetition interval is the prior interval scaled by the ease factor where recall quality clears the floor, else reset to one day.=IF(0.8>=0.6,6*2.5,1)

114 | Content Verification and Argumentation Tools(Reader / Researcher) | Chapter 100 | Check claims against grounded Fact Graph via CDRE contradiction logic; surface existing corroboration counts; assemble grounded position arguments via Scenario Graph, never fabricating unsupported content. | Structural rule: A surfaced contradiction cites the records establishing it; no conclusory characterization is generated that is not itself derivable from a record.=IF(TRUE()=TRUE(),"flag_with_citation","no_flag")

115 | Manifold-Based Trend Forecasting(Analyst) | §10.4 | Embed ingested time-series data into a nonlinear manifold; generate a forecast with an explicit confidence interval; render in a labeled region separate from grounded output. | forecast = f(manifold_embedding(time_series)), CI computed from embedding uncertainty=1.96*0.8/SQRT(16)

116 | Geometric Shape Matching / Grid-Cell Indexing(System) | §91.8–§91.9 | Compare candidate contours via the Registration Constraint Graph's three-pass solving using Geometry and Constraint Fact Units; index candidates on a fixed, non-learned hexagonal lattice for deterministic proximity search. | hex_coord = f(x,y), deterministic and fixed by construction=(ABS(1-0)+ABS(0-1)+ABS(-1-(-1)))/2

117 | Dual-Grounded Developmental World Model (The Box Principle)(System) | Chapter 59 | Build sensory-grounded and language-grounded concept streams independently; on co-occurrence of a spoken label and a visual object, confirm semantic consistency, then fuse via a binding edge; do not finalize a concept from one binding event — track repeated encounters via temporal consolidation before stabilizing. | A concept is finalized only after the consolidation count reaches the configured minimum; a single binding event is provisional.=IF(4>=3,"Concept consolidated","Provisional binding retained")

118 | External Sensory Hardware (Smell/Haptic)(Reader) | §92.7 | On explicit user action tied to a specific, authored sensory profile, trigger registered external scent/haptic hardware; never trigger from an inferred signal (e.g., detected emotion) under any configuration. | Structural rule: Output is triggered by an explicit user action tied to an authored profile; a system-inferred trigger is permanently excluded.=IF(AND(TRUE(),TRUE()),"eligible","blocked")

119 | Source Framing Signals(Reader / Researcher) | §100.6 | Surface a source's own documented editorial position and quantifiable citation-composition facts (count, funding overlap) as cited facts; never generate a conclusory bias judgment not grounded in the Fact Graph. | Source composition is reported as a quantified, cited ratio; no conclusory judgment is generated.=IF(3/12>0,"Report composition fact with citation","No composition fact available")

120 | VR-Specific Commerce Presentation(Operator / Reader) | §88.13 | Render matched commercial content in a spatially distinct VR region using the existing concept_id-triggered matching of §88.1; never use gaze, dwell-time, or spatial proximity as a selection signal. | Structural rule: Commercial content renders in a spatially distinct region; no gaze, dwell or attention signal is captured, derived or made available.

121 | Vehicle-Control AI Feed Monitoring(Analyst / Human Reviewer) | §96.8 | Ingest vehicle-control AI telemetry as AnalyzedOutput; check each entry against grounded reference facts (safe-operating envelope, geofence); on confirmed anomaly, generate an expedited, non-blocking alert to a named human reviewer. No control interface exists. | Structural rule: Telemetry analysis is read-only; no control interface to any vehicle system exists or is created.=IF(TRUE()=TRUE(),"generate_alert","log_only")

122 | Biased Competition for Object Saliency(System) | §53.8 | Compute a SaliencyWeight for each concurrently-detected Object Layer node from detection confidence and task relevance, normalized against co-present objects; never remove or exclude a node from the Fact Graph based on its weight. | SaliencyWeight_i = f(confidence_i, relevance_i) / ? f(confidence_j, relevance_j) for all j in scene=IF(SUM(0.9,0.7,0.6)=0,"undefined",0.9/SUM(0.9,0.7,0.6))

123 | Boltzmann-Style Diverse Hypothesis Sampling(Researcher) | §97.7 | Where enabled, run the Attractor Network's energy landscape through a stochastic update rule to sample multiple candidate hypotheses per input; label each as stochastically-generated; every result remains PROPOSED status only, gated by Chapter 97's human validation. | $p_i = 1 / (1 + e^{(??E_i)})$, $? = 1/T = 1/(1+EXP(-1*0.4))$

124 | Developmental Parallel — Verified Correspondence(System / Researcher) | §59.9 | Ground Chapter 59's Cross-Modal Binding and Temporal Concept Consolidation against real developmental psychology citations (Carey & Bartlett 1978; Yu & Smith 2007), stating explicitly where the correspondence is precise and where it is honestly incomplete. | Structural rule: The correspondence grounds existing mechanisms in cited developmental findings; it adds no mechanism and

alters no binding rule.

125 | Ego-Motion Compensation for Non-Stationary Capture(System) | §55.2.1 | Estimate camera ego-motion from the aggregate optical-flow field; subtract it from each region's raw motion vector before Stage 4 object/scene segmentation, improving detection accuracy under significant camera movement. | $v_{\text{compensated}} = v_{\text{raw}} - v_{\text{ego_estimated}}$

126 | Encyclopedia Content Ingestion(Author / Publisher) | §59.10 | Extract discrete factual claims from encyclopedia text as Fact Units under the fact/expression dichotomy (Feist v. Rural, 1991); images remain subject to Chapter 60's standard rights-aware ingestion, never exempted. | Structural rule: A discrete factual claim is extracted while the source's protected expression is not reproduced; accompanying images pass separate rights validation.

127 | Intersensory Redundancy Weighting(System) | §55.2.2 | Where a property is confirmed via temporally-synchronous signals across multiple concurrent modalities, elevate its Stage 3 confidence weighting relative to a single-modality-only property. | $\text{confidence_amodal} > \text{confidence_single_modality}$ when synchrony confirmed= $\text{IF}(0.78 > 0.62, \text{"Cross-modal weighting applied"}, \text{"Single-modality confidence retained"})$

128 | Object Permanence — Precise Correspondence(System) | §55.2.3 | Maintain a single Object Layer node's continuity when the object is fully occluded or temporarily unobservable within the same session, per Piaget's real definition; introduces no cross-session tracking. | Structural rule: A single object node retains continuity across full occlusion within a session; occlusion does not create a second node.

129 | Additional Disambiguation Heuristics for Cross-Modal Binding(System) | §59.11 | Apply gaze-direction inference, whole-object bias, and mutual exclusivity as evidence-overridable preferences during Cross-Modal Binding's co-occurrence resolution, alongside — never replacing — existing semantic-consistency confirmation. | Structural rule: Each disambiguation heuristic is evidence-overridable; a heuristic never overrides a directly-observed binding.

130 | Live-Session Erasure(Reader) | §36.7.1 | On user action, immediately clear the live Cognitive Context Window transcript, client- and server-side; log a Provenance Ledger entry recording that an erasure occurred without retaining the cleared content itself. | Structural rule: Clearing is immediate at client and server and produces no condensed retained form; the ledger entry records the erasure and none of its content.= $\text{IF}(\text{"clear_now"} = \text{"clear_now"}, \text{"transcript_cleared_immediately"}, \text{"no_action"})$

131 | Retention Policy (Retain-by-Default, Erasure as Option)(Reader / Operator) | §36.7.3, §36.7.5 | Retain SAFU records automatically at session end by default; provide session/work/account-level erasure (§36.7) and live-session erasure (§36.7.1) as fully available, explicit user-initiated options at any time. | Structural rule: Retention is the default and erasure is an explicit user action at session, work and account scope; neither is silently applied.= $\text{IF}(\text{"retained"} = \text{"retained"}, \text{"erasure_available_as_option"}, \text{"n/a"})$

132 | Verified Neuromorphic Correspondence Methodology(System / Researcher) | Chapter 101 | For a candidate neuroscience correspondence: (1) verify against primary publication, (2) check for implicit existing correspondence, (3) extract only the deterministic principle, excluding trained components, (4) extend existing mechanism over new invention, (5) formalize under a named, cited section. | Structural rule: A correspondence is adopted only after external verification, an implicit-correspondence check, deterministic extraction, extension over invention and named formalization.

133 | Legal Exposure Minimization Through Structural Compliance(Operator) | Chapter 102 | Reduce legal exposure via genuine regulatory compliance, structural non-custody (NFT, merchandise), rights discipline, protective audit trail, and consistent documented retention policy — never by reducing what is lawfully producible under a valid order. | Structural rule: Exposure is reduced through compliance, non-custody, rights discipline and consistent retention; never by reducing what is lawfully producible.

134 | Author Portal Book Upload(Author) | §61.1.1 / Ch. 60 | Receive an author-submitted work at the web portal bound to a verified author identity; validate rights status; queue the work for ingestion; return a receipt referencing the queued job. Ingestion does not begin until rights validation returns cleared. | Ingestion proceeds only where rights validation has returned a cleared status.= $\text{IF}(\text{TRUE}(), \text{"Queued for ingestion"}, \text{"Rejected - rights unvalidated"})$

135 | Probable Question Generation (PQG) | Ch. 41.6.3 / Ch. 10 | At ingestion, and before any query is received, traverse each committed Fact Unit together with its Ontology Mesh neighbours and generate candidate questions that those Fact Units answer; score each candidate by a coverage measure over the Fact Units it draws on; retain only candidates clearing the configured floor. | A candidate question is retained in the Question Index only where its coverage score clears the configured floor.= $\text{IF}(0.71 > 0.6, \text{"Retained in Question Index"}, \text{"Discarded"})$

136 | Pre-Computed Answer Set | Ch. 41.6.3 | For each retained question, assemble the grounded answer at ingestion time from exactly the Fact Units the question was derived from; store it as a Pre-Computed Answer Unit carrying a PRECOMPUTED_ANSWER provenance tag and the FUIDs relied upon, so a served answer is traceable to the same degree as a live one. | Pre-computed answer confidence is bounded by the least confident contributing Fact Unit.= $\text{MIN}(0.88, 0.79)$

137 | Question Index Reuse and Invalidation | Ch. 41.6.3 / §49.2 | On query, match against the Question Index before live traversal; serve the stored answer where match similarity clears the threshold and no contributing Fact Unit has been revised since computation; otherwise fall through to live retrieval. Revision of any contributing Fact Unit marks dependent entries stale and excludes them from service until recomputed. | A stored answer is served only where match similarity clears threshold and every contributing Fact Unit is unrevised since computation.= $\text{IF}(\text{AND}(0.83 > 0.75, \text{TRUE}()), \text{"Serve pre-computed answer"}, \text{"Recompute live"})$

138 | Author Supplemental Knowledge Layer(Author) | §82.1 | Accept author-supplied supplemental context at or after ingestion; convert it to Fact Units tagged AUTHOR_SUPPLEMENTAL with the author as attributed source; make those Fact Units available to every reader entitled to the work; regenerate the Question Index entries whose contributing Fact Units are affected. | Supplemental Fact Units carry author-attributed source reliability and trigger regeneration of every affected index entry.=IF(COUNTIF({"FU-101","FU-102"},"FU-101")>0,"Regenerate affected entries","No regeneration required")

139 | Attractor Load Regime Governor | §7.6.3 / §7.6.4 | Compute the Attractor Load Ratio $\alpha = P/N$ before admitting a further stable configuration; admit only where the resulting α stays below the configured operating ceiling; refuse admission and trigger EWC consolidation where it would not. Operation above the conservative ceiling requires a recorded deployment election; operation at or past the Retrieval-Phase Boundary is structurally refused. | $\alpha = P/N$ must remain below the configured operating ceiling, itself strictly below the Retrieval-Phase Boundary $\alpha_c \sim 0.138$ (AGS 1985, 1987).=IF(1000/8192<0.05,"Admit - conservative regime","Consolidate before admitting")

140 | Sampling Criticality Control | §97.7.1 | Expose the inverse-temperature parameter of the hypothesis-diversity sampler as an operator-set control with ordered, intermediate and disordered regimes; record the value in effect with the setting operator's identity so a hypothesis set is reproducible. No setting admits sampler output to the retrieval or generation path, and no process self-tunes the parameter. | Selection probability $p = 1 / (1 + e^{-(\beta * \Delta E)})$; β sets distance from the indifference point.=1/(1+EXP(-0.5*0.4))

141 | Concept Attribute Inheritance | §54.6.1 | Where a concept is connected to a superordinate concept by an IS-A edge and that superordinate carries a HAS-PROPERTY attribute, assign the attribute to the subordinate at traversal time, tagged with its originating concept and IS-A depth; do not write it back as stored state, so a revision at the superordinate propagates on the next traversal. | Inherited attribute confidence is the product of the traversed IS-A edge confidences and the attribute's own confidence, bounded above by the least of them.=MIN(0.97,0.99*0.95)

142 | Exception Override Resolution | §54.6.2 | Where a subordinate concept asserts an attribute conflicting with one it would inherit, the direct assertion governs for that concept and its subordinates while the superordinate's attribute is unmodified and still inherited by siblings; conflicts inherited at differing depths resolve to the lesser depth; conflicts at equal depth are refused and recorded in the contradiction index. | The attribute asserted at the lesser IS-A depth governs; equal-depth conflicts are not resolved and both values are reported.=IF(3>1,"Subordinate assertion governs","Inherited value retained")

143 | Inverse Edge Traversal | §54.6.3 / Appendix A | Traverse a typed edge in reverse by resolving the InversePredicate declared in the predicate schema at traversal time, storing no second edge; where no inverse is declared, refuse reverse traversal and report an absent inverse declaration rather than inferring a reverse reading. | Reverse traversal is available only where an InversePredicate is declared; stored edge count equals asserted relation count.=IF(COUNTIF({"parentOf","childOf"},"childOf")>0,"Reverse traversal available","Refused - no inverse declared")

144 | Wording-Operative Span Preservation | §16.7 | Where a Fact Unit is typed QUOTATION or carries a wording-operative marking, reproduce the source span in its source language and wording rather than regenerating it, accompanied in a target language by a regenerated gloss marked non-operative; exempt only the reproduced span from Stage 6 groundedness rewriting; withhold reproduction where Chapter 60 rights validation does not permit it and deliver the gloss alone. | A wording-operative span is reproduced, never regenerated; the accompanying gloss carries a non-operative marking.=IF(COUNTIF({"QUOTATION","CLAIM","INSTRUCTION"},"QUOTATION")>0,"Reproduce verbatim + non-operative gloss","Regenerate normally")

145 | Idiom Detection and Equivalence Mapping | §16.8 | Detect at ingestion a span whose compositional reading diverges from its attested reading beyond a configured threshold; record the attested meaning with an idiom-derived marking and the source span; realize it in a target language using a registered equivalent for the same attested meaning, and where none is registered render the meaning literally with an explicit note rather than selecting an equivalent by surface similarity. | An idiom mapping applies only where divergence clears the configured threshold and a target-language equivalent is registered.=IF(AND(0.82>=0.6,COUNTIF({"es:tomar-el-pelo"},"es:tomar-el-pelo")>0),"Apply registered equivalent","Render literal meaning with note")

146 | Linguistic Resource Registry | §4.9.1–§4.9.2 | Register every externally-sourced linguistic resource with identifier, version, licensor, licence instrument, permitted-use scope, redistribution permission, attribution requirement and ingestion mode before any extraction or realization stage consults it; refuse registration where the licence does not cover the declared use; propagate derivative conditions to artifacts derived from the resource, marking an artifact non-redistributable where two registered resources impose conflicting conditions. | A resource is consulted only where its registered permitted-use scope encompasses the operator's declared use.=IF(COUNTIF({"commercial-permitted","derived-only","refused"},"commercial-permitted")>0,"Register for full ingestion","Restrict or refuse")

147 | Derived-Artifact-Only Extraction | §4.9.3 | For a resource registered in derived-artifact-only mode, compute and retain aggregate quantities (frequency, collocation strength, part-of-speech distribution, morphological paradigm) while retaining no span of the resource's text in any Fact Unit, index or cache and reproducing none in output; verify by confirming the retained-span count is zero, enforcing the Feist fact/expression boundary already applied at §59.10. | Retained source spans attributable to a derived-artifact-only resource must equal zero.=IF(0=0,"Compliant - statistics retained, no source span","Refused - source span retained")

148 | External System Behavioural Probing | §96.9 | Submit controlled inputs to a registered external system and measure behaviour rather than content: repeat an identical input and score the proportion of outputs sharing a semantic fingerprint; reinject an output as input to a ceiling and classify the fingerprint sequence as converged, periodic, drifting or unresolved; perturb an input without changing its fingerprint and score output invariance. Every result carries QueryBook's own measured value on the identical probe or is incomplete. | A probe result is reportable only where a reference value measured on the identical probe is present.=IF(AND(8/10>=0.5,1=1),"Consistency measured against reference","Incomplete - reference value absent")

149 | Analysis Quarantine | §96.10 | Exclude AnalyzedOutput and ProbeResult Fact Units, and anything derived from them, from the scope description the Slice Resolver constructs for any query not directed at the analysis itself, so they are not traversable rather than filtered afterwards; exclude them equally from the question index and the Conceptual Memory Layer. | An analysis record is structurally ineligible as a grounding source; exclusion occurs at slice construction, not after traversal.=IF(COUNTIF({"AnalyzedOutput","ProbeResult"},"ProbeResult")>0,"Excluded from grounding slice","Eligible as grounding")

150 | Attestation Horizon Constraint | §103.3 | Admit a Fact Unit to a pre-event snapshot only where its T_attested (the source's documented creation date) precedes the event time less the horizon; do not consult TemporalAnchor for this test; exclude any record whose T_attested cannot be established from documented source metadata; enforce at Slice Resolver scope construction so an inadmissible record is not traversable rather than filtered afterwards. | Admissible iff T_attested < T_event - horizon; contested dates resolve to the latest candidate, so uncertainty resolves toward exclusion.=IF(DATE(1998,3,1)<DATE(2001,9,11)-90,"Admissible to pre-event snapshot","Excluded - attested after cutoff")

151 | Retrospective Forecast Scoring | §103.5 | Run Chapter 10 and §10.4 forecasting against the constrained pre-event snapshot alone within a Scenario Graph branch; record predicted outcome, assigned probability, predicted interval and every admitted FUID; on comparison with the recorded outcome, construct a PredictionError Fact Unit and derive accuracy by horizon and calibration by confidence band. | Mean squared error between assigned probability and realised binary outcome across the scored corpus.=((0.7-1)^2+(0.4-0)^2)/2

152 | Calibration Subject Class Gate | §103.6 | Admit only an institution, organisational process, system or aggregate group as the subject of a calibration event record; refuse construction of a PredictionError Fact Unit whose subject resolves to an individual natural person, at construction rather than by access restriction; report a demographic correlation as a property of the institutional process, cited, never restated as a property of individuals. | Construction is refused where the resolved subject class is an individual natural person.=IF(COUNTIF({"institution","system","group"},"individual")>0,"Refused - individual subject","Admissible - institutional subject")

153 | Event Type Catalogue | §104.2 | Register each event category with an immutable id, version, poly-hierarchical Ontology Mesh placement, a semantic definition stating both coverage and exclusions, and a MachineDefinition naming required, optional and forbidden fields; require an EVENT_DESCRIPTION Fact Unit to reference a registered id rather than a free-text category; refuse construction where the MachineDefinition is unsatisfied; permit a later version to narrow a required-field set but never widen it. | A record is admitted only where every field required by its event type's MachineDefinition is present.=IF(COUNTIF({"StartTime","EventTypeId","ActorRef"},"EventTypeId")>0,"Admit - required field present","Refused - MachineDefinition unsatisfied")

154 | Environment Vector Normalization | §104.3 | Store a world-state snapshot once as a content-addressed record carrying its own TemporalAnchor, typed Ontology-Mesh-resolved condition keys, confidence, provenance and T_attested; reference it from Fact Units through a reserved AnnotationSet key so no schema field is added; converge identical condition sets on one vector with two provenance branches; correct by supersession so every referencing record resolves to the corrected state on next traversal. | Identical canonicalized condition sets yield an identical content hash and therefore one vector, not two.=IF(COUNTIF({"a1f3c7","b2c9e1"},"a1f3c7")>0,"Reference existing vector","Create new vector")

155 | Causal Edge Temporal Validation | §104.4 | Admit a CAUSES edge only where the source TemporalAnchor does not begin after the target's begins; refuse otherwise and record the attempted assertion in the Chapter 17 contradiction index; permit concurrent and overlapping causation; never promote TEMPORAL_FOLLOWS to CAUSES automatically; express chains as Bundle membership with chain length, root and outcome computed at query time rather than stored. | CAUSES admitted iff T_start(source) <= T_start(target); duration is derived at query time, never stored.=IF(DATE(2020,3,1)<=DATE(2020,3,15),"CAUSES edge admitted","Refused - effect precedes cause")

156 | Bayesian ConfidenceScore Posterior Update | §41.6.2.1-.2 | Hold ConfidenceScore as Beta parameters alpha (supporting evidence) and beta (contradicting evidence) rather than a scalar; report CS as alpha/(alpha+beta); update in closed form by adding the observing source's reliability weight to whichever parameter the evidence supports, so revision magnitude falls as accumulated evidence grows without any rule conditioned on it; decay shrinks both parameters toward the prior, widening the credible interval while preserving the mean. | CS = alpha/(alpha+beta); posterior by conjugate addition alpha' = alpha + w*s, beta' = beta + w*f; order-independent.=18/(18+2+1)

157 | Prior-Strength Resistance Check | §41.6.2.3 | Initialise Beta parameters for a concept node from ConceptStabilityWeight, which sets the pseudo-count n0 = alpha+beta while the current confidence sets the ratio; a high-CSW node therefore requires multi-source contradicting evidence to revise, which is the behaviour §41.4.5 asserts and the additive rule does not produce. | Revision from a single contradiction against a strong prior must fall below the increment the additive

rule would have applied.=IF(0.9-18/(18+2+1)<0.05,"Strong prior resists single contradiction","Revision exceeds tolerance")

158 | Precision-Weighted Error Propagation | §41.6.4.2 | Compute an error signal's precision as inverse variance from the Beta parameters already held under §41.6.2; propagate a contradiction upward to World Brain concept attributes by a gain equal to the error's share of total precision, so a poorly-supported observation barely revises a settled concept while a well-established one substantially revises a weakly-held concept; replaces the ConfidenceScore-differential magnitude of §41.5.1. | $\text{gain} = \text{var_prior} / (\text{var_prior} + \text{var_error})$, equivalently $\text{pi_error} / (\text{pi_error} + \text{pi_prior})$; $\text{var} = \text{CS}(1-\text{CS})/(\alpha+\beta+1)$.=((180/200)*(20/200)/201)/(((180/200)*(20/200)/201)+((1/2)*(1/2)/3))

159 | Low-Precision Error Suppression | §41.6.4.3 | Withhold model revision entirely where the computed gain falls below a configured floor rather than applying an arbitrarily small revision; record each withheld contradiction in the Chapter 17 contradiction index so a pattern of individually-insignificant errors against one attribute stays queryable. | Revision is withheld where $\text{gain} < \text{configured floor}$; withheld contradictions are retained, not discarded.=IF(((180/200)*(20/200)/201)/(((180/200)*(20/200)/201)+((1/2)*(1/2)/3))<0.05,"Low-precision error suppressed","Model revision applied")

160 | Physical Constraint Enforcement | §53.9.2 | Evaluate every registered physical constraint whose applicability predicate a candidate scene completion satisfies; refuse the completion where the residual exceeds tolerance and regenerate with the implicated field left unspecified, so a completion with an acknowledged gap is returned in preference to a physically impossible one; constraints are evaluated before plausibility scoring and a refused completion never appears in the ranked output at any score. | Admissible iff $|\text{residual}| \leq \text{tolerance}$, tolerance taken relative to the magnitude of the conserved quantity.=IF(ABS((2*1+1*4)-(2*3+1*0))<=0.02*(2*3+1*0),"Completion admitted","Refused - conservation residual exceeds tolerance")

161 | Constraint Residual Computation | §53.9.2 | Compute the expected value of a constrained quantity from the registered closed-form residual expression in SI base units and compare against the completion's asserted value; record the refusal with constraint id, residual, tolerance and implicated field so a wrongly-parameterised constraint is detectable by its refusal rate rather than quietly suppressing correct inference. | Expected position under gravitational constraint $h_1 = h_0 - 0.5*g*t^2$ with $g = 9.80665 \text{ m/s}^2$; residual is the asserted value less h_1 .=-2-0.5*9.80665*0.4^2

162 | Public API Layer(Operator) | §105.1 | Expose seven programmatic interfaces (content, query, rights, layout, assessment, edition, semantic); each resolves a scope description before traversal, no parameter widens it, an out-of-scope record yields the response for a non-existent record, and the layer carries its own Chapter 8 safety gate as an operation layer; calls are metered per calling identity and a call over budget is refused rather than served at reduced priority. | A call is served only where the calling identity's metered budget is not exceeded; refusal is by budget, never by throttling.=IF(120<=150,"Within slice budget; served","Refused - slice budget exceeded")

163 | Plugin Architecture(Operator) | §105.2 | Register a third-party component against a declared capability set on the Source Connector Registry contract; the plugin reaches knowledge only through §105.1 interfaces within the executing identity's scope, holds no direct Fact Graph or Ledger access, is refused at registration if it declares any safety operation, cannot participate in the improvement pipeline, and its output is labelled, quarantined and not citable as grounding; activation and suspension require a named human operator. | Registration is refused where the declared capability set includes any safety-evaluation operation.=IF(COUNTIF({"read-output","declare-capability"},"write-factgraph")>0,"Activate plugin","Refused - capability not declared")

164 | Interface Versioning(Operator) | §105.3 | Version every interface; refuse a request omitting a version rather than defaulting it; deprecate for a configured support interval with a deprecation indication, then retire naming the successor; a field may be added or deprecated within a version, while removal or a change of type or meaning requires a new version; a citation issued under an earlier version remains valid. | A request is served only where the requested version is registered and not retired.=IF(AND(2>=1,2<=3),"Served under requested version","Refused - version retired")

165 | Collaborative Reading(Reader) | §105.4 | Establish a session shared by several readers over one work; present a shared view comprising the intersection of participants' scope descriptions, never the union; a record traversable by one participant and not another is absent and its absence is not reported; annotations keep their own ACLs; queries resolve against the intersection so no participant receives a citation another cannot open. | Shared view size is the intersection of participant scopes, not the union.=MIN(340,275)

166 | AR Overlay Presentation(Reader) | §105.5 | Anchor displayed Fact Units with citations to physical objects in a live camera feed as a presentation surface only, under the unchanged vocabulary constraint; exclude person and animal classifications entirely at any confidence; capture and derive no gaze, dwell, head-pose or attention signal and expose none to commerce or plugins; display no overlay where object identification does not clear its threshold. | A person- or animal-classified result is never anchored, at any confidence and in any configuration.=IF(COUNTIF({"object","person","animal"},"person")>0,"Excluded from overlay","Anchor overlay to object")

167 | Voice Navigation(Reader) | §105.6 | Accept spoken command and query input under explicit session-scoped revocable consent, distinct from prosodic analysis; route the transcription through the Language Command Router separating query from operational intent; request clarification below the confidence threshold rather than selecting an interpretation; discard audio after transcription; derive no voiceprint or biometric identifier; require second-modality confirmation for any state-changing instruction. | Intent is applied only where classification clears the configured confidence threshold under active

consent.=IF(AND(1=1,0.82>=0.7),"Routed as operational intent","Clarification requested")

168 | Spoiler Protection(Reader) | §105.7 | Record a reader progress marker and a narrative position for each Fact Unit of the work; bound the reader's scope description by the marker so a later-disclosed record is not traversable rather than masked in a rendered answer; state that an answer lies beyond the current position without disclosing its content; reader election with an author-configurable default, never applied to an instructor, researcher or author over their own work. | A Fact Unit is traversable only where its narrative position does not exceed the reader's progress marker.=IF(14<=22,"Within reader position; shown","Masked - beyond reader position")

169 | Learner Progress Tracking(Student / Instructor) | §105.8 | Record position reached, items attempted, responses submitted and outcomes scored against the Fact Units the items were generated from; compute mastery as the proportion of objective-attributable items answered correctly against a threshold; bound instructor visibility by a time-scoped class grant so expiry removes it without further action; generate no aptitude inference, forecast or comparative ranking, and admit no progress record as a calibration subject. | Mastery = correct items attributable to the objective / items attempted, against the configured threshold.=IF(17/20>=0.8,"Objective met","Continue practice")

170 | LMS Integration(Operator / Instructor) | §105.9 | Connect a learning management system through the Source Connector Registry under a declared capability set; refuse an undeclared operation at the boundary rather than attempting it; a roster read establishes instructor grant scope and confers no entitlement; a grade write transmits the outcome alone with no Fact Unit, response text or history; record every transmission with connector identity, capability exercised and fields sent. | An operation proceeds only where its capability is declared in the registered connector contract.=IF(COUNTIF({"roster-read","grade-write"},"grade-write")>0,"Grade passback permitted","Refused - capability undeclared")

171 | Multi-Edition Print Layout(Publisher) | §105.10 | Compose a work into paginated print-ready form per edition from common Fact Units through regeneration, so an edition is a rendering rather than a copy; perform rights validation for territory, quantity and interval before composition and carry a reference to it in the composed form; reproduce wording-operative spans verbatim with non-operative glosses marked; compose accessible equivalents at the same production stage; keep pagination deterministic so page references stay valid. | Composition proceeds only where rights validation permits print reproduction in the requested territory, quantity and interval.=IF(AND(1=1,3<=3),"Print export authorized","Refused - rights or edition count")

172 | Certification(Instructor / Student) | §105.11 | Issue a credential on both a measured outcome meeting the configured threshold and an identity established to the issuer's standard; carry as provenance the objectives assessed, items presented, source Fact Units, thresholds applied and issuing identity; permit third-party verification of issuance and non-revocation without disclosing the holder's responses; attest to the measured outcome and nothing further; record revocation rather than deleting. | Issuance requires threshold satisfaction AND identity establishment; neither alone suffices.=IF(AND(0.85>=0.8,1=1),"Credential issued","Withheld - threshold or identity unmet")

173 | Rights and Licensing Portal(Operator / Publisher) | §105.12 | Expose the licence and entitlement mechanism through a portal under the existing role model and time-scoped grants; the portal renders capability and confers none; rights holders view licences, outstanding entitlements and the audit record, operators administer seats and assignment within role scope, grantees hold a read-only view of grants over them; record every action with acting identity and stated reason; a revocation takes effect at the next query execution. | A portal action is available only where the acting role already holds that capability elsewhere.=IF(COUNTIF({"view","revoke"},"grant")>0,"Action available to role","Read-only view")

174 | Hyperedge Attention | §53.7.1 | Where a reasoning operation traverses a hyperedge linking several Fact Units, weight each constituent's contribution by its own ConfidenceScore and its degree of corroboration with the hyperedge's other constituents, rather than contributing equally regardless of reliability. Under §41.6.2 those two quantities are the Beta posterior mean and the accumulated evidence count, so the weight is the product of the two parameters the confidence representation already holds. A weighting refinement over the World Model Domain's existing hypergraph traversal, not a new traversal mechanism. | Constituent weight = $CS_i * (\alpha_i + \beta_i)$, normalized across the hyperedge's constituents.=0.92*20/(0.92*20+0.85*6+0.78*3)

175 | Attractor-Based Hyperedge Completion | §53.7.2 | Complete a partially-observed hyperedge — a geometric contour with observed points and gaps — by applying the Attractor Network's existing convergence dynamics (§7.2.4) to the constituent set as a whole, converging toward the most consistent complete configuration given the observed constituents and their §53.7.1 attention weights. Distinct from WorldModel.Infer, which completes a general scene by Schema, Physics, Affordance and Relation rules: this is geometric, hyperedge-granular and convergence-based. It introduces no dynamical system of its own; it is Claim 6's mechanism applied at hyperedge granularity. | Completion is admitted only where attention-weighted support for the converged configuration clears the configured floor.=IF(0.712>=0.6,"Hyperedge completed","Insufficient support - gap reported")

176 | Domain Transfer Matrix | §106.4 | Assign every ordered pair of domains a permission of ALLOW, DENY or TRANSFORM-REQUIRED; evaluate it at each boundary crossing; treat a pair with no entry as DENY; record every crossing in the Provenance Ledger with source domain, target domain and permission applied. The denials carry the accuracy guarantees: D12 to D4 denied so analysis never grounds an answer, D13 to D4 and D7 denied so commerce never influences retrieval or generation, D9 to D2 denied so no surface bypasses the slice, D11 to D0 denied so governance cannot amend the Prime Directive. | A crossing proceeds only where the matrix entry is ALLOW or a satisfied TRANSFORM-REQUIRED; an absent

entry is DENY.=IF(COUNTIF({"ALLOW","TRANSFORM-REQUIRED"},"DENY")>0,"Transfer permitted","Refused - DENY edge in matrix")

177 | Query/Ingestion Path Separation | §106.3 | Partition domains so the latency-critical query path (D8, D4, optionally D5, D7) and the batch ingestion path (D1, D2, D3, D6, D5) intersect only at D0 and D2; resolve contention at D0 by safety-category priority and at D2 by storage tiering; keep extraction out of the retrieval domain so a retrieval never blocks on an extractor and an extraction never blocks on attractor convergence. | Intersection of the query-path and ingestion-path domain sets must not exceed two (governance and substrate).=IF(2<=2,"Path separation holds","Contention risk - shared domain on both paths")

178 | Domain Slice Priority Ordering | §106.5 | Meter each domain's operations as bounded compute-accounting units carrying that domain's identifier and fix scheduling priority by domain: D0 highest without exception, D13 lowest so contention omits the commercial region rather than delaying the answer, and latency-path domains ahead of batch-path domains; contain each domain's failure to its own capability, with a D0 failure halting the operation rather than allowing it to proceed unevaluated. | Governance domain priority must strictly precede commerce domain priority in the scheduling order.=IF(0<13,"Safety slice served ahead of commerce slice","Priority inversion")

179 | Undiscriminated Hypothesis Detection | §97.8.1 | Detect continuously where two hypotheses concerning a common subject have posteriors within the configured separation floor while each exceeds a minimum support count; mark the pair undiscriminated and invoke §97.8.2. A pair in which one hypothesis is weakly supported is under-evidenced rather than undiscriminated and is routed to the existing evidence-seeking mechanism instead. | Undiscriminated where $|P(H1) - P(H2)| < \text{separation floor}$ and both accumulated evidence counts exceed the minimum.=IF(ABS(12/16-11/16)<0.15,"Undiscriminated - generate crucial instance","Separated; no instance required")

180 | Discriminating Instance Selection | §97.8.2 | For each obtainable candidate observation compute the magnitude of the difference between its likelihood under each hypothesis, and select the maximum; select irrespective of how probable either hypothesis considers the observation, so an observation both predict equally is never selected; withhold generation where no candidate clears the discrimination floor; emit a structured request stating both hypotheses' posteriors under presence and under absence, so the outcome is interpretable before the observation is obtained. | $\text{discrimination}(E) = |P(E|H1) - P(E|H2)|$; the selected instance maximises it and must clear the discrimination floor.=MAX(ABS(0.9-0.85),ABS(0.88-0.2),ABS(0.55-0.5))

181 | Hybrid and Edge Deployment Topology(Operator) | §107.1 | Add hybrid topology (substrate and access domains in customer infrastructure, media and immersive assets in operator infrastructure, scope resolved customer-side and travelling with the request) and edge topology (frequency-selected cache near a reader concentration, holding no authority, answering only where the requester's scope admits every cached record consulted and escalating otherwise, performing no ingestion and constructing no scope); the Chapter 106 domain partition is invariant across all five topologies. | An edge node serves from cache only where scope admits every cached record consulted and currency is verified; otherwise it escalates.=IF(AND(1=1,45<=80),"Edge node serves from local cache","Escalate to origin mesh")

182 | Media and Immersion Interfaces(Operator) | §107.3 | Extend the §105.1 interface set with a media interface returning deterministic transformations (semantic compression, expansion, colorization, resolution enhancement) carrying the provenance of the transformation, and an immersion interface returning scene descriptions assembled from spatially-detailed records; both resolve a scope before traversal, widen none, carry their own safety gate, capture no gaze or attention signal, and withhold rather than fabricate where detail is insufficient. | An interface call is served only where the requested interface is a registered member of the interface set.=IF(COUNTIF({"content","query","rights","layout","assessment","edition","semantic","media","immersion"},"media")>0,"Interface registered","Refused - unknown interface")

183 | Domain Module Replacement(Operator) | §107.2 | Permit a domain's implementation to be exchanged where the replacement implements every interface the replaced domain exposed at each version still within its support interval, satisfies every transfer-matrix entry in which that domain appeared as source or target, and is admitted through the improvement-proposal mechanism with a named human approval; Domain 0 is a member of the permanently-excluded set and is not replaceable. | Replacement admitted only where interface-version conformance and transfer-matrix conformance both hold.=IF(AND(2>=2,3<=3),"Module replacement admitted","Refused - interface contract unmet")

184 | Parallel Multilingual Presentation(Reader) | §107.4 | Present two or more language realizations of a work simultaneously, aligned by knowledge record rather than by page or sentence position, each generated from the same records so alignment is a property of shared provenance and cannot drift; a citation in either realization resolves to the same record; a record without a realization in one language shows as a gap rather than being filled from the other; wording-operative spans stay in source language in every realization with glosses rendered per target language. | Alignment confidence is bounded by the least confident of the aligned realizations.=MIN(0.94,0.91)

185 | Tenant-Scoped Key Custody(Operator) | §107.5 | Scope each encryption key to a tenant or deployment so a key of one tenant decrypts no content of another; hold key material within a hardware or equivalent custody boundary where configured, so the system holds a handle and every use is a call across the boundary rather than a read of key material; the zero-knowledge reader-history election and the no-operator-key message channel classification are unchanged and are not weakened by the presence of a hardware module. | A decryption proceeds only where a key scoped to the requesting tenant resolves.=IF(COUNTIF({"tenant-A","tenant-B"},"tenant-A")>0,"Key resolved for tenant","Refused - no tenant key")

186 | Signed Provenance Ledger Verification(Operator) | §107.5 | Sign each provenance chain node under a deployment key in addition to its keyed checksum, so a third party holding the verification key can confirm that a recorded operation and its recorded safety evaluation are as written, without access to the producing system; verification establishes fidelity of the record and nothing about the correctness of the operation recorded. | A ledger entry is admitted to a verification report only where its signature verifies against the deployment key.=IF(0.88>=0.8,"Signature verified; entry admitted","Refused - ledger signature mismatch")

187 | Reasoning Mode Classification | §108.1 | Classify every inference as entailed (a conclusion following necessarily from admitted records) or conjectural; admit an entailed conclusion as a knowledge record with a derived-inference type and a confidence equal to the product of the contributing edge confidences; withhold a conjectural conclusion from admission and emit it with status PROPOSED, with no automated process configured to transition it to VALIDATED. Deduction may write to the knowledge store; induction and abduction may not. | An inference is admitted as a record only where it is entailed; a conjectural inference is emitted PROPOSED and creates nothing.=IF(COUNTIF({"entailed","conjectural"},"conjectural")>0,"Refused - emit as PROPOSED","Admit as fact")

188 | Inductive Support Threshold | §108.3 | Operate the evidence-integration layer: accumulate corroborating and contradicting evidence as posterior parameters, weight corroboration by source independence, elevate confidence for cross-modal agreement within a temporal window, hold a concept provisional until the consolidation count is met, and retain every contradiction too weak to revise the model so a pattern of individually-insignificant negatives stays queryable. A generalization is emitted only where subordinate support and evidential strength both clear their floors. | A generalization is proposed only where the subordinate count clears the minimum and the supporting evidence exceeds the contradicting evidence.=IF(AND(4>=3,0.78>0.62),"Inductive support sufficient; hypothesis proposed","Insufficient - no proposal")

189 | Observation Diversity Measure | §41.6.5.2 | Attribute every increment of the confidence parameters to a registered source class and compute the effective number of contributing classes as the reciprocal of the summed squared weight proportions; admit a corroboration from an already-dominant class at reduced weight; require a diversity floor as well as a count floor before proposing a generalization; attenuate precision-weighted revision where a contradiction's diversity approaches unity; and disclose concentration in an output relying on a record below the diversity threshold. | $D = 1 / \sum_i (w_i / \sum_j w_j)^2$ over registered source classes; $D=1$ where all weight is in one class.=1/((8/20)^2+(4/20)^2+(4/20)^2+(2/20)^2+(2/20)^2)

190 | Context Modulation | §109.2 | Modulate an already-admissible result for device class, network condition, session type and referenced environment record: select among entitled editions, offer an available media transformation, adapt layout, and supply a temporal or edition scoping hint. Where a context signal would select an unentitled edition, fall back to an entitled one and disclose the fallback. Rights enforcement is not an output of this layer; the Slice Resolver and rights validation decide availability before any modulation runs. | A modulation is applied only where it leaves the admissible record set unchanged; otherwise it is refused.=IF(AND(1=1,0=0),"Modulation applied within admissible set","Refused - would alter admissible set")

191 | Perspective Modulation | §109.3 | Modulate framing by the role a requester elects for the session — reader, author, instructor, student, researcher, publisher, operator — selecting which relationships among admitted records surface first and how explanations are framed; a student sees learning pathways, an author revision pathways, a publisher rights pathways over identical records. Electing a perspective confers no capability: it frames the records the requester's own entitlement already admits. | A perspective lens is applied where the elected role is registered; it alters ordering and framing, never the admitted set.=IF(COUNTIF({"reader","instructor","operator"},"instructor")>0,"Role lens applied","Default lens")

192 | Ergonomic Modulation | §109.4 | Modulate density, layout, legibility and emphasis from reading pace, interaction pattern, display and input characteristics, accessibility settings and the §50.4 ergonomic preference signals. A stated preference always takes precedence over an inferred signal. Compression selects among admissible records and shortens their expression without summarising beyond what they support. An inferred cognitive-load signal modulates presentation only, produces no record, and reaches no other layer, commerce or interface. | Stated preference governs wherever a stated and an inferred signal bear on the same dimension.=IF(COUNTIF({"stated","inferred"},"stated")>0,"Stated preference governs","Inferred signal applied")

193 | Cultural Modulation | §109.5 | Modulate for locale, regional convention, idiom and localized examples, selecting among admissible expressions and entitled territory-specific editions. The cultural profile is declared by the requester, by an operator, or by an explicitly selected locale — never inferred from a name, network address, query language, device locale or behaviour. Where no equivalent is registered for a locale, render plainly rather than substituting an unregistered equivalent. | A cultural modulation applies only from a declared profile; an inferred profile is refused.=IF(COUNTIF({"declared","inferred"},"inferred")>0,"Refused - cultural profile must be declared","Declared profile applied")

194 | Social Modulation | §109.6 | Modulate pacing, shared position, shared annotation surfaces, register and formality for class, group-reading, collaborative-authoring and instructor-guided settings, operating strictly within the intersection of participants' scope descriptions so no participant is shown a record another's entitlement excludes and no absence is disclosed. A social annotation keeps its author's identity and its own ACL, which this layer does not widen; a group setting confers no entitlement. | Social modulation operates within the intersection of participant scopes, never their union.=MIN(340,275)

195 | Interpretive Composition Order | §109.7 | Compose the five modulations in fixed order over an already-admissible

result — perspective, cultural, social, ergonomic, context — with Prime Directive review preceding and following the whole composition and overriding every layer. Composition is deterministic: identical result plus identical configuration yields identical modulated output. A stated preference overrides an inferred signal at any stage, and where two layers modulate one dimension incompatibly the later stage governs form and the earlier substance. | Governance review precedes stage 1 and follows stage 6; a stated preference overrides an inferred signal at any stage.=IF(0<1,"Safety modulation precedes all interpretive layers","Precedence inversion")

196 | Response Feedback Admission(Reader) | §110.2 | Record every delivered response with its query, scope description, cited records and their confidence at delivery; admit a reader's report on that response as a knowledge record through the ordinary extraction path with the reporting identity as attributed source. A report against a cited record increments that record's contradicting parameter by the source's weight — it does not set confidence, supersede the record or remove it. The reporting identity counts as a source class, so ten reports from one identity weigh less than ten from ten. A report is never admissible as grounding for the subject matter it concerns. | A contradicting report increments beta by the reporting source's reliability weight; confidence follows from §41.6.2 arithmetic.=18/(18+2+0.85)

197 | Outcome Pairing and Subject Gate | §110.3 | Admit an outcome observation recording what subsequently occurred in respect of a matter a response addressed, pair it with the response record, and refuse construction where the subject resolves to an individual natural person so no per-individual predictive record accumulates from operational use; institutional, organisational, system and aggregate-group subjects are admissible. | Construction is refused where the resolved subject class is an individual natural person.=IF(COUNTIF({"institution","process"},"individual")>0,"Refused - individual subject","Outcome admitted")

198 | Delivered-Answer Calibration Score | §110.4 | Compute over paired responses and outcomes the mean of squared differences between the confidence stated at delivery and the realised outcome, per domain and per confidence band, so a system well calibrated in one domain and poorly in another is not concealed by an aggregate; a finding of poor calibration is as reportable as one of good calibration and identifies where a confidence value should not be offered at its current precision. | score = sum (c_i - o_i)^2 / n over paired responses and outcomes.=((0.9-1)^2+(0.85-1)^2+(0.8-0)^2+(0.75-1)^2+(0.7-0)^2)/5

199 | Confidence Band Calibration Gap | §110.4 | Compute per confidence band the difference between the proportion of realised outcomes and the mean confidence stated in that band; raise a calibration finding where the magnitude reaches the configured tolerance; emit the finding as an improvement proposal to a named human reviewer rather than applying it, with the parameters governing the measurement itself held in the permanently-excluded set so the mechanism cannot adjust the terms on which it is judged. | gap = observed outcome rate - mean stated confidence within the band; |gap| >= tolerance raises a finding.=IF(ABS((1+1+0+1+1)/5-(0.9+0.85+0.8+0.88+0.92)/5)>=0.05,"Calibration finding raised","Within tolerance")

200 | Expressive Register Selection | §111.1 | Select exactly one expressive register per response from a governed, versioned set — formal, plain, academic, instructional, narrative, supportive, authoritative — resolved before generation as an element of the Generation Directive and distinct from audience level and discipline mode; refuse an unregistered register rather than approximating it. A register governs voicing only: it alters no admitted record, no confidence value and no permissible vocabulary, so the authoritative register never removes a hedge a confidence value requires and the supportive register never softens an assertion the records state plainly. | A register is admitted only where it is a member of the registered set; voicing is governed by register, assertion by confidence.=IF(COUNTIF({"formal","casual","academic","instructional","narrative","supportive","authoritative"},"academic")>0,"Register admitted","Refused - unregistered tone")

201 | Affective Register Gate | §111.2 | Select the supportive register only on a stated preference, a configured default for an accessibility or discipline configuration, or the existing consent-gated prosodic signal; never by inference from word choice, punctuation, query length, error rate, retry behaviour or interaction timing, and construct no record of an inferred affective state. Where in effect it acknowledges the requester's stated position, reduces density and avoids imperative construction, while withholding no answer the records support and altering no confidence value. | The affective register requires a stated preference, a configured default, or a consented signal; an inferred affective state is refused.=IF(COUNTIF({"stated","consented-signal"},"inferred-from-text")>0,"Affective register applied","Refused - affective register requires a stated or consented signal")

202 | Media-Integrated Expression | §111.3 | Carry media assets inside a grounded response rather than alongside it, including an asset only where a knowledge record of the stabilized set references it and delivering it with a citation resolving it to that record; withhold an asset supporting no admitted record and an asset whose referencing record the scope description excluded, so no asset conveys what the text could not assert. A transformed asset carries its transformation provenance; an interpretive modulation selects among includable assets and cannot render one includable; an accessible equivalent accompanies each included asset at the same production stage. | An asset is included only where it is referenced by an admitted record within the scope description, and is delivered with that record's citation.=IF(AND(1=1,0.91>=0.6),"Media asset admitted with citation","Refused - asset does not support a cited record")

203 | Sentence Aggregation | §112.2 | Realize propositions sharing a subject and joined by a conjunctive Narrative Graph edge as one sentence with a coordinated predicate, up to a configured maximum; refuse aggregation where confidence values differ beyond tolerance so a weakly-supported proposition is never qualified by coordination with a well-supported one; carry every constituent proposition's citation on the combined sentence so each clause remains resolvable to its record. | Aggregate

where shared-subject propositions number at least two and their confidence values fall within the configured tolerance.=IF(AND(3>=2,1=1),"Aggregate into one sentence","Realize separately")

204 | Referring Expression Selection | §112.3 | Realize an entity in full registered form at first mention within a discourse unit and in reduced form thereafter; restore full form where a same-type entity intervenes or a new discourse unit begins. Ambiguity is decided by Ontology Mesh type rather than surface form, and full form is used wherever the test is indeterminate, since an unnecessary repetition costs a reader nothing and an ambiguous pronoun costs them the sentence. | Reduced form is permitted only where no same-type entity intervenes since the last full mention within the discourse unit.=IF(COUNTIF({"first-mention","same-type-intervening"},"same-type-intervening")>0,"Full form required","Reduced form permitted")

205 | Connective Emission | §112.4 | Emit a connective marking the relation type of the Narrative Graph edge joining two sequentially-realized propositions — causal, adversative, expository, illustrative, sequential, concessive — selecting within the class by expressive register and sentence position so no connective recurs mechanically; emit none where no typed edge joins the propositions, since asserting an unbacked relation would assert what no record supports. | A connective is emitted only where a typed edge joins the two propositions; the class follows the edge type.=IF(COUNTIF({"CONTRASTS","CAUSES","ELABORATES"},"CONTRASTS")>0,"Emit contrastive connective","No connective")

206 | Given-Before-New Ordering | §112.5 | Order constituents within a sentence so that material already realized in the preceding sentence precedes material new to the response, computed from the response's own prior content rather than from general familiarity; yield absolutely to target-language syntax, and exempt wording-operative spans, which are reproduced in source order because reordering would alter the wording that carries force. | A constituent realized in the preceding sentence is ordered ahead of one not yet realized, where target-language syntax permits.=MIN(4,MAX(1,3))

207 | Fingerprint-Bounded Lexical Variation | §112.6 | Substitute a registered synonym for a recurring term only where the proposition's semantic fingerprint is unchanged by the substitution, making variation verifiable rather than a source of drift; draw synonymy from resources registered under §4.9 so every admitted substitution is licensed and recorded; never vary a term inside a wording-operative span, a registered canonical entity name, or a governed discipline term. | A substitution is admitted only where the canonicalized triple's semantic fingerprint is preserved.=IF(0.94>=0.9,"Substitution admitted; fingerprint preserved","Refused - fingerprint would change")

208 | Commercial Eligibility Determination(Reader) | §113.1 | Determine eligibility of a commercial item from four already-established facts about the request — registered rights, recorded territory, delivered edition, device class — and from no other input; inferred ergonomic, cognitive-load and affective signals remain unavailable to commercial selection, a declared cultural profile may bound eligibility where an item's registration is locale-specific; where the eligible set is empty the commercial region is omitted and the omission is not disclosed, since a region that reports having nothing to show conveys information about the requester. | Eligibility is computed from declared request facts only; any inferred personal signal is barred from the determination.=IF(COUNTIF({"rights","territory","edition","device"},"cognitive-load")>0,"Refused - inferred signal barred from commerce","Eligibility computed from declared facts")

209 | Placement Class and Density Ceiling(Reader) | §113.2 | Present a commercial item only in a registered placement class — inline between sections, edition-aligned, media-adjacent, or spatial within an immersive presentation — each a labelled region never interleaved with response propositions so position distinguishes answer from commerce; bound placements per response and per session by a configured density ceiling, omitting rather than deferring an item that exceeds it; capture and derive no gaze, dwell, head-pose or attention signal in any class. | A placement is admitted only where the running count remains within the configured per-response and per-session ceiling.=IF(AND(1=1,3<=5),"Placement admitted within density ceiling","Refused - density ceiling exceeded")

210 | Advertiser Provenance Record(Operator) | §113.3 | Register each commercial item with the presenting party's verified identity, the rights permitting presentation, covered territories, validity interval and registering operator, refusing registration where any is absent; record each presentation in the Provenance Ledger with item, placement class, occasioning concept identifier, eligibility facts and computed score, so a presentation is auditable against the rules in force at the time. A commercial item is not a knowledge record: not citable, not traversable, contributing to no confidence value, and any assertion within it is labelled as the presenting party's. | Selection score = weighted sum of relevance, source-priority and bid factors with weights summing to unity; recorded per presentation.=0.4*0.9+0.3*0.7+0.3*0.6

211 | Territory Price Registration(Operator) | §113.5 | Register each commerce object — edition, media asset, immersive experience, bundle, institutional or classroom licence, supplemental pack — with the rights permitting sale, the territories in which it may be sold, and a price per territory and currency; withhold the offer in a territory for which no price is registered rather than offering at a default. | An offer is made only where a price is registered for the request's territory and currency.=IF(COUNTIF({"US","CA","UK"},"US")>0,"Territory price applied","Refused - no price registered for territory")

212 | Transaction Reconciliation(Operator) | §113.5 | Perform purchase, rent, subscribe, unlock, upgrade or transfer against a registered commerce object, producing a transaction record carrying acting identity, object, action, price and currency, territory, rights validation relied upon and a ledger reference; supersede rather than delete a prior entitlement on upgrade; refuse a transaction that cannot be reconciled against an entitlement change rather than recording it pending, so entitlement state and transaction history cannot diverge; a token whose state disagrees with the entitlement record raises a divergence finding, the

entitlement remaining authoritative. | A transaction is recorded only where the entitlement change reconciles exactly; an unreconcilable transaction is refused.=IF(0=0,"Ledger balanced; transaction recorded","Refused - transaction not reconcilable")

213 | Commandable Equipment Registration(Operator) | §114.1 | Register each commandable unit of the capture and inspection class — pan-tilt-zoom head, lighting fixture, bounded-volume capture drone, inspection arm, mobile capture base, positioning stage — with its kinematic envelope, velocity and force limits and operating volume; refuse emission to a unit missing any of these and to any class outside scope. The §96.8 vehicle prohibition is unchanged: vehicle telemetry remains analysis-only and no vehicle control interface exists. | A command is emitted only to a registered unit of an in-scope class whose registration carries envelope, limits and volume.=IF(COUNTIF({"pan-tilt-head","capture-drone","inspection-arm"},"vehicle")>0,"Refused - class outside authorized scope","Equipment class in scope")

214 | Session Authorization for Actuation(Operator) | §114.3 | Grant command authority per session rather than per command, recorded with the authorizing operator's identity, covered units, permitted volume, permitted task classes and expiry; halt emission on expiry rather than prompting mid-motion; effect revocation before the next command rather than at plan end; confer no capability over unnamed equipment; record in the Provenance Ledger so every command traces to its authorization and grantor. | Emission proceeds only while the session authorization is unexpired, unrevoked, and covers the unit, volume and task class.=IF(AND(1=1,120<=300),"Session authorization valid","Refused - authorization expired")

215 | Pre-Emission Constraint Veto | §114.5 | Evaluate every command before emission against operating volume, kinematic envelope, velocity and force limits, minimum obstacle clearance, human exclusion distance, inter-unit swept-volume overlap and recording-restricted zones; withhold any violating command unconditionally — not emitted at reduced priority, not emitted with a warning, and not automatically modified into a permissible command. | A command is emitted only where every constraint test passes; a single violation withholds emission outright.=IF(AND(1=1,0.42<=0.6),"Command emitted","Vetoed - clearance below minimum")

216 | Safety-Only Human Proximity Detection | §114.5 | Detect persons solely to compute exclusion clearance along a commanded trajectory; retain no detection beyond the instant of evaluation, admit none to any result set, make none queryable and expose none to any other mechanism. This preserves the standing exclusion of person classifications from result sets while allowing a machine to avoid people: a system that could not see a person could not avoid one, and one that remembers seeing them is a surveillance capability. | Governing clearance is the lesser of measured distance to a detected person and the configured exclusion distance.=MIN(0.35,0.6)

217 | Halt Path Independent of Reasoning | §114.6 | Emit a halt on clearance or proximity breach, stale equipment state, authorization expiry or revocation, sensor-stream loss, or explicit request; evaluate the halt directly from equipment state and constraint set without traversing retrieval, stabilization, generation or interpretive modulation, so it cannot be delayed by a query in progress; place the unit in its registered safe state and require an explicit resume; record every halt so repeated halts on a unit or task class are queryable. | A halt triggers where measured clearance falls to or below the configured minimum, evaluated outside the reasoning path.=IF(0.42<=0.6,"Halt - clearance breach","Continue")

218 | Assistive Contact Registration and Hazard Analysis(Operator) | §115.1 | Register a unit making intentional contact with a person under an authorization class disjoint from Chapter 114's, in one of three classes — servant (carries or hands over), assistant (supplements movement), carrier (bears weight); refuse registration where a hazard analysis for the intended task, user and environment has not been completed and recorded with the assessor's identity, since every later requirement is meaningless if the hazards were never enumerated. A device registered under both chapters is governed by the authorization under which a given command was emitted. | Registration is refused absent a recorded hazard analysis; the two authorization classes confer nothing on one another.=IF(AND(1=1,1=1),"Contact authorized","Refused - both authorizations required")

219 | Two-Party Contact Authorization(Operator / Reader) | §115.2 | Require a deployment authorization from an identified operator and a contact consent from the assisted person concurrently, neither substituting for the other: deployment establishes contact is permissible, consent establishes it is wanted. Consent is bounded to task class and interval and does not persist across sessions. Withdrawal by the assisted person through any registered channel halts within the registered stopping interval and cannot be overridden by an operator, a representative or a task in progress; an operator halt is not a withdrawal and does not require re-consent. | Contact proceeds only while both authorizations are held; a withdrawal by the assisted person governs absolutely.=IF(COUNTIF({"assisted-person","operator"},"assisted-person")>0,"Halt - withdrawal by assisted person","Continue")

220 | Contact Force and Pressure Limits | §115.5 | Replace the exclusion-distance constraint with a contact constraint set: force and pressure limits registered per body region so a within-limit force over a reduced area is not thereby permitted, registered contact regions, a contact velocity below free-motion velocity, consent currency, and absence of any configuration from which the person could not be released; a violation halts rather than adjusts. Registered values derive from the hazard analysis and the governing standard's biomechanical limits rather than from figures stated in this specification. | Measured force and pressure at each contact region must not exceed the limit registered for that region.=IF(28<=65,"Within registered limit for contact region","Halt - force limit exceeded")

221 | Release as Default Failure State | §115.6 | Design and register every unit so that power loss, control-signal loss, safety-path fault or any halt releases the person rather than retaining them — a gripper that holds on power loss traps whom it

was assisting. A weight-bearing unit is excepted only where its registration records redundant support: its safe state is controlled retention with an alert and a bounded interval for transferring support. The release path does not traverse the reasoning path and must operate where that path is unavailable entirely. | Any fault or halt yields release; retention is permitted only for a registered weight-bearing unit with redundant support.=IF(0=0,"Safe release executed","Fault - grip retained")

222 | Support Redundancy for Weight-Bearing | §115.5 | Maintain support for a person whose weight is borne through at least two independent means so no single failure withdraws support; a unit registered without redundant support may not bear weight. Contact is never initiated from inference — only on request by the assisted person, by an operator under held consent, or within a consented task sequence — and no record of the person's body, movement, capability or dependency is constructed, retained or made queryable. | A weight-bearing command requires at least two independent support means confirmed present.=IF(2>=2,"Redundant support confirmed","Refused - single point of support")

223 | Energy Descent and Termination | §116.1 | Ground convergence in the monotonic decrease of $E = -1/2 \sum$ over distinct pairs of $w_{ij} s_i s_j$, so termination follows from the descent rather than from an iteration bound and a given initial configuration under given weights reaches an identical stable state every execution; distinguish termination from uniqueness, the capacity bound making a spurious state improbable rather than impossible. | $E = -1/2 \sum_{i<j} w_{ij} s_i s_j$; each accepted update strictly decreases E, so no configuration recurs.= $-0.5*(2*1*1*1+2*-1*1*1+2*1*1*1+2*1*1*1+2*-1*1*1+2*1*1*1)$

224 | Bounded Frame Reconstruction | §116.3 | Complete a partially-observed sensor frame by converging observed regions to the nearest stored configuration; bound the completed region's confidence above by the lesser of the observed regions' confidence and the stored configuration's; render it inadmissible as grounding so it supports no assertion and moves no confidence value; produce nothing where convergence fails within the iteration ceiling and report the gap instead. | Reconstructed confidence = MIN(observed confidence, stored-configuration confidence).=MIN(0.94,0.88)

225 | Reconstruction Marking | §116.3 | Mark every reconstructed region as reconstructed so it is never presented as observed; a frame carrying an acknowledged hole is truthful, a frame whose hole has been filled with an inference presented as observation is not. | A region is presented as observed only where it was observed; a converged region carries the reconstructed marking.=IF(COUNTIF({"observed","reconstructed"},"reconstructed")>0,"Marked reconstructed; not presented as observed","Observed region")

226 | Difference-Preserving Alignment | §116.4 | Align timelines, sequences and editions by convergence only where the divergence falls below the configured threshold, treating that as extraction variance; preserve a divergence at or above threshold as a genuine difference and preserve any divergence that cannot be classified, since erasing a real difference costs a reader the fact while preserving a spurious one costs a footnote; record incompatible content between preserved editions in the contradiction index. | Converge where divergence < threshold; preserve where divergence >= threshold or is unclassifiable.=IF(0.31>=0.15,"Genuine edition difference - preserved","Extraction variance - converged")

227 | Control Oscillation Fault | §116.5 | Damp oscillation and jitter in a control signal by convergence, evaluating the damped command against the constraint set identically to an undamped one so damping never brings a command within a limit it would otherwise exceed; where control-error sign changes reach the configured frequency across a window, raise a control fault, cease damping and invoke the halt path, because a persistently damped oscillation is a concealed fault rather than a stabilised loop. | Sign changes of the control error across the window >= configured frequency raises a fault and halts.=IF(5>=4,"Oscillation - raise control fault","Control stable")

228 | One-Hop Concept Adjacency Set(Reader) | §113.8.1 | Supply commercial selection with the query's resolved concept identifier together with concepts one edge away, computed at the point of supply after retrieval, stabilization and generation complete; restrict to adjacency-bearing edge types — subsumption, part-whole, registered general association — excluding causal, temporal and contradiction edges, since a concept related as a cause or a contradiction is not thereby one the reader wants offered; perform no traversal from an adjacent concept, so two-edge concepts never enter the set. | An adjacent concept is admitted only where its connecting edge type is registered as adjacency-bearing and lies within one hop.=IF(COUNTIF({"IS-A","PART-OF","RELATED-TO"},"CAUSES")>0,"Edge type permitted for adjacency","Refused - edge type not in adjacency set")

229 | Adjacency Scope Confinement(Reader) | §113.8.1 | Exclude from the adjacency set any concept lying outside the requester's scope description, and do not signal the exclusion, so the existence of a concept beyond a requester's entitlement cannot be inferred from what was advertised to them; the adjacency set is recomputed per request and retained no longer than the request. | An adjacent concept enters the set only where the requester's scope description admits it.=IF(COUNTIF({"in-scope","out-of-scope"},"out-of-scope")>0,"Refused - adjacent concept outside requester scope","Adjacent concept admitted")

230 | Attenuated Adjacency Relevance(Reader) | §113.8.2 | Let an adjacent concept participate in the relevance factor alone and never in eligibility, so an ineligible item does not become eligible by matching a neighbour; attenuate relevance computed through adjacency relative to relevance against the query concept itself, so a direct match outranks a neighbour match; leave the three existing weights unchanged and summing to unity. | Score = $0.4*relevance + 0.3*source-priority + 0.3*bid$, with adjacency-derived relevance attenuated below direct relevance.= $0.4*0.72+0.3*0.7+0.3*0.6$

231 | Convergence Regime Disclosure | §116.2.1 | Operate each network under exactly one recorded convergence regime and return with every derived output an indication of which was in effect: a spectral-norm contraction guaranteeing a unique

equilibrium by construction, or a capacity bound below the retrieval-phase boundary rendering a differing converged state improbable rather than impossible. The two support different statements about an identical output and a recipient cannot distinguish them from the output itself. The election is a member of the permanently-excluded subsystem set, changed only by an operator and recorded with that operator's identity, so no internally-generated proposal can elect the weaker regime to obtain freer weight updates. | Exactly one regime is in effect per network; the output discloses which, and the election cannot be altered from within the system.=IF(COUNTIF({"unique-equilibrium","capacity-bounded"},"unique-equilibrium")>0,"Disclose: unique equilibrium guaranteed","Disclose: capacity-bounded convergence")

232 | Trial Budget and Scheduling(Operator) | §117.1 | Meter every trial as a bounded compute-accounting slice in a trial category with a per-interval budget; schedule trial slices below safety evaluation without exception, below requester-serving slices and below ingestion; defer a trial exceeding its allocation rather than serving it at reduced priority, so a campaign cannot degrade the system it measures. Trials are free of physical consequence — nothing wears, is damaged or crashes — but not free of cost; the cost shifts from equipment and risk to compute and scheduling, and is recorded so an operator decides with a figure in front of them. | A trial executes only where the trial-category budget for the interval is not exceeded; otherwise it defers.=IF(AND(2400<=5000,1=1),"Test budget within allocation","Deferred - would contend with production slices")

233 | Trial Isolation and Non-Application(Operator) | §117.2 | Execute trials against an isolated corpus and configuration: construct no production knowledge record, adjust no production confidence value, emit no command to physical equipment outside simulation, deliver no output to a requester. Record results as trial records excluded from ordinary scope descriptions. Apply nothing — a finding is emitted as a proposal requiring a named human approval, and ten million trials produce the same class of output as one: evidence for a person to act on. Regression blocks rather than reports: a case passing before a change and failing after it withholds the change. | A change is applied only where no previously-passing trial case regresses; a regression withholds the change.=IF(COUNTIF({"passed","regressed"},"regressed")>0,"Refused - previously-passing case now fails","Regression clear")

234 | Trial Space Coverage(Operator) | §117.5 | Enumerate the trial space per layer rather than sampling it, execute cases from the enumeration, and record coverage as the executed proportion together with an enumeration of the cases not executed, so the residue is stated rather than left as an unexamined assumption. Exhaustive mode is where the absence of physical cost pays: a space prohibitive to explore with equipment can be explored completely. | Coverage = executed cases / enumerated trial space, with uncovered cases listed individually.=847/920

235 | Determinism Verification(Operator) | §117.5 | Measure determinism as the proportion of repeated trials over an identical corpus state and configuration yielding an identical semantic fingerprint; a measurement spanning a corpus that changed between trials is not a determinism measurement, since confidence legitimately moves as evidence accumulates. Any value below unity is a defect rather than a tolerance. The \$96.9 probes applied inward measure the claimed property directly rather than a proxy. | Determinism = identical-fingerprint repeats / total repeats over an unchanged corpus state; the only passing value is 1.0.=IF(1000/1000>=0.999,"Determinism verified across trial set","Defect - determinism not reproduced")

236 | Simulation Is Not Clearance(Operator) | §117.6 | Treat a trial result concerning equipment as evidence about a model, never as clearance to operate: clearance comes from hazard analysis, verification and certification on the specific unit in its configuration. Where a simulation and a physical trial disagree, the physical trial governs and the disagreement is recorded as a finding about the simulation. A campaign may contribute to a hazard analysis as one input; it may not conclude one. | Operation is cleared only by unit certification; a simulation pass, at any scale, does not substitute.=IF(COUNTIF({"simulation-pass","unit-certification"},"unit-certification")>0,"Cleared for operation","Refused - simulation is not clearance")

237 | Graspable Object Registration(Operator) | §120.1 | Register each graspable object with mass, permitted grip force, fragility limit and admissible contact regions; refuse to grasp an unregistered object rather than inferring grip force from appearance, since an inference about how hard to squeeze something errs destructively and immediately. Apply the least of the unit's force ceiling, the object's fragility limit, and the computed requirement raised by the configured margin — the margin applied before the limits, so it can never carry force above one. | Applied force = MIN(unit ceiling, fragility limit, required force x (1 + margin)).=MIN(18,15,12.4*1.15)

238 | Slip Response and Controlled Release | §120.2 | Detect slip from divergence between commanded and observed object position during a held grasp; increase applied force toward the lesser of unit ceiling and fragility limit and no further; where slip continues at that bound, release under control rather than tightening, recording object, force at release and observed slip rate. The fragility limit is a limit, not a target: an object that will not be held within it is an object not to be held. | On slip, force rises to MIN(ceiling, fragility); continued slip at that bound triggers controlled release.=IF(COUNTIF({"within-limit","at-limit"},"at-limit")>0,"Release under control","Increase force toward limit")

239 | Gripper Opens on Fault | §120.3 | Register every manipulator so power loss, control-signal loss, safety-path fault or any halt opens the gripper. Plan each grasp so the fall path is unobstructed before the grasp is made, refusing trajectories whose release path is obstructed; on an obstruction entering during a held grasp, move to the nearest pose with a clear path and, failing that, retain and raise a fault. A dropped object is a cost; a crushed object, a trapped hand and an object no one can command are worse. | Any fault or halt opens the gripper; retention is permitted only where the release path is obstructed and no clear pose exists.=IF(0=0,"Gripper opens; object released","Fault - object retained")

240 | Handover Release Condition(Reader) | §120.4 | Perform handover under the Chapter 115 authorization, presenting the object within reach and applying no force toward the person; release on detected support sustained across the configured interval, and on timeout lower the object under control to a registered surface rather than releasing into the air. Never pull, never resist a person taking the object, never re-grasp what a person has taken, and never press an object against anyone. | Release occurs only on sustained detected support; absent it, controlled lowering rather than release.=IF(COUNTIF({"stable-support","no-support"},"stable-support")>0,"Release to recipient","Retain - controlled lowering on timeout")

241 | Navigable Region Registration(Operator) | §121.1 | Register a navigable region — travelable surfaces, boundaries, excluded zones — in place of a fixed operating volume, since free space in a shared corridor changes faster than a plan can be recomputed. Maintain a continuously-updated clearance model as session-scoped working state, never admitted as a knowledge record, and halt motion where the model exceeds its staleness bound. | Motion proceeds only within the registered navigable region and only on a clearance model within its staleness bound.=IF(COUNTIF({"corridor-A","atrium","loading-bay"},"restricted-ward")>0,"Region admitted","Refused - outside registered navigable region")

242 | Clearance-Bounded Speed | §121.2 | Bound commanded speed by the lesser of the unit's registered maximum and a configured coefficient times the clearance ahead, so the unit slows as space closes rather than holding speed until an obstacle forces a stop; stopping distance at every permitted speed stays below the clearance that permitted it. No task parameter, priority or deadline raises the coefficient — a machine that hurried when late would be fastest exactly when its planning was worst. | Permitted speed = MIN(registered maximum, coefficient x clearance).=MIN(1.5,0.6*0.6)

243 | Unconditional Yield | §121.3 | Yield to a person on every path conflict: stop or withdraw along the path already travelled, never routing around at speed, never signalling for accommodation, never computing which party should give way — a person who cannot predict a machine cannot avoid it. Resolve unit-to-unit conflicts by registered ordering rather than negotiation, faulting both on persistent deadlock. Never maintain a following distance behind a person. | Where a person's path conflicts with the unit's, the unit yields without exception.=IF(COUNTIF({"machine","person"},"person")>0,"Yield - person has right of way","Proceed")

244 | Motion-Only Clearance Envelope | §121.4 | Compute the occupied envelope for a person from observed position and velocity over a short window and from nothing else: no identity, no history beyond the window, no retention past the interval governed, and no availability to any other mechanism or interface. Where observed motion does not support an envelope, treat the full reachable region as occupied rather than assuming continuation, since an assumption about where someone will go is what produces a collision when they do otherwise. | The envelope derives from observed motion alone; identity may not inform it.=IF(COUNTIF({"observed-motion","identity"},"identity")>0,"Refused - identity may not inform the envelope","Envelope computed from motion alone")

245 | Stop Rather Than Relax | §121.5 | Where no path satisfies the constraint set, stop: do not reduce a clearance requirement to find a path, do not proceed slowly through a sub-minimum clearance, do not select a least-violating trajectory. Report the obstruction with the clearance model after the configured interval and await an operator. A gate reached often is a gate under pressure to become a preference, which is why this is restated for a condition that will invoke it frequently. | A path is admitted only where its clearance meets the configured minimum; otherwise the unit stops.=IF(0.28>=0.35,"Path admitted","Stop - no path satisfies the clearance minimum")

246 | Predictive Level Registration | §122.1 | Maintain four hierarchically ordered expectation levels — lexical (the next element within a record), sequential (the next record given a sequence), conceptual (an instance attribute given its concept), invariant (a regularity of the world model) — each predicting within its own scope and never predicting what a level above it predicts. | An expectation is evaluated only at a registered level of the hierarchy.=IF(COUNTIF({"lexical","sequential","conceptual","invariant"},"conceptual")>0,"Level registered","Refused - unregistered expectation level")

247 | Lowest-Level Error Attribution | §122.2 | Attribute a prediction error to the lowest level whose residual accounts for the configured share of total residual, applying revision at that level alone and passing nothing upward; a mis-parsed sentence says nothing about physics. Without this rule a mis-read unit propagates until it perturbs a physical constraint, and a system revising invariants on parsing errors becomes less reliable the more it reads. | Level share = level residual / total residual; attribution goes to the lowest level clearing the configured share.=0.42/(0.42+0.31+0.08+0.05)

248 | Escalation Gate | §122.3 | Escalate to a level only where every level below it fails the share test, so a conceptual revision requires lexical and sequential residuals below the share and an invariant revision requires all three below it; an invariant is revised only where the observation survived every cheaper explanation. Revision magnitude at the attributed level remains governed by precision weighting, so attribution fixes where and precision fixes how far. | A level is attributed only where each lower level's residual falls below the configured share.=IF(0.4884>=0.4,"Attribute to lowest accounting level","Escalate to next level")

249 | Unattributed Error Retention | §122.3 | Where no level clears the share, record the observation in the contradiction index and apply no revision anywhere — an error no level accounts for is a finding about the hierarchy, not an instruction to it; surface repeated unattributed errors against a common expectation as a pattern, since one is noise and a hundred is a missing level. No level revises a registered physical constraint; an apparently contradicting observation enters the index rather than

relaxing it. | An unattributed error is recorded and applied nowhere; physical constraints are never revised by observation.=IF(COUNTIF({"attributed","unattributed"},"unattributed")>0,"Recorded in contradiction index; no revision applied","Revision applied at attributed level")

250 | Edge-Class Traversal Cost | §123.1 | Assign each edge class a traversal cost and bound pre-activation by a budget in cost units rather than by hop count: semantic edges cheap because meaning stays informative over distance, editorial dearer because relevance decays across version boundaries, temporal dearest because adjacency in time is weak evidence of relatedness and a long walk gathers noise. Costs are registered configuration recorded with the response, never adjusted by the system from its own experience. | Budget consumed = sum over classes of (hops x cost); traversal halts when the registered budget is exhausted.=IF(12*1+2*3+0*8<=24,"Within propagation budget","Refused - budget exceeded")

251 | Propagation Reach by Class | §123.1 | Derive each edge class's reach from its cost against the traversal budget, so a query may reach widely through meaning and shallowly through time at the same cost as the reverse; traversal is bounded by construction rather than by a stopping heuristic, and the same query under the same cost table pre-activates the same set on every execution. | Reach of a class = budget / cost per hop for that class.=MIN(24/1,24/3,24/8)

252 | Tri-Temporal Posture | §123.2 | Resolve a query in exactly one of three postures and state which produced the answer: attested past under the attestation-horizon cutoff, live present reading the live layer over the committed layer, or projected future carrying a credible interval. Past and present results are ordinary knowledge records; a projected result is never admitted as a record and never grounds a subsequent generation. Where a response draws on more than one posture each proposition carries its own, so a difference between answers reads as a difference of posture rather than an inconsistency. | A result is admissible as grounding only where its posture is attested-past or live-present; projected-future is labelled and withheld.=IF(COUNTIF({"past","present","future"},"future")>0,"Forecast - labelled, not admitted as record","Attested state - admissible")

253 | Six-Dimensional Coherence | §124.1 | Compute a coherence value across semantic, temporal, spatial, editorial, media and evidential dimensions, each derived from mechanisms already specified, and take the least of the six as governing rather than the mean, since a mean lets five strong dimensions conceal the one weak dimension that is the actual finding. | coherence = MIN over the six dimensional values; the governing dimension is named with the result.=MIN(0.94,0.88,0.91,0.62,0.90,0.86)

254 | Coherence Classification | §124.1 | Classify by the governing value against configured floors — stable, drifting, misaligned — always naming the dimension that produced the classification; distinguish two further states with different causes and remedies: incomplete, where a dimension cannot be computed for want of data, and contradictory, where two records assert incompatible content and a specific pair is recorded. Coherence informs and never gates: it adjusts no confidence, admits or excludes no record, and withholds no output. | Stable at or above the upper floor; drifting between floors; misaligned below the lower

floor.=IF(MIN(0.94,0.88,0.91,0.62,0.90,0.86)>=0.85,"Stable",IF(MIN(0.94,0.88,0.91,0.62,0.90,0.86)>=0.70,"Drifting - reported","Misaligned - reported with dimension named"))

255 | Context Lock and Key | §124.2 | Enumerate the context dimensions whose values must agree for the determinism guarantee to hold — device class, entitlement set, locale, expressive register, audience level, temporal posture, traversal cost table, convergence regime — compute a key over them and return it with the output. Identical keys mean the outputs must be identical and a difference is a defect; differing keys mean nothing is claimed. The enumeration is in the permanently-excluded set, since a mechanism able to unlock a dimension could restore apparent determinism by narrowing what determinism means. | Determinism is asserted between two executions only where the context key computed over the locked dimensions is

identical.=IF(COUNTIF({"device","entitlement","locale","register","audience","posture","costs","regime"},"session-time")>0,"Locked dimension - determinism claimed","Unlocked - determinism not claimed across this difference")

256 | Relevance Gradient | §124.3 | Compute at traversal time the rate of change of relevance across a record's neighbours within an edge class, and order traversal to prefer directions of lower gradient, since slowly-changing relevance marks a region of stable meaning while a sharp change marks a boundary beyond which traversal gathers material of a different kind. Computed from values the records already carry, stored nowhere, adding no schema field, and altering the order in which the §123.1 budget is spent rather than what is reachable within it. | gradient = |relevance(a) - relevance(b)| across neighbours in an edge class; lower gradient is traversed first.=ABS(0.91-0.62)

257 | Substrate Output as a Modality(Operator) | §4.10.1 | Register a computational substrate — neuromorphic processor, memristive array, multi-electrode array — as a source under a declared capability and its output as a modality with its own independently-failing extractor, so a substrate that becomes unavailable removes one modality and leaves the rest operating; reduce its output to concept-level records before admission, retain no raw episode, pass the safety gate, calibrate confidence against the substrate's registered reliability, and treat the substrate as a source class so many readings from one array do not weigh as readings from many. | A substrate output is ingested only where its modality and extractor are registered.=IF(COUNTIF({"text","image","audio","video","sensor","spike-train"},"spike-train")>0,"Registered modality; extractor dispatched","Refused - unregistered modality")

258 | Registered Substrate Interpretation | §4.10.2 | Admit a substrate output as a measurement of that substrate's behaviour; admit it as bearing on any other subject only through an interpretation registered in advance with its own provenance

and reliability, revisable by an operator. Absent a registered interpretation the output is a measurement of the substrate and nothing more, since a response to a stimulus is evidence of what the substrate produced and not of what it was asked about. | Derived confidence = MIN(substrate reliability, interpretation reliability) — no better than the weaker of instrument and reading.=MIN(0.71,0.55)

259 | Substrate Stimulation Dispatch(Operator) | §4.10.3 | Dispatch a stimulus to a registered substrate through its connector under a declared capability, refusing an undeclared capability at the boundary and recording each stimulus with its dispatching identity; emit no stimulus on an inference that stimulation would be useful, and let no substrate response adjust the parameters governing when it may be stimulated. Where a substrate comprises living tissue, the operating institution's governance and any review its jurisdiction requires apply; this specification authorises no substrate's operation. | A substrate output is admitted as calibrated evidence only where it clears the configured confidence floor.=IF(0.71>=0.6,"Substrate output admitted as calibrated evidence","Below floor - recorded, not admitted")

260 | Analog Convergence Realization | §7.7.1 | Perform the threshold update by the settling of a variable-conductance array rather than by computing it: activations applied as row potentials, weights as element conductances, the weighted sum as column current, the stable state read by comparing current against a reference. The mapping is exact rather than analogical — Ohm's and Kirchhoff's laws produce the summation — and the duration of one settling does not scale with network size, where a digital update of an 8,192-unit network performs 67,108,864 multiply-accumulates. | Digital multiply-accumulates per update = N^2 ; one analog settling performs the same summation as a single physical operation.=8192*8192

261 | Unique-Equilibrium Precondition | §7.7.2 | Permit analog realization only under the §116.2 monotone parameterization, refusing an array operated under the capacity bound alone. Conductance elements cannot be manufactured exactly, so two arrays hold weight matrices differing by fabrication tolerance; under a capacity bound both converge but may settle to different stable states while each behaves correctly, whereas under a unique equilibrium both settle to the same one. Uniqueness ceases to be a strengthening and becomes the precondition. | Analog operation is permitted only where the network's registered convergence regime is unique-equilibrium.=IF(COUNTIF({"unique-equilibrium"},"capacity-bounded")>0,"Analog realization permitted","Refused - analog realization requires the unique-equilibrium regime")

262 | Cross-Array Agreement | §7.7.3 | Register each array with its dimensions, conductance range, fabrication tolerance and read tolerance; resolve a column falling within tolerance of the threshold by a further settling rather than by rounding, and report unresolved with fallback to the digital path where it does not resolve; require two arrays on a common input under a common weight specification to agree within tolerance, raising a fault on divergence rather than averaging, since averaging two disagreeing results produces a state neither array reached and a state no array reached is not an attractor of either. | Two arrays agree where $|state_A - state_B| \leq$ registered read tolerance; beyond it a fault is raised.=IF(ABS(0.42-0.415)<=0.05,"Chips agree within tolerance","Divergence - device variation exceeds tolerance")

263 | Cross-Array Agreement Under Variation | §116.2 | Parameterize by spectral norm below unity rather than by diagonal dominance: a diagonal-dominance constraint collapses the state toward zero where a unit's sign is near-ambiguous, yielding a unique equilibrium carrying no information, whereas a contraction retains the structure the weights encode. Simulated over 64 units holding six patterns, two matrices differing by 5% conductance variation settled differently on 30.5% of inputs under the capacity bound and 0.5% under a contraction at spectral norm 0.9 — recorded as a simulation result, not a hardware measurement. | Disagreement rate between two varied weight matrices is materially lower under a spectral contraction than under the capacity bound.=1-0.005/0.305

264 | Engine-Specific Prompt Composition(Operator) | §126.2 | Compose a prompt from knowledge records through the regeneration mechanism under the vocabulary constraint, adding semantic markers naming the concepts to address, extraction instructions specifying the return form, and alignment constraints stating the target record schema; compose per engine from a registered profile recording accepted input form, context bound, declared capabilities and reliability. Composition is deterministic, so an identical request under an identical profile yields an identical prompt and a differing reply is attributable to the engine rather than to the asking. | Confidence of a converted record = MIN(engine reliability, extraction-instruction reliability).=MIN(0.88,0.62)

265 | Reply Conversion to Knowledge Records(Operator) | §126.3 | Dispatch through a registered connector under a declared capability, refusing an unregistered engine or undeclared capability at the boundary; rate-limit and batch under compute accounting, deferring over-budget dispatches; terminate on failure, malformed reply or over-bound reply without producing a partial record. Normalize and segment the reply into individual assertions, convert each to a candidate record validated against its type's machine definition, and assign a temporal anchor from elicitation time, a semantic fingerprint and a provenance reference naming prompt, engine, version and position within the reply. | An assertion is converted only where it validates against the machine definition of its type; otherwise it is refused at construction.=IF(COUNTIF({"human-authored","ai-derived"},"ai-derived")>0,"Citation discloses model origin","Ordinary citation")

266 | Derivation Marking and Disclosure(Operator) | §126.5 | Carry the engine identity as an unmodifiable, unremovable field of the record schema, and disclose it in any citation resolving to that record, so a reader following a citation is told the assertion was generated by a named engine from a recorded prompt rather than asserted by an authored source. Prevents the failure in which the authority behind a cited claim turns out to be that a model produced it when asked. | A record cannot be

re-typed to shed its derivation marking; every citation to it discloses the marking.=IF(COUNTIF({"human-authored","ai-derived"},"ai-derived")>0,"Citation discloses model origin","Ordinary citation")

267 | Self-Corroboation Bar(Operator) | §126.5 | Treat each registered engine as one source class, so repeated elicitation raises accumulated evidence without raising diversity — three assertions from one engine yield an effective source-class count of 1.0 against 3.0 for three distinct engines — and refuse a corroboration offered by a record derived from the same engine as the record it corroborates. Prevents confidence manufactured by repetition, where a hundred elicitations from one model would otherwise report near-certainty for a claim no source ever made. | Effective source classes $D = 1/\text{sum of squared weight proportions}$; all weight in one engine yields $D = 1. = 1/((3/3)^2)$

268 | Citation Verification of a Reply(Operator) | §126.6 | Verify every citation contained in a model-derived assertion against the record store, recording it as verified where it resolves and unverified where it does not; admit the assertion carrying that recording rather than withholding it, so a fabricated citation becomes a recorded property of the assertion rather than an undetected defect. Record contradictions with authored records in the contradiction index, noting the derivation difference so a reviewer knows one side originated from a model. | A corroboration is admitted only where the corroborating record derives from a different source class than the record corroborated.=IF(COUNTIF({"model-A","model-A","model-A"},"model-A")>1,"Refused - self-corroboration within one model","Independent corroboration")

269 | Competitive Candidate Resolution | §21.4.3 | Where a query admits several mutually-inconsistent interpretations, activate the candidate records supporting each together rather than evaluating them in sequence; reduce each candidate's activation in proportion to the summed activation of the candidates inconsistent with it; iterate to a stable state under the existing threshold update; suppress candidates below the retention floor. The competition is among candidates rather than among domains, since a domain owns exactly one decision and two domains proposing competing answers would contradict the partition. Where two candidates survive at comparable activation the set does not resolve and clarification is requested rather than the higher being taken. | Suppressed activation = $a_i - k * \text{sum of activations of candidates inconsistent with } i$; retained where above the floor.= $0.82-0.35*(0.61+0.44)$

270 | Specialized Feed Registration(Operator) | §127.1 | Register a live data source with publication authority, refresh interval, staleness bound, value class, unit and dimension, restatement policy and redistribution entitlement; a registration omitting any of these is connectable but not admissible as a query source, since each governs a decision the system would otherwise make by assumption. Refuse an arithmetic combination of two values of differing dimension rather than computing it. | A feed is admissible as a query source only where all seven registration properties are present.=IF(COUNTIF({"publication","extraction"},"publication")>0,"Admitted at publication authority; calibration bypassed","Calibrated from extraction")

271 | Publication Authority Admission | §127.2 | Admit a non-forecast value at its publisher's authority without applying extraction confidence, modality coefficient or source reliability, since calibration would invent an uncertainty that does not exist — a published price is not 87% likely to be the price. The record asserts that the authority published the value, not that it is correct. A forecast is admitted carrying the uncertainty its publisher states. Where a publisher states an interval, retain the interval and compute any required point by a registered rule rather than selecting one. | Point value from a published interval = registered rule over the interval; the interval is retained alongside.= $0.5*(18.2+21.6)$

272 | Staleness Bound | §127.3 | Serve a value only where its age at query time is within the registered staleness bound; beyond it state the age and publication time rather than serving, extrapolating or substituting the nearest available. A feed value does not decay — it expires, since a quoted value does not grow less certain with age but becomes superseded, and answering from it is not answering with lower confidence but answering wrongly. Absence of publication within a multiple of the refresh interval marks the feed interrupted, distinguishing a stopped feed from an unchanged value. | Serve where (query time ? publication time) ? staleness bound; otherwise return the age and the publication time.=IF($47 \leq 60$,"Within staleness bound - served","Stale - refused with age stated")

273 | Restatement Distinct From Correction | §127.4 | Record a publisher's revision of an already-reported period as a restatement related to the prior value, retaining the prior value unmarked and queryable as what was published at its publication time — because it was not wrong, it was correct on the information then available. Reserve the supersession chain and the erroneous marking for a revision issued as a correction. A revision arriving outside the registered restatement policy is held for a named human rather than classified by inference. | A revision is a restatement or a correction by registered policy; the relation determines whether the prior value is marked erroneous.=IF(COUNTIF({"restatement","correction"},"restatement")>0,"Prior value retained as correct-when-published","Prior value superseded as erroneous")

274 | Redistribution Entitlement(Reader) | §127.5 | Express what a requester may do with a served value — display, derive, delayed, or redistribute — as a dimension distinct from what they may reach. Enforce at composition rather than at export: a response that would carry a value beyond its entitlement is composed without it and the entitlement is stated, so the requester is told rather than left to infer. A derived value carries the source feed's entitlement unless a registered derivation discharges it, so an entitlement cannot be escaped by computing on a value and redistributing the result. | A value enters a response only where the requester's redistribution entitlement admits the intended use.=IF(COUNTIF({"view","redistribute"},"redistribute")>0,"Refused - redistribution not entitled","Value may be displayed")

275 | Agent Definition Completeness(Operator) | §128.1 | Register a defined agent only where all six phases are complete

— purpose and boundary, operational function with deterministic decision rules and error handling, scope description as an entitlement, runtime with duration bound and termination condition, declared capability set, and a review with a named human approval. Refuse a definition lacking any element rather than supplying a default, since an agent whose behaviour is not fully determined before it runs cannot be evaluated before it runs. An agent without a termination condition is not registrable. | Registration proceeds only where every element of all six definition phases is present.=IF(AND(1=1,1=1,1=1,1=1,1=1,1=1),"Agent admitted to registration","Refused - definition incomplete")

276 | Agent Capability Bounding(Operator) | §128.1 | Bound an agent's declared capability set above by the capabilities of the identity registering it, so an agent cannot be given authority its author does not hold; refuse an undeclared capability at the boundary at execution time. Re-perform the governance evaluation at each execution rather than at registration, so an agent does not accumulate authority by having been approved once. | An operation executes only where it lies within the agent's declared capability set.=IF(COUNTIF({"read","compose"},"write")>0,"Refused - capability not declared","Within declared capability")

277 | Agent Lifecycle Enforcement(Operator) | §128.4 | Enforce rather than track the lifecycle across six stages: initialization, execution within window and duration bound, non-disableable monitoring, advisory reporting, termination, archival. Terminate on lifecycle expiry, owner or operator request, governance exclusion, capability violation or exceeding a configured error rate, taking effect before the next execution. Renewal is a fresh registration with a fresh review and approval, so an agent cannot persist by inattention. | An agent terminates where elapsed lifetime reaches the registered lifecycle bound, irrespective of attention.=IF(14>=14,"Lifecycle expired - agent terminated","Within lifecycle")

278 | Agent Decision Trace | §128.5 | Write on each execution a trace of the trigger fired, scope resolved, records read, branch taken by each rule, outputs, every refusal with its cause, and compute consumed. Write it by the execution mechanism rather than by the agent, so an agent cannot omit from its own trace what it would prefer not to record, and append it to the provenance chain under the keyed-checksum discipline so later alteration is detectable. Record refusals and terminations with equal prominence to successes. | A deductive conclusion an agent may admit carries confidence bounded by the least of the traversed edge confidences.=MIN(0.62,0.55)

279 | Advisory Distinguished From Approval(Operator) | §128.6 | Adopt the agent's self-analysis and warnings, which block registration on an unresolved finding, while refusing its self-confirmation of compliance: a mechanism cannot establish its own compliance, and a confirmation so produced would be relied on precisely where reliance is least warranted. Compliance is established by the governance evaluation performed upon the agent and by a named human approval. During operation an agent reports but does not decide — it does not renew itself, widen its scope, extend its lifecycle or clear its own findings. | A registration or continuation proceeds on a recorded human approval; an agent's own report is advisory only.=IF(COUNTIF({"advisory","approval"},"approval")>0,"Named human approval recorded","Advisory only - not an approval")

280 | Skill Inventory Layer | §53.10 (Pass 113) | Store procedural knowledge — how to perform a task, manipulate an object, or execute a multi-step operation — as provenance-tracked procedural concept models distinct from declarative Fact Units, formed from repeated observation across the ordinary ingestion modalities including direct sensory experience and ingested video, consolidating on a count; retrievable, citable, and bounded by the requester's entitlement. | A procedural concept informs Scenario Graph simulation and equipment-control command compilation; it never itself emits a command, every emission remaining subject to the constraint evaluation and authorization classes specified for equipment control.

281 | Epistemic–Aleatory Uncertainty Decomposition | §3.8.1 (Pass 115) | Classify residual uncertainty on the evidence pair deterministically as epistemic (reducible by further observation) or aleatory (irreducible variability of the observed world), carrying source-reliability weighting, measurement precision, and consensus breadth as explicit, queryable sub-parameters. | The Baconian loop iterates while residual uncertainty is predominantly epistemic and stands down, discarding no evidence, when the residue is aleatory.

282 | Assertion Polarity Field | §3.8.2 (Pass 115) | Mark every Fact Unit explicitly affirmative or negative, making a negative finding a first-class record; polarity participates in the semantic fingerprint. | Records agreeing on the same assertion core with opposite polarity are contradictory by field comparison; the absence of any record obliges refusal under the Without-Information Rule, while a known negative is an answer.

283 | Native N-ary Assertion Structure | §3.8.3 (Pass 115) | Carry one assertion with multiple typed argument slots — subject, value, unit, instrument, method, condition, observer — under one FUID, one semantic fingerprint computed over the complete argument structure, and one evidence pair. | An n-ary assertion is one atomic record, never a constellation of binary statements joined by synthetic nodes.

284 | Applicability Scope Field | §3.8.4 (Pass 115) | State in the record the conditions under which the assertion holds — temporal interval, jurisdiction, edition, parameter range, population — as the record-side complement of the context lock. | Scope participates in retrieval matching; assertions under disjoint scopes do not conflict and are reported by condition rather than as contradiction.

285 | Baconian Adjustment History Layer | §3.8.5 (Pass 115) | Append an immutable history entry on every feedback-loop pass — the observation, its source class, the fractional reliability weighting applied, and the resulting movement of the evidence pair and credible interval. | The history layer is metadata only and never touches semantic identity; a record's confidence is deterministically replayable from first observation to present state.

286 | Cryptographic Integrity Chain | §3.8.6 (Pass 115) | Hash each record's immutable core and carry the superseded

record's hash on every supersession link, making version chains tamper-evident end to end by standard hashing alone. | Integrity machinery is internal to the Fact Graph: no distributed ledger, no token, no ownership identity node — the declined blockchain ownership layer remains declined.

287 | Canonical Compact Encoding | §3.8.7 (Pass 115) | Serialize the Fact Unit to a defined lossless binary form — required fields at fixed positions, optional fields as tagged extensions — with FUID and semantic fingerprint identical in either form. | The compact form is the same record, not a summary of it; it round-trips without change and grounds micro-embedded and offline deployment and record-level semantic media compression.

288 | Record-Fused Canonical Encoding | §3.9.1 (Pass 116) | Carry the compact canonical encoding of §3.8.7 as the Fact Unit's own at-rest and wire form — required fields at fixed offsets, optional fields tagged, FUID and semantic fingerprint identical to the full form — so the encoded record is self-describing without an external schema. | The encoding is fused to the record, not to any store; it is the same record in memory, on disk, or in flight.

289 | Opaque-Blob Storage Contract (Database-Neutral Boundary) | §3.9.2 (Pass 116) | Store an encoded Fact Unit in any host database, repository, or key-value store as an opaque value the store never parses, indexes internally, or interprets; expose only the FUID and fingerprint as opaque keys where an index is wanted. | The store's only obligations are to accept and return bytes unchanged, so any database qualifies and no proprietary format or query-engine capability is imposed on it.

290 | Internal Fact Query with Edge Translation | §3.9.3 (Pass 116) | Express queries over Fact Units internally in terms of the record (identifiers, fingerprint, typed adjacency, confidence, applicability scope); where external material is needed, an adapter translates the request into the host store's native language. | The external store never receives the internal fact-query form; it is a source and repository, never the reasoning engine, so the guarantees are independent of it.

291 | In-Transit and At-Rest Record Confidentiality and Integrity | §3.9.4 (Pass 116) | Carry the integrity hash chain (§3.8.6) and the in-record safety classification and access-control list (§3.1) within the encoded blob across the storage boundary; under the zero-knowledge and key-custody provisions the encoded record is unreadable by the host store. | Any store will hold the record and no store can read, reorder, or leak its meaning; database neutrality is preserved and strengthened.

292 | Fact Unit Language | §3.10 (Pass 117) · Provisional claim 342 | Express every Fact Unit in a formal, language-independent assertion language whose vocabulary is the governed predicate catalogue, whose grammar is the record schema, and whose identity function is the semantic fingerprint; refuse a non-conforming candidate at admission. | Two records asserting the same proposition in different natural languages are the same statement; retrieval, contradiction detection, and regeneration operate on the assertion language, not on natural-language text. Recited as provisional claim 342.

293 | Contextual Value Models | §109.9 (Pass 118) | Store cultural norms, social values, and ergonomic preferences as registered, versioned, provenance-carrying configuration records — declared, never inferred — that only the interpretive layer's ergonomic, cultural, and social modulations read. | A Contextual Value Model modulates expression only; it never widens or narrows scope, admits or excludes a record, alters a confidence value, or extends vocabulary, and is not a grounding source. It is distinct from a Fact Unit (a fact about a culture) and from a World-View Model concept; context proper is carried by the context lock.

Appendix B — Filing Notes for Counsel

[0124] This appendix is a working note and is not part of the disclosure. It should be removed before filing.

Item | Note

Fee | Provisional filing fee only: \$325 large entity, \$130 small, \$65 micro (37 C.F.R. § 1.16(d)). No claim fees apply at § 111(b) irrespective of the 128 claims included.

Page count | A size fee under § 1.16(s) applies above 100 sheets. This document exceeds that with Appendix A included; the fee is \$450/\$180/\$90 per additional 50 sheets and is worth paying rather than trimming the appendix, which carries the function-level enablement.

Filing method | File through Patent Center in DOCX. Paper filing adds \$200 and a separate non-DOCX surcharge applies.

Drawings | Informal figures are acceptable at § 111(b). Formal sheets meeting § 1.84 are required only on conversion.

12-month bar | A non-provisional claiming benefit under § 119(e) must be filed within 12 months. Diarise the date on filing.

New matter | Once filed, § 132(a) prohibits adding new matter. Any subject matter not in this document takes the date of the later filing that first discloses it.

Public disclosure | Establish whether any mechanism disclosed here has been described publicly. A public description may start the § 102(b)(1) one-year window independently of this filing date.

Claim selection on conversion | Do not carry all 128 claims into one non-provisional. See the note at the head of the Claims section.

Unverified material | Five formal-method statements referenced in the specification are marked unverified and have not been

executed in a proof environment. They are presented as illustrative and nothing in this disclosure depends on them.